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Medicinal plants in diabetic nephropathy: bioactive compounds, mechanisms, and therapeutic potential

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Diabetic nephropathy (DN) is known as one of the most critical microvascular complications associated with diabetes mellitus, and is a primary cause of chronic kidney disease and end-stage renal disease worldwide. Renal injury in DN is primarily driven by persistent hyperglycemia, which sequentially triggers oxidative stress, activation of innate immune signaling pathways, pro-inflammatory cytokine production, accumulation of advanced glycation end products, activation of the renin–angiotensin–aldosterone system (RAAS), and extracellular matrix deposition. Exisitng treatments, such as glycemic control, antihypertensive agents, and RAAS inhibitors, are efficacious in slowing disease progression. Alongside these standard therapies, medicinal plants and their bioactive constituents have garnered substantial attention as potential complementary approaches. The aim of this review is to highlight the therapeutic potential of medicinal plants and their phytoconstituents, leading to the prevention and management of DN, while elucidating associated pharmacological mechanisms. Several *in vitro* and *in vivo* studies have shown that various medicinal plants exert nephroprotective effects through antihyperglycemic, antioxidant, anti-inflammatory, anti-apoptotic, and lipid-lowering mechanisms, as well as by inactivating innate immune pathways. Notable phytoconstituents, including quercetin, rutin, curcumin, berberine, mangiferin, gallic acid, and thymoquinone, have demonstrated promising potential in enhancing renal biomarkers, reducing oxidative stress, suppressing inflammatory mediators, and preserving renal tissue structure. Altogether, medicinal plants emerge as promising multi-target therapeutic candidates for DN, warranting rigorous clinical trials to better evaluate their future therapeutic translation.

KEYWORDS
alternative therapy, cytokines, diabetic nephropathy, medicinal plants, pathophysiology, phytoconstituents, renoprotective mechanism

1 Introduction

Diabetes mellitus (DM) is characterized by chronic hyperglycemia resulting from impaired insulin secretion, insulin resistance, or a combination of both (1). Persistent hyperglycemia leads to a broad spectrum of complications due to progressive damage imposed on several organs and systems. Complications related to DM can be classified as

microvascular and macrovascular. Microvascular complications, including diabetic retinopathy, neuropathy, and nephropathy, account for major contributions to morbidity in the diabetic population (1). In contrast, macrovascular complications, entailing coronary heart disease, peripheral vascular disease, and ischemic stroke, prevail as the leading cause of mortality among diabetic patients (2). Notably, as a primary microvascular complication, diabetic nephropathy (DN) prevails as a major driver of end-stage renal disease (ESRD) and a defining stage in the progression of renal microvascular damage (3).

DM stands as the leading cause of ESRD at present times, which often requires treatments like dialysis or kidney transplantation. However, it is important to highlight that numerous individuals may succumb to cardiovascular diseases before reaching ESRD, stemming from the well-established link between DN and cardiovascular morbidity (2). The clinical presentation of DN is characterized by elevated arterial blood pressure, chronic albuminuria, and a progressive decline in glomerular filtration rate (GFR). Exacerbated hypertension and ongoing proteinuria aggravate these clinical features. Further acceleration of renal deterioration is driven by additional factors, including dysregulated glycemic control and abnormal lipid metabolism, though these risk factors warrant further characterization (4).

The pathophysiology of DN remains complex and incompletely elucidated despite advances in medical science; this ambiguity limits the efficacy of clinical treatment, resulting in elevated healthcare costs and suboptimal clinical outcomes (5). The existing therapies focus primarily on managing metabolic and hemodynamic disturbances, including glycemic control and blood pressure regulation (6). However, the therapeutic landscape of diabetic kidney disease (DKD) has evolved considerably. Current management extends beyond conventional glycemic and blood pressure control. Renin–angiotensin–aldosterone system (RAAS) blockade remains a component of therapy, while sodium-glucose cotransporter-2 (SGLT2) inhibitors have emerged as key kidney and cardiovascular-protective therapies for eligible patients with type 2 diabetes mellitus (T2DM) and chronic kidney disease (CKD) (7). Glucagon-like peptide-1 (GLP-1) receptor agonists have also gained a role because of their demonstrated cardiovascular benefits and emerging evidence of kidney-protective effects (8). In addition, the non-steroidal mineralocorticoid receptor antagonist finerenone has demonstrated kidney and cardiovascular benefits in patients with T2DM and CKD, representing another important disease-modifying option (7, 9). Despite these advances, DKD continues to carry a substantial residual risk of kidney disease progression and cardiovascular complications, indicating that existing interventions do not eliminate all aspects of disease burden (10). Therefore, there remains a rationale for investigating approaches that may provide additional nephroprotective effects alongside established pharmacological therapies rather than replacing them.

Such limitations mandate greater attention to alternative and complementary therapeutic strategies, particularly in therapeutic agents derived from natural sources and medicinal plants. Plant-derived bioactive compounds are gaining traction for their efficacy, safety, and accessibility in the treatment of DM and its complications (11). For example, phytoconstituents have been investigated for their ability to modulate pathways including NOD-like receptor family pyrin domain containing 3 (NLRP3), nuclear factor kappa B (NF-κB), and transforming growth factor beta-1 (TGF-β1)-associated signaling which are implicated in inflammation, oxidative stress and renal fibrosis (12, 13).

Importantly, several of these effects overlap with pathological pathways already influenced by contemporary DKD therapies. This mechanistic overlap does not imply that medicinal plants can replace therapies; rather, it provides a scientific basis for investigating whether plant-derived compounds may complement existing treatment strategies through additional or overlapping mechanisms. Recent reviews have proposed phytochemicals and medicinal plants as potential adjunctive approaches in DKD management while emphasizing the need for stronger clinical evidence before their routine incorporation into clinical practice (13, 14).

The affordability and availability of herbal remedies in many developing countries provide distinct advantages over conventional medicines. Nevertheless, the therapeutic translation of plants remains challenging due to variations in phytochemical composition, difficulties in standardization, limited bioavailability, uncertainty about optimal dosage, potential drug–herb interactions, and the limited number of adequately designed clinical trials (15). Medicinal plants and their phytoconstituents, therefore, offer an avenue for the complementary management of DKD, but their potential should be evaluated alongside, rather than in opposition to, established evidence-based therapies.

The primary aim of this review is to investigate the potential of medicinal plants and their bioactive compounds in the prevention and management of DN. This article will also explore the mechanisms underlying their nephroprotective effects and discuss their potential future applications as complementary or adjunctive therapeutic agents in contemporary DKD management.

2 Literature search methods

This review article was written by conducting literature searches covering the period between 1990 and 2026 across seven databases: Web of Science, PubMed, Google Scholar, ScienceDirect, Scopus, the Cochrane Library, and Google. The search employed keywords including diabetes mellitus, diabetic nephropathy, medicinal plants, traditional plants for kidney dysfunction, and alternative therapies.

Following preliminary screening, an initial pool of 514 potentially relevant records was identified; and 257 articles were scrutinized and included to construct the primary focus of this review detailing the botanical therapies. Articles were included if they provided experimental (*in vitro* or *in vivo*) or clinical evidence on the antidiabetic or renoprotective effects of medicinal plants and isolated phytochemicals in DM and DN models. Studies were excluded if they met any of the following criteria: duplicate records, conference proceedings, book chapters, meta-analyses, case studies and case series, or studies failing to evaluate relevant renal biomarkers or metabolic parameters.

3 Epidemiology

The incidence of DM is globally rising at an alarming rate (6). At present, DM affects ~537 million people worldwide, accounting for 10% of the overall population. This estimation is expected to rise to ~783 million by 2045 (1). Nearly 90% of all occurrences of DM are T2DM, which typically affects those who are ≥40 years old. Roughly 10% of diabetic people have type 1 diabetes mellitus (T1DM), commonly referred to as juvenile onset DM, which is brought on by the autoimmune destruction of the pancreatic beta cells (2, 3). DKD is now the most common cause of ESRD worldwide. An increased incidence risk of ESRD is reported mostly in older patients aged between 45 and 64 years (4). Furthermore, individuals with T1DM have a higher risk of developing nephropathy compared to those with T2DM. However, due to the larger number of people with T2DM, there are more cases of nephropathy in this group overall (5).

Although other factors can lead to kidney damage in individuals with DM, a diagnosis of DN is strongly indicated by specific key features. Renal biopsies are commonly used to identify the underlying cause of kidney disease, but are typically reserved for suspected non-diabetic conditions rather than for patients with diabetes (16). The presence of both albuminuria and retinopathy indicates an increased likelihood of DN, which is often associated with the development of Kimmelstiel-Wilson lesions, a form of nodular glomerulosclerosis (6). Clinical guidelines indicate that CKD can be linked to DM when either macroalbuminuria (>300 mg/24 hour) or microalbuminuria (30–300 mg/24 hour) is present, alongside diabetic retinopathy or a history of diabetes for more than 10 years (17). On the other hand, if retinopathy and autonomic neuropathy are absent at the time of diabetes diagnosis, it suggests a non-diabetic origin for persistent albuminuria in individuals with DM (18).

4 Etiology

The progression of DN in T1DM and T2DM is linked to increased urine albumin excretion, initially presenting as microalbuminuria (30–300 mg/g creatinine in a spot urine sample) (17). Risk factors include a family history, uncontrolled blood glucose, duration of hyperglycemia, high blood pressure, alcohol consumption, smoking, poor diabetic diet knowledge, physical inactivity, obesity, and secondary diseases (11). Elevated glycated hemoglobin (HbA1c) levels decrease GFR, and effective glucose control reduces albuminuria risk (19). Diabetic patients are twice as likely to develop hypertension (20), and the coexistence of hypertension and diabetes for 10–15 years leads to nephropathy due to sodium retention and increased vascular resistance (21). Dyslipidemia accelerates DN via increased low-density lipoprotein (LDL), very low-density lipoprotein (VLDL), triglyceride (TG), and decreased high-density lipoprotein (HDL) (22), promoting glomerulosclerosis and nephropathy (18). A high body mass index is associated with the development of DN and the progression to ESRD due to increased GFR, elevated capillary hydrostatic pressure, and proteinuria, all of which contribute to the onset of ESRD (23). In addition, smoking promotes microalbuminuria, arterial stiffness, oxidative stress, and renal vasoconstriction in diabetic patients (17, 24). Diabetic retinopathy is prevalent in patients with proteinuria or dialysis, reflecting systemic microvascular damage and declining renal function (25). Elevated oxidative stress markers and increased levels of proinflammatory cytokines contribute to the development of DN and accelerate the decline in renal function (26). Matrix metalloproteinases (MMPs) modulate extracellular matrix (ECM) dynamics in DN, and abnormal MMP activity disrupts ECM, promoting renal hypertrophy and fibrosis (27).

5 Pathophysiology

CKD is characterized by a variety of hemodynamic, metabolic, cellular, and molecular abnormalities. In case of DN, the majority of these changes are predominantly caused by persistent hyperglycemia and may lead to other pathological abnormalities (28).

A hallmark histological feature of DN is the increased accumulation of ECM in both the glomerulus and interstitium. This excessive buildup leads to glomerulosclerosis (scarring of the glomeruli) and arteriolosclerosis (thickening of arterioles), which result in reduced glomerular perfusion and a loss of blood flow auto-regulation. However, evidence also suggests that early hemodynamic changes precede glomerular fibrosis and play a role in the initial stages of DN development (29).

DN is triggered by glomerular hyperfiltration, where high blood glucose triggers the release of vasoactive mediators, including prostaglandins, nitric oxide (NO), insulin-like growth factor-1 (IGF-1), glucagon, and vascular endothelial growth factor (VEGF). These mediators lead to the dilation of afferent arterioles. Alongside these vascular changes, early DM also involves alterations in renal tubular function, which are linked to the degree of glycemic control. The tubuloglomerular feedback mechanism reduces sodium chloride levels reaching the macula densa in the distal tubules, causing further dilation of afferent arterioles. Concurrently, elevated angiotensin II levels induce constriction of the efferent arterioles, resulting in disrupted auto-regulation and the development of glomerular hypertension (30). This excessive angiotensin II activity reflects an overactivation of the RAAS, a central driver of progressive renal damage. Given their role in disease progression, clinicians rely on angiotensin-converting enzyme inhibitors (ACEIs) and angiotensin II receptor blockers (ARBs) to block the RAAS, control blood pressure, and reduce proteinuria (31).

Irreversible reactive advanced glycation end products (AGEs) form irreversibly due to the non-enzymatic glycation of proteins and lipids under hyperglycemic conditions. Their accumulation in kidney tissues is linked to harmful effects in the glomeruli, such as increased vascular permeability, higher production of cytokines and growth factors, and upregulation of matrix proteins (32). In DN, AGEs cause damage by triggering receptor-mediated cellular responses and altering extracellular protein structure and function. Unlike in non-diabetic renal diseases, the interaction between AGEs

and their receptor (RAGE) leads to distinct pathological outcomes that are differentially regulated in DN (33). Glomerular endothelial cells, podocytes, and mesangial cells have shown an upregulation of RAGE expression in DN (34). Additionally, the AGE–RAGE interaction in podocytes is believed to play a major role in activating oxidative stress and innate immune responses, contributing to the progression of DN (Figure 1) (35).

Oxidative stress plays a significant role in mediating the early stages of cellular damage seen in diabetes-related complications. Reactive oxygen species (ROS) are generated when elevated intracellular glucose levels are oxidized through the citric acid cycle. This occurs due to an increase in electron donors, such as NADH and $FADH_2$, entering the electron transport chain, which disrupts normal ATP synthesis in the mitochondria (36). The addition of electrons to oxygen generates highly unstable ROS. These radicals oxidize proteins, lipids, carbohydrates, and DNA, causing structural damage and dysfunction to cells. Concurrently, ROS induce cellular dysfunction by promoting lipid peroxidation (LPO), activating NF-κB and protein kinase C (PKC), increasing peroxynitrite production, and triggering apoptosis (37). Furthermore, oxidative stress directly activates the TGF-β1, Smad, and mitogen-activated protein kinase (MAPK) signaling pathways. This activation suppresses MMP production, driving the abnormal accumulation of ECM proteins. Ultimately, this disrupted ECM regulation leads to mesangial matrix expansion, tubular fibrosis, and glomerulosclerosis (Figure 1) (38).

In DN, proteinuria activates the RAAS. Angiotensin II causes changes that lead to glomerular sclerosis by enhancing the production of chemotactic factors, adhesion molecules, and fibrogenic cytokines, while also disrupting the balance between extracellular matrix formation and degradation. Increased levels of angiotensin II have been reported in DN as a result of proteinuria and ROS (1). Podocytes accumulate damaged mitochondria and oxidized proteins due to angiotensin II, which stimulates the production of ROS. This process triggers autophagy in the cells and leads to an increase in glomerular filtration (Figure 1) (35).

Nephropathy was initially considered a non-immune disease. However, recent research indicates that immune responses and inflammation are key drivers of its onset and progression (1). Studies suggest that inflammatory cytokines contribute to microvascular complications, including DN, retinopathy, and neuropathy. Hyperglycemic conditions activate innate immune pathways that produce these cytokines and exacerbate inflammation (39). Elevated levels of pro-inflammatory cytokines such as interleukin-1 (IL-1), interleukin-6 (IL-6), interleukin-18 (IL-18), and tumor necrosis factor-alpha (TNF-α) in both local and invading glomerular cells are linked with DN (Figure 1) (35).

More specifically, in DN, innate immune activation mainly depends on pattern recognition receptors (PRRs), especially Toll-like receptors (TLRs) and NOD-like receptors (NLRs) (40). Chronic hyperglycemia and oxidative stress lead to the buildup of damage-associated molecular patterns (DAMPs), which in turn activate TLR4 and its downstream adaptor, MyD88 (41). This cascade not only maintains NF-κB activation but also activates the NLRP3 inflammasome. The formation of the NLRP3 inflammasome then triggers the cleavage and maturation of pro-inflammatory cytokines such as interleukin-1 beta (IL-1β) and IL-18, fueling a cycle of macrophage infiltration and intense tubulointerstitial inflammation (40, 41).

6 CKD versus DN

Declining kidney function is a common physiological aspect of aging, but CKD remains a distinct clinical condition that is highly prevalent among the elderly. Notably, more than 50% of those aged 80 and older have an estimated GFR below 60 mL/min/1.73 m^2 (42). The development of CKD is influenced by several risk factors, including hypertension, obesity, physical inactivity, smoking, and genetic predispositions (43). Consequently, the condition significantly increases the risk of cardiovascular events and overall

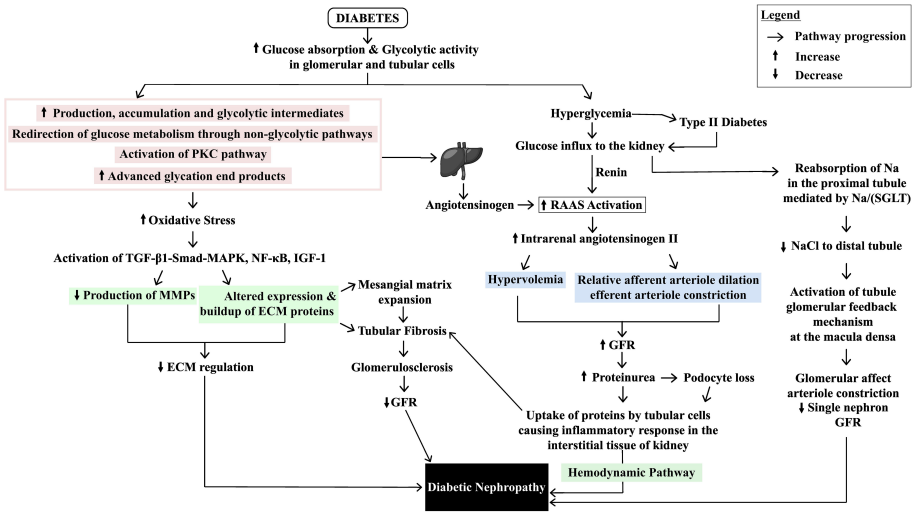


FIGURE 1
Pathophysiology of DN.

mortality (44). Additionally, drug-induced kidney injury serves as another separate cause of renal decline. Exposure to nephrotoxic drugs like certain antibiotics, nonsteroidal anti-inflammatory drugs (NSAIDs), and ACEIs can cause direct vascular and structural damage. This toxicity can result in severe kidney issues, such as glomerular lesions, acute interstitial nephritis, tubular necrosis, osmotic nephropathy, and acute nephrocalcinosis (45).

In contrast, DN is a specific, gradually worsening microvascular complication primarily caused by persistent hyperglycemia. It affects about 40% of people with diabetes and is a leading cause of ESRD worldwide, often leading to dialysis or kidney transplants. Unlike general age-related decline or sudden drug toxicity, DN follows a distinctive clinical progression. It usually appears 5 to 15 years after diabetes begins, with microalbuminuria as the earliest sign. As the disease advances, proteinuria grows and becomes less selective, often reaching nephrotic levels. This increase in proteinuria closely correlates with a steady decline in GFR. Once microalbuminuria or overt proteinuria develops, patients typically experience continuous kidney function loss, culminating in ESRD (46).

7 Alternative therapies for DN

Medicinal plants have served as a valuable adjunctive approach to synthetic drugs exhibiting diverse health benefits, including lowering blood glucose levels, promoting wound healing, improving blood circulation and pressure, and treating infections, among other conditions (47). The rising global burden of chronic pathologies poses profound health challenges and severely compromises patients' quality of life. Conventional synthetic drugs have demonstrated considerable therapeutic efficacy in the treatment and management of DN. To build upon these established regimens, there is growing interest in evaluating plant-derived extracts and isolated compounds as potential complementary therapeutic options (48).

In recent years, herbal medicines have gained prominence as viable alternatives to traditional single-chemical medicines. Plant extracts contain rich phytoconstituent profiles and structural diversity that act on different pharmacological targets (Table 1). These attributes stem from its multiple therapeutic actions and the potential for synergistic effects, in which distinct phytochemicals cooperatively enhance bioactivity (49). The presence of bioactive components in these plant extracts has significantly contributed to the progression in both *in vitro* and *in vivo* studies. Furthermore, the plant extracts provide widely accessible options, though their clinical safety profiles require thorough validation compared with conventional drugs (50). Under these conditions, natural product formulations hold a considerable promise for the management of DN, especially in developing countries.

7.1 *Abelmoschus esculentus* (L.) Moench

As a staple in the Malvaceae, *Abelmoschus esculentus* (okra) is cultivated across tropical and subtropical regions and is valued for its pharmacological potential, particularly against T2DM and oxidative stress. Ethanolic, methanolic, and aqueous extracts reduce glucose absorption and post-prandial levels by inhibiting α-glucosidase and α-amylase (51). Methanolic extracts increase superoxide dismutase (SOD) and catalase (CAT) activities, decrease malondialdehyde (MDA) levels and NO levels, and improve kidney function by lowering blood urea nitrogen (BUN) and creatinine (52). The active compounds linalool and eugenol enhance glucose uptake and protect renal tissue by reducing oxidative stress and downregulating the NF-κB/TNF-α pathways (53, 54).

7.2 *Abrus precatorius* L.

Abrus precatorius, or jequirity bean, from the Fabaceae family, is used in traditional medicine to treat metabolic disorders and support kidney health. Its methanolic and ethanolic extracts improve glucose tolerance, stimulate insulin secretion, and lower blood glucose (55, 56). Active flavonoids, such as luteolin, enhance insulin sensitivity and protect against renal apoptosis. It also lowers serum creatinine, urea, and uric acid levels, indicating improved kidney function (57).

7.3 *Acacia arabica* (Lam.) Willd.

Acacia arabica (babul tree) from the Fabaceae family has antidiabetic, antioxidant, and renoprotective properties. Studies show *A. arabica* extracts enhance glucose uptake, inhibit α-amylase, and reduce oxidative stress, improving glucose regulation and post-prandial levels (58). Flavonoids such as quercetin and kaempferol boost antioxidant enzymes and mitigate oxidative stress and inflammatory cytokines (TNF-α, IL-1β), protecting the kidneys and supporting metabolic health (59, 60).

7.4 *Acacia catechu* (L.f.) Willd.

Acacia catechu, a deciduous tree native to South and Southeast Asia, has been used in traditional medicine to manage blood glucose and kidney health. Ethanolic extracts inhibit enzymes such as α-glucosidase and α-amylase, and scavenge 2,2-diphenyl-1-picrylhydrazyl (DPPH) radicals, thereby reducing glucose absorption and oxidative stress. They also boost antioxidant enzyme activity and reduce ROS (61). Studies show improved kidney tissue, with less tubule dilation and glomerular congestion (62). Compounds like chrysin, umbelliferone, and naringenin enhance antioxidant defenses, lower inflammatory markers (TNF-α, interleukin-33 [IL-33], IL-1β), and reduce serum creatinine and urea, indicating therapeutic potential (63–65).

7.5 *Achillea millefolium* L.

Achillea millefolium (yarrow) is a perennial herb in the Asteraceae family, native to temperate regions of the Northern Hemisphere, with a long history of use for its antidiabetic and renoprotective effects. Aqueous extracts lower serum glucose, creatinine, and urea, and improve red blood cell (RBC) and hemoglobin counts (66). Luteolin enhances insulin sensitivity, reduces renal apoptosis, and lowers cholesterol and TG levels, thereby supporting metabolic health (57). Apigenin further benefits kidney function by

TABLE 1 Medicinal plants and their pharmacological effectiveness against diabetes and DN.

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
Abelmoschus esculentus L. Moench	Ethanol, methanol, aqueous	*In vitro*	50, 100, 150, 200, 250 μg/mL	Inhibition assay for α-glucosidase, α-amylase, and DPPH radical scavenging activity	↓ Glucose absorption rate, ↓ post-prandial glucose levels, and reduced oxidative stress	(51)
Abrus precatorius L.	Methanol	*In vivo*	200 mg/kg body weight	Assay for insulin secretion, oral glucose tolerance test (OGTT)	↓ Blood glucose levels, ↑ insulin secretion	(55)
Acacia arabica (Lam.) Willd.	Hot water	*In vivo* and *in vitro*	250 mg/5 mL/kg body weight	Glucose diffusion tests, glucose uptake tests, inhibition assay for α–amylase, and DPPH radical scavenging activity	↓ Glucose diffusion rate, ↑ glucose uptake into skeletal and adipose tissues, enhanced insulin secretion, ↓ post-prandial blood glucose levels, and reduced oxidative stress	(58)
Acacia catechu (L.f.) Willd.	Ethanol	*In vitro*	10, 20, 30, 40, and 50 μg/mL	Inhibition assay for α-glucosidase, α-amylase, and DPPH radical scavenging activity	↓ Glucose absorption rate, ↓ post-prandial glucose levels, ↓ oxidative stress	(61)
	Ethanol	*In vivo* and *in vitro*	75 mg/kg body weight	Analysis of biochemical and oxidative stress assessment parameters, histological analysis	↑ Antioxidant enzyme activity of SOD and GSH, ↓ ROS generation, showed a reduction in tubule dilation and degeneration with normal glomerular space, ↓ glomerular congestion	(62)
Achillea millefolium L.	Aqueous	*In vivo*	90 and 270 mg/kg body weight	Biochemical and kidney function assay	↓ Serum glucose levels, improved RBC and hemoglobin count, and declined creatinine and urea levels	(66)
Achyranthes aspera L.	Methanol, petroleum ether	*In vitro*	160, 250 μg/mL	Inhibition assay for α-glucosidase, α-amylase, and DPPH radical scavenging activity	↓ Glucose absorption rate, ↓ post-prandial glucose levels, and ↓ oxidative stress	(68)
Aegle marmelos (L.) Corrêa	Aqueous	*In vivo*	100, 250, and 500 mg/kg body weight	OGTT, lipid profile	↓ Fasting blood glucose levels, ↑ HDL, ↓ LDL, and TG levels	(71)
Aframomum melegueta K.Schum.	Ethanol, ethyl acetate, water	*In vitro*	30–240 μg/mL	Inhibition of hemoglobin glycosylation, α-amylase, α-glucosidase, and DPPH free radical scavenging activities	↓ HbA1c levels, delayed carbohydrate digestion, ↓ post-prandial glucose levels, and ↓ oxidative stress	(78)
Ageratum conyzoides L.	Aqueous	*In vivo*	100, 200, and 300 mg/kg body weight	OGTT, glucose uptake	↓ Fasting blood glucose levels, enhanced insulin secretion	(82)
Allium cepa L.	Aqueous	*In vivo*	200, 250, and 300 mg/kg body weight	Lipid profile assay, OGTT	↓ Fasting blood glucose levels, ↓ total serum lipids	(84)
Allium sativum L.	Aqueous	*In vivo*	200, 250, and 300 mg/kg body weight	Lipid profile assay, OGTT	↓ Fasting blood glucose levels, ↓ total serum lipids	(87)
Aloe vera Burm.f.	Methanol	*In vitro*	5 mg/mL	Inhibition assay of α-amylase, α-glucosidase, and pancreatic lipase	↓ Advanced glycation end products formation, ↓ starch digestion, and ↓ post-prandial glucose levels	(90)
Alstonia scholaris (L.) R. Br.	Total alkaloids	*In vivo* and *in vitro*	Oral treatment for 6 weeks; vallesamine (5 mg/kg)	Oxidative stress, NO and LDH assays, RT-qPCR and Western blotting	↓ Proteinuria, urinary protein-to-creatinine ratio, hyperglycemia and hyperlipidemia; improved serum protein and albumin levels; reduced renal injury and matrix hyperplasia	(205)
Amaranthus tricolor L.	Methanol	*In vitro* and *in vivo*	200 and 400 mg/kg body weight	Ferric reducing ability of plasma assay, DPPH assay, lipid profile assay	↓ Blood glucose levels, ↑ hemoglobin levels, ↑ HDL levels, ↓ serum TG, LDL, and total cholesterol, ↓ and oxidative stress in the kidneys	(93)
Anacardium occidentale L.	Methanol	*In vivo*	200 mg/kg body weight	Lipid profile and lipid peroxidative assay, creatinine and urea tests, plasma protein tests,	↓ Blood glucose, ↓ total cholesterol, TGs, atherogenic index (LDL/HDL ratio), ↓ urea, and creatinine levels, ↓ and plasma MDA	(95)
Ananas comosus (L.) Merr.	Ethanol	*In vivo*	400 and 500 mg/kg body weight	Determination of urea, uric acid, creatinine, serum protein, HbA1c levels	↓ Blood glucose, ↓ HbA1c levels, ↓ creatinine, urea, uric acid, and serum protein levels	(97)
Andrographis paniculata (Burm.f.) Wall.	Chloroform	*In vivo*	50, 100, and 150 mg/kg body weight	Analysis of blood glucose, protein, albumin, and creatinine levels	↓ Blood glucose, protein, urinary albumin, and creatinine levels; inhibited the induction of albuminuria, proteinuria, and uremia	(99)
Annona squamosa L.	Hot water	*In vitro* and *in vivo*	250 mg/5 mL/kg body weight	DPP-IV assays, glucose diffusion, starch digestion, glucose uptake, protein glycation, OGTT	↓ Insulin glycation, enhanced insulin secretion, ↓ blood glucose levels, delayed glucose absorption, improved glucose tolerance	(101)

(Continued)

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TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
Arachis hypogaea L.	Essential oil from seeds	*In vivo*	500 and 1000 mg/kg body weight	OGTT, insulin level, liver glycogen content, HbA1c, lipid profile test	↓ Blood glucose levels, ↑ liver glycogen, and insulin levels, ↓ HbA1c, improved overall lipid profile (↑ HDL and ↓ LDL)	(104)
Arctium lappa L. (*Fructus arctii*)	*Fructus arctii* extract	*In vivo* and *in vitro*	100 mg/kg/day body weight	CCK-8, EdU, Western blotting, IHC, RT-qPCR and renal histology	↓ Blood glucose, proteinuria and renal collagen deposition; ↑ Apoh and PPAR-γ expression; reduced renal cell proliferation and fibrosis through the Apoh/PPAR-γ pathway	(206)
Artemisia absinthium L.	Ethanol	*In vivo*	250, 500, and 1,000 mg/kg body weight	Lipid profile, serum creatinine, urea, total urinary protein	↓ Blood glucose levels, ↓ serum creatinine, urea, and total urinary protein significantly; enhanced insulin secretion, ↓ total cholesterol levels	(106)
Artocarpus heterophyllus Lam.	Acetone	*In vitro* and *in vivo*	400 mg/kg body weight	DPPH radical scavenging activity, assay of α-amylase and α-glucosidase activity, antioxidant enzyme activity, liver glycogen content	↓ Fasting blood glucose levels, ↑ DPPH radical scavenging activity, ↓ oxidative stress, ↓ glucose absorption, ↑ liver glycogen content, ↑ antioxidant enzyme activities in the liver	(108)
	Aqueous, methanol	*In vivo*	100, 200 mg/kg body weight	Biochemical assay, kidney function tests, histological analysis	↑ RBC count and hemoglobin concentration, ↓ serum creatinine, urea, and uric acid; improved renal tissue architecture	(207)
Asparagus racemosus Willd.	Ethanol	*In vivo*	200 and 400 mg/kg body weight	Blood glucose determination, lipid profile, antioxidant enzyme activity	↓ Fasting blood glucose levels, ↓ TG and total cholesterol levels, ↑ antioxidant enzyme activity of GSH, SOD, and CAT	(111)
Azadirachta indica A. Juss.	Ethanol	*In vivo*	200, 400, and 800 mg/kg body weight	OGTT	↓ Blood glucose, improved glucose tolerance	(113)
Berberis aristata DC.	Ethanol	*In vivo*	100 and 200 mg/kg body weight	Analysis of hypoglycemic activity	↓ Blood glucose levels, stimulated insulin release	(115)
Berberis vulgaris L.	Aqueous	*In vivo*	175 and 625 mg/kg body weight	Serum lipid profile, hypoglycemic activity	↓ Blood glucose levels, ↓ total cholesterol, and TGs in serum significantly	(119)
Biophytum sensitivum (L.) DC.	Aqueous	*In vivo*	200 and 400 mg/kg body weight	Analysis of blood glucose, albumin, total urinary protein and lipid profile	↓ Blood glucose levels, ↓ urinary albumin and total urinary proteins, ↓ total cholesterol, TGs, and LDL and ↑ HDL levels	(121)
Bixa orellana L.	Aqueous	*In vivo*	200 mg/kg body weight	Assay of biochemical and kidney function parameters	↓ Blood glucose levels, ↓ serum creatinine, urea, and uric acid content	(124)
Boerhavia diffusa L.	Ethanol	*In vivo*	500 mg/kg body weight	Analysis of blood glucose, kidney function, antioxidant activity	↓ Blood glucose levels, improved ionic balance and renal Na^+–K^+ ATPase activity, ↑ SOD, CAT, GSH, and GPx activity	(208)
Bombax ceiba L.	Ethanol	*In vivo*	70, 140, and 280 mg/kg body weight	Analysis of blood glucose, serum biochemical parameters, OGTT, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ serum insulin, ↓ HbA1c, improved glucose tolerance, ↓ total cholesterol, TG, LDL, ↑ HDL, ↑ SOD activity, and ↓ MDA	(126)
	Ethanol	*In vivo*	80, 120 mg/kg body weight	Assay of fasting blood glucose, kidney function tests, histological analysis	↓ Fasting blood glucose levels, ↓ urinary albumin, serum creatinine, and BUN; prevented oxidative stress in mesangial cells and improved renal injury	(127)
Brassica juncea (L.) Czern.	Aqueous	*In vivo*	250, 350, and 450 mg/kg body weight	Assay of serum insulin and blood glucose	↑ Insulin secretion and ↓ blood glucose levels significantly	(130)
	Methanol	*In vivo*	450 mg/kg body weight	Biochemical assay, antioxidant enzyme activity assay	↓ Blood glucose, TG, and cholesterol levels, enhanced SOD, GSH, and CAT activity in kidneys, ↓ renal MDA, and NO levels	(131)
Brassica oleracea L.	Ethanol	*In vivo*	200, 400, and 800 mg/kg body weight	OGTT, effect of fasting assay of serum insulin, hemoglobin, and HbA1c	↓ Fasting blood glucose levels, improved glucose tolerance, enhanced insulin secretion, ↑ hemoglobin levels, and ↓ HbA1c levels	(133)
	Aqueous	*In vivo*	1 g/kg body weight	Biochemical assay, kidney function tests, antioxidant enzyme assay	↓ Blood glucose levels, restored blood weight loss, enhanced SOD, GSH, and CAT activity, ↓ serum creatinine, urea, and renal MDA levels	(134)
Caesalpinia decapetala (Roth) Alston	Hydromethanol	*In vivo*	300 and 500 mg/kg body weight	Analysis of serum lipid profile, urea, creatinine	↓ Blood glucose levels, ↓ total cholesterol, serum TGs, ↓ serum creatinine, and urea	(135)
Camellia sinensis (L.) Kuntze	Hot water	*In vitro* and *in vivo*	250 mg/5 mL/kg body weight	Insulin-releasing effect, starch digestion, DPP-IV activity, OGTT, glucose uptake, and glucose diffusion assay	↑ Insulin secretion from the pancreas, delayed starch digestion, inhibited DPP-IV activity, improved glucose tolerance and ↑ glucose uptake, and ↓ blood glucose levels	(137)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
Carica papaya L.	Aqueous	*In vivo*	0.75, 1.5, and 3 g/100 mL	Lipid profile test, NO concentration	↓ Blood glucose levels, ↓ total cholesterol, TGs, LDL, ↑ HDL, and enhanced NO concentration	(139)
	Aqueous	*In vivo* and *in vitro*	150 and 300 mg/kg body weight	Renal biomarkers assay, antioxidant assay	↓ Urinary urea, serum creatinine, uric acid, and renal MDA levels, ↓ oxidative stress in the kidneys	(209)
Catharanthus roseus (L.) G.Don	Dichloromethane: methanol	*In vivo* and *in vitro*	500 mg/kg body weight	Glucose metabolism, glycogen content, effect on GSH	↑ GSH concentration and ↓ oxidative stress, enhanced glucose metabolism, ↓ blood glucose levels, ↑ liver glycogen content	(140)
Centella asiatica (L.) Urb.	Aqueous	*In vivo*	50 mg/kg body weight	Determination of blood glucose, lipid profile	↓ Blood glucose levels, ↓ total cholesterol, TGs and LDL, ↑ HDL levels	(143)
Cinnamomum zeylanicum Blume	Methanol, ethanol, acetone	*In vivo*	5, 50, 300 and 2000 mg/kg body weight	Lipid profile, DPPH radical scavenging activity	↓ Blood glucose levels, ↓ oxidative stress, ↓ total cholesterol, and LDL levels	(147)
Coccinia grandis (L.) Voigt	Aqueous	*In vivo*	0.75 g/kg body weight	Analysis of glycemic parameters and insulin levels	↓ Blood glucose levels, ↓ HbA1c, enhanced insulin secretion, and ↑ serum insulin levels	(150)
	Ethanol	*In vivo*	750 mg/kg body weight	Analysis of blood glucose levels, antioxidant enzyme activity, kidney function tests	↓ Blood glucose levels, enhanced antioxidant activity in the liver and kidneys, and ↓ creatinine levels	(151)
Coccinia indica Wight & Arn.	Ethanol	*In vivo*	200 mg/kg body weight	Analysis of blood glucose, antioxidant activity, lipid profile, creatinine, BUN levels and proteinuria, OGTT	↓ Blood glucose levels, improved glucose tolerance, ↓ intestinal glucose absorption, ↓ total cholesterol and TGs, ↓ BUN, proteinuria and creatinine content	(152)
Coriandrum sativum L.	Aqueous	*In vivo* and *in vitro*	0·25–10 mg/mL	Assay of insulin secretion, glucose transport and metabolism	↓ Hyperglycemia, enhanced insulin secretion, improved glucose uptake into skeletal muscles, enhanced glucose oxidation and glycogenesis	(153)
Curcuma longa L.	Aqueous, methanol, hexane	*In vivo*	400 mg/kg body weight	Effect of weight, food and water intake, determination of fasting blood glucose levels	↓ Fasting blood glucose levels, ↓ food and water intake, ↓ body weight of diabetic rats	(155)
Eriobotrya japonica (Thunb.) Lindl.	Ethanol	*In vivo*	250 mg/kg and 500 mg/kg body weight	Determination of glucose, urea, creatinine, total urinary protein, albumin	↓ Blood glucose levels, ↓ serum creatinine, urea, total urinary proteins and urinary albumin	(160)
Euphorbia hirta L.	Ethanol	*In vivo*	400 mg/kg body weight	Determination of blood glucose, serum lipid profile	↓ Blood glucose levels, ↓ total cholesterol, LDL, VLDL, ↑ HDL	(161)
Ficus benghalensis L.	Ethanol	*In vivo*	150, 300, and 500 mg/kg body weight	Determination of blood glucose, carbohydrate, and lipid metabolism	↓ Blood glucose levels, enhanced insulin secretion, improved carbohydrate and lipid metabolism	(162)
	Ethanol	*In vivo*	100, 500, and 100 mg/kg body weight	Analysis of lipid profile, kidney function test	↓ Total cholesterol, TGs, LDL levels, ↑ HDL levels, ↓ urea, and creatinine content	(163)
Ficus racemosa Willd.	Ethanol	*In vivo*	200 and 400 mg/kg body weight	Lipid profile, determination of blood glucose, albumin, creatinine, sodium, and BUN levels	↓Blood glucose levels, ↓ serum TGs, LDL, ↑ HDL levels, ↓ serum creatinine, urinary albumin, BUN, and sodium levels	(164)
Fragaria ananassa Duchesne	Aqueous	*In vivo*	2 g/kg body weight	Assay of glucose level, lipid profile, oxidative stress, and kidney function parameters	↓ Blood glucose levels, ↓ serum cholesterol, LDL, VLDL TG levels, ↓ serum creatinine content, enhanced renal CAT activity, and ↓ MDA levels	(166)
Garcinia kola Heckel	Methanol	*In vivo*	100 and 200 mg/kg body weight	LPO analysis, blood glucose estimation, and antioxidant activity	↓ LPO, ↓ blood glucose levels, ↑ GSH levels, and ↓ oxidative stress	(167)
Ginkgo biloba L.	Aqueous	*In vivo*	100, 200, and 300 mg/kg body weight	Determination of serum glucose, lipid profile, oxidative stress markers	↓ Fasting blood glucose levels, ↓ total cholesterol, TGs, and LDL, ↑ HDL levels, inhibited the formation of MDA, ↑ GSH, SOD, and CAT activity	(210)
Glycine max (L.) Merr.	Aqueous seed extract	*In vivo*	100, 200, and 400 mg/kg body weight	Serum glucose estimation, OGTT	↓ Blood glucose levels and improved glucose tolerance	(169)
Gundelia tournefortii L.	Aqueous	*In vivo*	5, 10, 20, and 40 mg/kg body weight	Assay of biochemical and kidney function parameters	↓ Blood glucose levels, ↓ serum creatinine, and urea content	(174)
Heritiera fomes Buch.-Ham.	Hot water	*In vitro* and *in vivo*	250 mg/5 mL/kg body weight	OGTT, insulin-releasing study, starch digestion, DPP-IV activity, glucose diffusion assay	↓ Blood glucose levels, enhanced insulin secretion, improved glucose tolerance, delayed starch digestion, inhibited DPP-IV enzyme activity, inhibited glucose diffusion, and ↓ glucose absorption	(176)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
	Ethanol	*In vitro* and *in vivo*	250 and 500 mg/kg body weight	DPPH radical scavenging activity test, OGTT, lipid profile tests, kidney function assessment	↓ Blood glucose levels, improved glucose tolerance, inhibited ROS generation, ↓ creatinine, uric acid, and BUN, ↓ cholesterol, TGs, and LDL levels, and improved HDL level	(177)
Hibiscus rosa-sinensis L.	Hydromethanol	*In vivo*	400 mg/kg body weight	Analysis of blood glucose, uric acid, creatinine, total urinary protein, cholesterol, and oxidative stress parameters	↓ Blood glucose levels, enhanced insulin secretion, ↓ serum creatinine, uric acid, total urinary proteins, cholesterol, ↑ SOD, and CAT activity	(178)
Jatropha curcas L.	Methanol	*In vivo*	150 and 250 mg/kg body weight	Determination of blood glucose, plasma lipid levels, hematological parameters	↓ Blood glucose levels, enhanced insulin secretion, exhibited hematopoietic activity, ↓ TG, total cholesterol, and LDL levels, and ↑ HDL levels	(180)
	Ethanol	*In vivo*	200 mg/kg body weight	Kidney function tests, assessment of oxidative stress	↓ Serum glucose levels, enhanced CAT, SOD, and GSH in liver and kidneys, ↓ renal MDA levels, ↓ urea and creatinine content, and serum electrolytes (sodium and potassium)	(181)
Linum usitatissimum L.	Aqueous	*In vivo*	100 mg/kg, 200 mg/kg and 400 mg/kg body weight	Analysis of glycemic parameters, urea, creatinine, and oxidative stress parameters	↓ Fasting blood glucose levels, ↓ HbA1c levels, serum creatinine, and urea, ↓ lipid peroxidase levels, and ↑ SOD and CAT activity	(183)
Lycopus lucidus Turcz. var. *hirtus* Regel (*Lycopi herba*)	Crude extract	*In vivo* and *in vitro*	50, 100 mg/kg body weight	HPLC-DAD-ELSD, histology, ELISA, RT-qPCR, Western blotting, immunofluorescence and molecular docking	↓ Fasting blood glucose and renal TNF-α, IL-6 and IL-1β; improved renal function and histopathology; inhibited NF-κB/TNF-α/STAT3 inflammatory signaling and STAT3/NF-κB activation	(211)
Mangifera indica L.	Ethanol	*In vitro*	200 μg/mL	Assay of glucose uptake, DPPH radical scavenging activity, NO production, and α-amylase activity	↓ Blood glucose levels, inhibited α-amylase activity and delayed starch and sucrose digestion, ↓ oxidative stress, inhibited NO production, ↑ glucose uptake	(184)
Matricaria chamomilla L.	Aqueous	*In vivo*	100 and 200 mg/kg body weight	Analysis of blood glucose, insulin, urea, creatinine, uric acid, and lipid profile	↓ Fasting blood glucose, ↑ serum insulin levels, ↓ serum creatinine, urea, and uric acid, ↓ total cholesterol, TGs, and LDL levels	(212)
Mentha x piperita L.	Aqueous	*In vivo*	100 and 250 mg/kg body weight	Determination of blood glucose, lipid profile, and OGTT	↓ Blood glucose levels, improved glucose tolerance, ↓ total cholesterol, TGs, and LDL and VLDL levels	(213)
Mimosa pudica L.	Ethyl acetate	*In vivo*	50 mg/kg and 100 mg/kg body weight	OGTT, analysis of lipid profile, antioxidant enzyme activity, LPO, and anti-inflammatory activity	↓ Blood glucose levels, improved glucose tolerance, ↓ TGs, total cholesterol and LDL levels, ↑ HDL levels, ↑ SOD, CAT, and GPx activity, ↓ MDA formation, and pro-inflammatory cytokines in the kidneys	(214)
Momordica charantia L.	Aqueous	*In vitro* and *in vivo*	20 mg/kg body weight	Determination of blood glucose, LPO, DPPH radical scavenging activity, kidney function parameters	↓ Blood glucose levels, ↓ renal MDA formation, inhibited LPO, inhibited free radical production and ↓ oxidative stress, ↓ serum creatinine and urea levels	(187)
Momordica dioica Roxb. ex Willd.	Methanol	*In vivo*	300 mg/kg body weight	Analysis of blood glucose, insulin, and urea levels, antioxidant enzyme activity	↓ Blood glucose levels, ↑ serum insulin levels, ↓ urea levels, ↑ renal SOD, CAT, GSH, and GPx activity, and ↓ oxidative stress in the kidneys	(215)
Moringa oleifera Lam.	Methanol	*In vivo*	250 mg/kg body weight	Determination of plasma glucose, creatinine, total urinary protein, urea, albumin, effect on LPO, antioxidant, and anti-inflammatory activity	↓ Blood glucose levels, ↓ serum creatinine, urea, total urinary protein, and urinary albumin, ↓ LPO, ↑ SOD, CAT, GSH, and GPx activity, and ↓ inflammatory cytokines in the kidney	(216)
Murraya koenigii (L.) Spreng.	Aqueous, methanol, petroleum ether	*In vivo*	200 and 400 mg/kg body weight	OGTT, analysis of serum glucose, insulin, hemoglobin, albumin, and lipid profile, oxidative stress parameters	↓ Blood glucose levels, ↑ serum insulin, improved glucose tolerance, ↑ hemoglobin levels, ↓ urinary albumin levels, ↑ SOD, CAT activity, and ↓ lipid peroxidase	(217)
Musa sapientum L.	Stem juice	*In vivo*	25, 50, and 100 mg/kg body weight	Analysis of blood glucose levels, HbA1c, insulin and lipid profile	↓ Fasting and post-prandial glucose levels, ↓ HbA1c levels, ↑ serum insulin levels, and improved overall lipid profile	(218)
Nigella sativa L.	Aqueous	*In vivo*	2, 6.4, 21, 33, and 60 g/kg body weight	Analysis of blood glucose, insulin, urea, creatinine, and albumin levels, lipid profile	↓ Blood glucose levels, ↑ serum insulin levels, ↓ urea, creatinine, and urinary albumin content, ↓ total cholesterol, TGs, and LDL levels	(189)
Ocimum sanctum L.	Aqueous	*In vivo*	250 mg/kg body weight	Analysis of blood glucose, HbA1c, serum urea, and creatinine	↓ Fasting blood glucose levels, ↓ HbA1c levels, ↓ serum creatinine, and urea content	(219)
Olea europaea subsp. *europaea*	Aqueous	*In vitro* and *in vivo*	200 mg/kg body weight	Assay of α-amylase and α-glucosidase activity, DPPH radical scavenging activity, analysis of blood glucose, lipid profile, and kidney function parameters	↓ Blood glucose levels, delayed starch digestion and ↓ glucose absorption, ↓ creatinine and urea content, ↓ TGs, total cholesterol, and LDL levels	(220)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
Panax ginseng f. *trifoliolatus* Makino	Ethanol, butanol	*In vivo*	150 mg/kg body weight	OGTT, analysis of blood glucose and serum insulin, and cholesterol levels	↓ Fasting blood glucose levels, improved glucose tolerance, ↑ serum insulin levels, and ↓ plasma cholesterol levels	(221)
Phaseolus vulgaris L.	Aqueous	*In vivo*	400 mg/kg body weight	Analysis of blood glucose, total urinary protein, albumin, and cholesterol and TG levels	↓ Blood glucose levels, ↓ total urinary protein, urinary albumin, cholesterol, and TG levels	(192)
Phyllanthus amarus Schumach. & Thonn.	Aqueous	*In vivo*	200 mg/kg body weight	Assay of biochemical parameters, lipid profile, and antioxidant activity	↓ Blood glucose levels, ↓ TGs, LDL, VLDL, and total cholesterol, ↑ HDL level, prevented LPO, enhanced renal SOD, CAT, GSH activity, and ↓ oxidative stress, and ↓ total urinary protein content	(222)
Phyllanthus niruri L.	Polyphenol-rich leaf extract	*In vivo*	50, 75, 100 and 125 mg/kg body weight	Renal antioxidant enzymes, MDA, KIM-1 and NGAL mRNA; serum creatinine, urea and uric acid	↓ Serum creatinine, urea and uric acid; ↑ renal SOD and catalase; ↓ renal MDA and expression of kidney injury markers KIM-1 and NGAL	(223)
Piper betle L.	Leaf powder in 2% gum acacia	*In vivo*	75 and 150 mg/kg body weight	Analysis of blood glucose, hemoglobin, HbA1c, and kidney function parameters	↓ Fasting blood glucose levels, ↑ hemoglobin level, and ↓ HbA1c, ↓ levels of total urinary protein, serum creatinine, urea, and uric acid	(224)
Plantago ovata Phil.	Hot water	*In vivo*	0.5 g/kg body weight	OGTT, analysis of blood glucose, glucose absorption, lipid profile	↓ Post-prandial glucose levels, improved glucose tolerance, ↓ intestinal glucose absorption, ↓ total cholesterol, TGs, and non-esterified fatty acids	(225)
Pterocarpus marsupium Roxb.	Ethanol	*In vivo*	150 mg/kg body weight	Analysis of blood glucose, plasma insulin, HbA1c, serum protein, albumin, and antioxidant enzymes lipid profile	↓ Blood glucose levels, ↑ plasma insulin, ↓ HbA1c, urinary albumin, ↓ total cholesterol, TGs, LDL and VLDL, improved SOD, CAT, GSH, and GPx activity	(226)
Punica granatum L.	Aqueous	*In vivo*	250 and 500 mg/kg	Analysis of blood glucose, lipid profiles, LPO, and antioxidant enzyme activity	↓ Fasting blood glucose levels, ↓ total cholesterol, TGs, LDL and VLDL levels, ↓ tissue lipid peroxidase levels, improved SOD, CAT, GSH, and GPx	(195)
	Hydromethanolic	*In vivo*	50, 100, and 200 mg/kg body weight	Analysis of biochemical and kidney function parameters, OGTT	↓ Fasting blood glucose levels, improved glucose tolerance, improved overall lipid profile, ↓ urinary albumin, total urinary protein, serum creatinine, and BUN levels	(196)
Rehmannia glutinosa var. *makinoi* Matsuda	Hot water	*In vivo*	100 mg/kg body weight	Analysis of blood glucose, plasma insulin levels, and glucose-6-phosphatase activity	↓ Blood glucose levels, ↑ plasma insulin levels, ↑ glucose-6-phosphatase activity, and hepatic glycogen content	(227)
	Ethanol	*In vivo*	360 mg/kg body weight	Assessment of kidney function parameters, lipid profile test	↓ Serum creatinine, urea, BUN, and urine protein levels, ↓ TGs, total cholesterol, and LDL levels	(228)
Rheum ribes L.	Hydroethanol	*In vivo*	75 and 150 mg/kg body weight	Assay of biochemical and kidney function parameters, lipid profile, and histopathological analysis	↓ Blood glucose levels, ↓ LDL, TGs, and ↑ HDL levels, ↓ serum creatinine and urea content, improved tubular necrosis	(229)
Salvia miltiorrhiza var. *miltiorrhiza*	Aqueous	*In vivo*	500 mg/kg body weight	Assay of biochemical and kidney function parameters, analysis of antioxidant and anti-inflammatory activity, lipid profile	↓ Blood glucose levels, BUN, urinary protein excretion and creatinine content, ↓ LDL, TG levels, ↓ expression of AGEs, pro-inflammatory cytokines, and oxidative stress	(230)
Santalum album L.	Petroleum ether	*In vivo*	10 μg/ kg body weight	Analysis of blood glucose, HbA1c, lipid profile, and assessment of kidney function parameters	↓ Blood glucose levels, ↓ HbA1c levels, ↓ total cholesterol, TG and LDL levels, ↓ serum urea and creatinine content	(231)
Scoparia dulcis L.	Ethanol	*In vivo*	100 and 200 mg/kg body weight	Analysis of blood glucose, OGTT, assay of DPPH radical scavenging activity	↓ Fasting blood glucose levels, improved glucose tolerance, inhibited free radical formation, and ↓ oxidative stress	(198)
Sesamum indicum L.	Ethanol	*In vivo*	500 mg/kg body weight	Assay of biochemical kidney function parameters	↓ Blood glucose levels, ↓ total urinary protein, albumin and globulin, BUN, serum creatinine, and uric acid levels	(232)
Swertia chirayita (Roxb.) H.Karst.	Aqueous	*In vitro* and *in vivo*	25 g/L	Assay of blood glucose, insulin secretion, glucose uptake, and protein glycation	↓ Blood glucose levels, ↑ cellular glucose uptake, enhanced insulin secretion, and ↓ protein glycation	(233)
	Methanol	*In vitro* and *in vivo*	100–200 mg/kg body weight	Assessment of kidney function, antioxidant enzyme activity	↓ Creatinine and urea content, enhanced SOD, CAT, and GSH activity, and ↓ renal MDA levels	(234)
Syzygium cumini (L.) Skeels	Ethanol	*In vivo*	1.25 and 2.5 g/kg body weight	Analysis of blood glucose and lipid profile	↓ Fasting blood glucose levels, ↓ LDL levels, and ↑ HDL levels	(235)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Renoprotective effect models associated with DN						
Tamarindus indica L.	Ethanol	*In vivo*	300 and 500 mg/kg body weight	Analysis of serum glucose and lipid profile	↓ Fasting blood glucose levels, ↓ total cholesterol, TGs, LDL and VLDL, and ↑ HDL levels	(236)
Terminalia chebula Retz.	Methanol	*In vivo*	50, 100, 200, and 400 mg/kg body weight	Analysis of blood glucose, albumin, creatinine, total urinary protein, and lipid profile	↓ Blood glucose levels, ↓ urinary albumin, serum creatinine, total urinary proteins, ↓ serum TGs, LDL, VLDL, and ↑ HDL levels	(201)
Tinospora cordifolia (Willd.) Miers ex Hook.f. & Thomson	Ethanol	*In vivo*	200 mg/kg body weight	Analysis of blood glucose, urea, creatinine, and uric acid	↓ Blood glucose levels, restored urea, creatinine, and uric acid levels to normal levels	(237)
Trigonella foenum graecum L.	Methanol	*In vivo*	500 mg/kg and 1 g/kg body weight	Analysis of blood glucose, renal function test, antioxidant enzyme activity	↓ Blood glucose levels, ↓ creatinine, urinary albumin, BUN, and total urinary protein levels, ↓ MDA, ↑ SOD, and CAT activity, ameliorated renal hypertrophy, nodular glomerular sclerosis, hyaline arteriolosclerosis, and glomerulonephritis	(238)
	Methanolic fenugreek seed extract	*In vivo*	100, 200 and 400 mg/kg body weight	DPPH assay, renal function markers, urinary albumin, AGE-related assessments, inflammatory markers and oxidative stress/antioxidant assays	↓ Urine output, urinary albumin excretion, albumin-to-creatinine ratio, serum creatinine, BUN and MDA; ↑ serum and renal antioxidant capacity; attenuated renal dysfunction and oxidative damage	(239)
Tripterygium wilfordii Hook.f. L.	Aqueous	*In vivo*	8 or 16 mg/kg body weight	Biochemical and kidney function parameters, anti-inflammatory activity	↓ Blood glucose levels, ↓ levels of BUN, serum creatinine, and urinary protein excretion; suppressed the generation of inflammatory cytokines in the kidney	(240)
Urtica dioica L.	Ethanol	*In vivo*	100 mg/kg body weight	Analysis of kidney function parameters and antioxidant activity	↓ BUN and serum creatinine content, enhanced GSH activity, and ↓ renal MDA levels	(203)
Vaccinium macrocarpon Aiton	Hydroethanol	*In vivo*	200 and 400 mg/kg body weight	Analysis of glycemic parameters, lipid profile, BUN, urea, and creatinine content	↓ Fasting blood glucose levels and HbA1c, ↑ insulin levels, ↓ cholesterol, TGs, BUN, urea, and creatinine content	(241)
Vaccinium myrtillus L.	Fruit extract	*In vivo*	54, 162, and 486 mg/kg body weight	Analysis of lipid profile, glycated proteins, inflammatory markers	↓ Glycated serum protein (GSP) and LDL, improved glucose/lipid metabolism, suppressed pro-inflammatory cytokines via NF-κB	(242)
Zingiber officinale Roscoe	Ethanol	*In vivo*	400 and 800 mg/kg	Analysis of blood glucose, HbA1c, BUN, serum creatinine, urea, and urinary albumin, lipid profile, antioxidant activity	↓ Blood glucose levels, ↓ HbA1c, BUN, serum creatinine, urea, and urinary albumin; improved overall lipid profile, ↓ oxidative stress, and pro-inflammatory cytokines	(204)
Other renoprotective effect models						
Abelmoschus esculentus L. Moench	Methanol	*In vitro* and *in vivo*	50, 100, 200, and 400 mg/kg body weight	DPPH and Ferric reducing ability of plasma assay, tests for BUN and creatinine	↑ SOD and CAT activity, ↓ MDA and NO levels, ↓ BUN, and creatinine content	(52)
Abrus precatorius L.	Ethanol	*In vivo*	150, 300 mg/kg body weight	Tests for kidney function parameters	↓ Urea, uric acid, serum creatinine, total urinary protein, and NO levels	(56)
Achyranthes aspera L.	Ethanol	*In vivo*	200 mg/kg body weight	Analysis of kidney function parameters, histological analysis	↓ Urea, uric acid, serum creatinine, BUN, and total urinary protein levels; ↓ tubular damage	(69)
Aegle marmelos (L.) Corrêa	Hydro-ethanol	*In vitro* and *in vivo*	200, 400 mg/kg body weight	Assay of biochemical and kidney function parameters, analysis of antioxidant enzyme activity	↑ Antioxidant activity of SOD, GSH, and CAT enzymes, ↓ BUN, serum creatinine, and urea	(72)
Aerva javanica Juss.	Aqueous	*In vivo*	400 mg/kg body weight	Assay of biochemical and kidney function parameters, analysis of antioxidant activity	↓ Blood urea, serum creatinine, total urinary protein and albumin levels, enhanced GSH activity, and ↓ LPO	(74)
Aframomum melegueta K.Schum.	Aqueous	*In vivo*	100 and 200 mg/kg body weight	Biochemical assay of antioxidant enzymes, kidney function tests, histological analysis	↑ GSH activity, ↓ serum creatinine, urea, and renal MDA content; suppressed renal cell apoptosis and improved renal tissue architecture	(79)
Ageratum conyzoides L.	Ethanol	*In vivo*	100 mg/kg body weight	Biochemical assay, kidney function tests	↑ Hemoglobin concentration, ↓ total urinary protein, urea, serum creatinine, urinary albumin, and globulin levels	(83)
Allium cepa L.	Ethanol	*In vivo*	200 and 400 mg/kg body weight	Kidney function tests, histological analysis	↓ Serum creatinine, urea, and total urinary protein levels, kidney weights, and body weights; improved kidney tissue architecture	(85)

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10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Other renoprotective effect models						
Allium sativum L.	Ethanol	*In vivo*	150, 300 mg/kg body weight	Biochemical assay of antioxidant enzymes, kidney function tests	↑ SOD, GST, and CAT activity, ↓ serum creatinine, urea, uric acid, and BUN levels	(88)
Aloe vera Burm.f.	Aqueous	*In vivo*	200 mg/kg, 400 mg/kg, and 600 mg/kg body weight	Biochemical assay, kidney function tests, histological analysis	↑ Hemoglobin concentration, enhanced CAT activity, ↓ serum creatinine and BUN, ↓ MDA levels, and ameliorated pyknotic nuclei in the renal epithelial cells	(91)
Annona squamosa L.	Ethanol	*In vitro* and *in vivo*	200 and 400 mg/kg body weight	Biochemical and antioxidant parameters assay, kidney function tests, histological analysis	↑ SOD, GSH, and CAT activity, ↓ serum creatinine, urea, uric acid, and BUN levels, regenerated damaged renal tissue	(102)
Apium graveolens L.	0.3% aqueous carboxyl methylcellulose (CMC) suspension	*In vivo*	500 and 1000 mg/kg body weight	Assay of kidney function parameters and serum electrolytes	↓ Serum creatinine, urea, BUN, and uric acid levels, ↓ serum K^+, Cl^-, ↑ Na^+, and Ca^{2+} levels, and ↓ total urinary protein content	(103)
Arachis hypogaea L.	Aqueous, ethanol, and methanol	*In vivo*	50 and 100 mg/kg body weight	Biochemical assay of kidney function parameters, histological assay	↓ Serum creatinine, urea, and urinary albumin levels; improved renal tissue architecture	(105)
Asparagus racemosus Willd.	Ethanol	*In vivo*	500 mg/kg body weight	Kidney function tests, assay of antioxidant enzyme activity histological analysis	↑ SOD, GSH, and CAT, ↓ renal MDA, plasma urea, and creatinine content; ameliorated damage to kidney tissues	(112)
Azadirachta indica A. Juss.	Methanol	*In vivo*	500 mg/kg body weight	Biochemical assay, kidney function tests	↑ SOD, GSH, GST, and CAT activity, ↓ renal MDA and NO levels, ↓ serum creatinine, urea, and uric acid levels, ↓ oxidative stress on the kidneys	(114)
Berberis vulgaris L.	Aqueous	*In vivo*	25, 50, 100, and 150 mg/kg body weight	Kidney function tests, antioxidant enzyme activity assay, histological analysis	↓ Serum creatinine, urea, and uric acid, ↓ renal MDA levels, ↑ SOD, GSH, GPx, and CAT activity, improved glomerular degeneration and hypercellularity	(120)
Berberis aristata DC.	Aqueous	*In vitro*	100–1000 μg/mL	Assay of anti-urolithiatic activity, antioxidant activity assay	↓ Nucleation and aggregation of calcium oxalate crystals; inhibited inflammation in the kidneys and ↓ oxidative stress	(116)
Camellia sinensis (L.) Kuntze	Aqueous	*In vivo*	100 mg/kg bodyweight	Biochemical assay, kidney function tests	↑ Hemoglobin concentration, ↓ serum creatinine, urea, and uric acid levels, and enhanced SOD activity in the kidneys	(138)
Catharanthus roseus (L.) G.Don	Ethanol	*In vitro* and *in vivo*	500 mg/kg body weight	DPPH radical scavenging activity, hematological parameters assay, antioxidant assay	↑ RBC count and hemoglobin concentration, ↓ ROS generation and oxidative stress, enhanced SOD activity in kidneys, ↓ renal MDA levels	(141)
Centella asiatica (L.) Urb.	Ethanol	*In vivo*	200 mg and 400 mg/kg body weight	Analysis of antioxidant enzyme activity and kidney function tests	↑ SOD and CAT activity, ↓ urea, BUN, serum, and urine creatinine	(144)
Cinnamomum zeylanicum Blume	Ethanol	*In vivo*	100 and 200 mg/kg body weight	Analysis of antioxidant enzyme activity and kidney function tests	↑ SOD, GSH, and CAT activity in kidneys, ↓ serum urea, serum creatinine, uric acid, and total urinary protein levels	(148)
Coriandrum sativum L.	Ethanol	*In vivo*	200 mg/kg body weight	Assay of antioxidant enzyme activity, kidney function tests	↑ GSH levels in kidney, ↓ MDA levels, ↓ urea and serum creatinine levels	(154)
Curcuma longa L.	Methanol	*In vivo*	400, 800 and 1000 mg/kg body weight	Kidney function tests, histological assay	↓ BUN, uric acid, and creatinine levels, improved tissue architecture in kidney due to damage caused by tubular necrosis	(156)
Emblica officinalis Gaertn.	Hydroethanol	*In vivo*	100 mg/kg, 200 mg/kg and 400 mg/kg body weight	Assay of biochemical and kidney function parameters, antioxidant enzyme activity	↑ RBC and hemoglobin count, ↓ serum creatinine and BUN content, enhanced renal CAT, SOD, and GPx	(158)
Euphorbia hirta L.	Ethanol	*In vivo*	400 mg/kg body weight	Assay of antioxidant enzyme activity, kidney function tests	↑ SOD, GSH, CAT, GPx, GST activity in the kidney, ↓ BUN, serum creatinine, urea, and uric acid levels, and ↓ renal MDA levels	(243)
Ficus hispida Blanco	Methanol	*In vitro* and *in vivo*	250 and 500 mg/kg body weight	LPO assay, NO scavenging activity, determination of BUN, serum creatinine, urea, and total urinary proteins	↓ Oxidative stress, significant ↓ MDA levels, ↓ serum creatinine, urea, total urinary proteins, and BUN	(244)
Ficus religiosa L.	Methanol	*In vivo*	200 mg/kg body weight	Analysis of antioxidant activity, assay of kidney function parameters	↓ Serum creatinine and urea content, enhanced SOD, CAT, and GSH activity in kidneys, and ↓ renal MDA levels	(245)
Garcinia kola Heckel	Methanol	*In vivo*	10 mg/kg body weight	Assessment for kidney function, analysis of antioxidant activity	Enhanced antioxidant activity in kidneys, ↓ serum urea, creatinine, and urine protein levels	(246)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 1 Continued

Species	Extract	Study	Dose	Assay	Pharmacological effects	Ref.
Other renoprotective effect models						
Ginkgo biloba L.	Ethanol	*In vivo*	200 mg/kg and 400 mg/kg body weight	Kidney function test, histological analysis	↓ Serum urea, creatinine, BUN, serum urea, improved tissue damage in tubular epithelial and glomerular cells	(247)
Glycine max (L.) Merr.	Aqueous seed extract	*In vivo*	2.5 and 5 mL/kg body weight	Kidney function test, analysis of antioxidant enzyme activity	↑ SOD, CAT, GPx activity in kidneys, ↓ serum BUN, creatinine, urea, uric acid, and urine volume, kidney weight, urinary sodium and potassium, and total urinary protein	(170)
Glycyrrhiza glabra L.	Aqueous	*In vivo*	30, 90, and 270 mg/kg body weight	Biochemical and kidney function assay, analysis of antioxidant enzyme activity	↑ RBC count, ↓ serum creatinine, and urea content, enhanced SOD and CAT activity in the kidneys	(172)
Lycopersicon esculentum Mill.	Tomato extract	*In vivo*	30 mg/kg body weight	Assay of biochemical, kidney function parameters, and serum electrolytes	↓ Serum creatinine, urea, and BUN levels, ↓ serum K^+ levels, and ↑ serum Na^+ levels	(248)
Mangifera indica L.	Ethanol	*In vivo*	100 and 500 mg/kg body weight	Assessment of kidney function, analysis of antioxidant activity	↑ SOD, CAT, and GSH, ↓ serum creatinine, and urea levels	(185)
Mentha x piperita L.	Ethanol	*In vivo*	200 mg/kg body weight	Assessment of kidney function parameters	↓ BUN, serum creatinine, creatinine clearance, serum uric acid, serum electrolytes, urinary volume, urinary creatinine, urinary proteins, urinary alkaline phosphatase, and urinary lactate dehydrogenase (LDH)	(249)
Musa sapientum L.	Ethanol	*In vivo*	200, 600, and 2000 mg/kg body weight	Biochemical and kidney function parameters assay, antioxidant enzyme activity	↑ Renal SOD, CAT, GSH, GST, and GPx, and ↓ levels of BUN, creatinine, uric acid, total urinary proteins, cholesterol, and urinary albumin	(250)
Nasturtium officinale R.Br.	Hydroethanolic	*In vivo*	250 and 500 mg/kg body weight	Assay of kidney function parameters, LPO, and histopathological analysis	↓ Serum creatinine, urea, and creatinine clearance, ↓ renal MDA levels, mitigated tubular epithelial cell desquamation, swelling, and tubular dilatation	(251)
Panax ginseng f. *trifoliolatus* Makino	Aqueous	*In vivo*	100 and 200 mg/kg body weight	Assessment of kidney function and biochemical parameters,	↓ Urea, creatinine, and BUN levels, ↓ blood glucose levels, cholesterol, and TG levels	(252)
Raphanus sativus var. sativus (L.) Schmalh.	Methanol	*In vivo*	100 mg/kg body weight	Analysis of urine profile, antioxidant enzymes, and kidney function parameters	↓ Albumin, urobilinogen, and protein levels of urine, ↓ levels of serum creatinine, urea, and renal LDH, enhanced renal SOD, CAT, GR, GST, GSH, and GPx activity	(253)
Scoparia dulcis L.	Ethanol	*In vivo*	400 mg/kg body weight	Assessment of kidney function parameters	↓ Serum creatinine, BUN, and LPO levels, and improved creatinine clearance	(199)
Syzygium cumini (L.) Skeels	Aqueous	*In vivo*	200 mg/kg body weight	Kidney function test, antioxidant enzyme activity assay	↑ CAT, GPx activity in kidneys, ↓ urea, and creatinine content	(254)
Tamarindus indica L.	Hydroethanol	*In vivo*	200 mg/kg body weight	Kidney function test, histological assay	↓ BUN, serum urea, uric acid, and creatinine, serum potassium and calcium, improved renal structure and function	(255)
Tanacetum parthenium (L.) Sch.Bip.	Methanol	*In vivo*	40, 80, and 120 mg/kg body weight	Assay of kidney function parameters and antioxidant activity	↓ Levels of urinary albumin, creatinine, urea, and uric acid, enhanced renal SOD and GPx activity, ↓ MDA levels, and hence, prevented LPO	(256)
Tribulus terrestris L.	Aqueous	*In vivo*	840 mg/kg and 1680 mg/kg body weight	Biochemical and kidney function assay	Inhibited rise in BUN, serum creatinine, and uric acid levels	(257)
Vigna angularis (Willd.) Ohwi & H.Ohashi	Aqueous	*In vivo*	10 mg/kg body weight	Assay of biochemical and kidney function parameters, lipid profile	↓ Blood glucose levels, ↓ serum creatinine, urea, and uric acid, ↓ TGs, VLDL, and ↑ HDL levels	(258)

↓, decrease; ↑, increase.

10.3389/fimmu.2026.1964602

decreasing BUN and creatinine, reducing oxidative stress, and suppressing inflammation, protecting against renal damage (67).

7.6 *Achyranthes aspera* L.

Achyranthes aspera, known as prickly chaff flower, belongs to the Amaranthaceae family. Its methanolic and petroleum ether extracts inhibit α-glucosidase, α-amylase, and DPPH radicals, thereby reducing glucose absorption, post-meal glucose levels, and oxidative stress (68). Ethanolic extracts decrease kidney damage by lowering urea, uric acid, serum creatinine, and BUN, while improving tissue integrity (69). Compounds like oleanolic acid boost renal function by lowering urinary albumin, urea, and creatinine, and enhancing GFR (70).

7.7 *Aegle marmelos* (L.) Corrêa

Aegle marmelos, or bael, is a sacred tree in the Rutaceae family, valued in traditional Indian medicine for its antidiabetic and kidney-protective effects. Aqueous extracts improve glucose tolerance, lipid profiles, and lower fasting blood glucose, LDL, and TGs, while raising HDL (71). Hydroethanolic extracts boost antioxidants (SOD, glutathione [GSH], CAT), reduce BUN and serum creatinine, and improve renal function (72). Compounds like rutin and eugenol help lower blood sugar, improve kidney health, and reduce oxidative stress, LPO, and TNF-α (54, 73).

7.8 *Aerva javanica* Juss.

Aerva javanica, a desert plant from the Amaranthaceae family, has been traditionally used in folk medicine for renoprotection. Aqueous extracts decrease blood urea, serum creatinine, urinary protein, and albumin levels, while increasing GSH activity and reducing LPO (74). Compounds like rosmarinic, gallic, and vanillic acids improve kidney function by lowering oxidative stress and inflammation, mainly TNF-α, suppressing LPO, and boosting antioxidant enzymes such as SOD, CAT, GSH, and glutathione peroxidase (GPx) (75–77).

7.9 *Aframomum melegueta* K. Schum.

Aframomum melegueta (Grains of Paradise) is a perennial plant in the Zingiberaceae family, used in African medicine for diabetes and kidney health. Its ethanolic, ethyl acetate, and water extracts inhibit hemoglobin glycosylation, α-amylase, α-glucosidase, and DPPH radicals, delaying carbohydrate digestion and lowering post-meal glucose (78). Aqueous extracts boost renal antioxidants (SOD and GSH), reduce serum creatinine and urea, and suppress renal cell apoptosis (79). Compounds like 6-gingerol and 6-shogaol improve renal function by lowering MDA levels, boosting antioxidants, and protecting kidneys from oxidative damage (80, 81).

7.10 *Ageratum conyzoides* L.

Ageratum conyzoides (Billy goat weed) is a herbaceous plant in the Asteraceae family, known for its traditional use in treating diabetes and kidney health. Aqueous extracts boost insulin secretion, improve glucose tolerance, and lower fasting blood glucose (82). Ethanolic extracts support kidney health by increasing hemoglobin and reducing urinary protein, urea, serum creatinine, urinary albumin, and globulin (83). Compounds like kaempferol aid glucose metabolism and renal function by reducing oxidative stress and inflammation (60).

7.11 *Allium cepa* L.

Allium cepa (onion), a prominent member of the Amaryllidaceae, has a long history of culinary and ethnopharmacological use for its antidiabetic and renoprotective effects. Aqueous extracts significantly reduce blood glucose and serum lipids, improving lipid and glucose balance (84). Ethanolic extracts support kidney health by reducing serum creatinine and urea levels, decreasing urinary protein excretion, and improving tissue structure (85). Allicin, a key compound, boosts kidney health by reducing serum creatinine and BUN, urinary protein, and MDA, while increasing renal GSH and SOD (86). Quercetin increases renal SOD, CAT, and GSH, reduces MDA and LPO, and improves kidney function (59).

7.12 *Allium sativum* L.

Allium sativum (garlic) is a perennial plant in the Amaryllidaceae family, used for its medicinal properties, particularly its antidiabetic, antioxidant, and nephroprotective effects. Aqueous extracts lower fasting blood glucose and serum lipids, enhancing lipid profile and glucose tolerance (87). Ethanol extracts boost antioxidant enzymes (SOD, glutathione-S-transferase [GST], CAT) and improve kidney function by decreasing serum creatinine, urea, uric acid, and BUN (88). Allicin, a key compound, enhances kidney health by lowering these markers and urinary albumin levels, while increasing renal GSH and SOD (86). Diallyl sulfide improves kidney function by reducing serum creatinine, BUN, urinary protein, inhibiting ROS, and oxidative stress (89).

7.13 *Aloe vera* Burm.f.

Aloe vera (the Aloe plant) in the family Asphodelaceae is traditionally used for its therapeutic benefits. Methanol extracts inhibit glycation, starch digestion, and post-meal glucose by blocking α-amylase, α-glucosidase, and pancreatic lipase (90). Aqueous extracts boost kidney function, raise hemoglobin, and improve CAT activity, lowering serum creatinine, BUN, and MDA, and reducing renal damage (91). Emodin, a compound in *Aloe vera*, enhances kidney health by decreasing urinary albumin, creatinine, uric acid, BUN, and increasing renal SOD and GSH (92).

7.14 *Amaranthus tricolor* L.

Amaranthus tricolor (Joseph's coat) is an annual from the Amaranthaceae family, known for its medicinal benefits, especially in managing diabetes and kidney health. Methanol extracts lower blood glucose and oxidative stress in kidneys, boost hemoglobin and HDL cholesterol, and decrease serum TGs, LDL, and total cholesterol (93). Ferulic acid in *A. tricolor* improves kidney

function, lowers BUN and creatinine, and prevents oxidative stress and inflammation in kidneys (94). Quercetin has antioxidant effects and enhances kidney health by reducing blood glucose, increasing renal SOD, CAT, and GSH, and lowering MDA and LPO (59). Vanillic acid reduces glycemic levels, serum creatinine, urea, uric acid, and urinary microproteinuria, while improving creatinine clearance and reducing pro-inflammatory cytokine TNF-α (77).

7.15 *Anacardium occidentale* L.

Anacardium occidentale (cashew tree), part of the Anacardiaceae family, has notable antidiabetic and renoprotective effects. Methanol extracts improve lipid profiles by lowering blood glucose, cholesterol, TGs, and atherogenic index, while also reducing urea, creatinine, and plasma MDA levels (95). Naringenin, a flavonoid, enhances kidney function, reduces LPO, urea, uric acid, and bilirubin, urinary albumin, and serum thiobarbituric acid reactive substrates (TBARS), and boosts SOD, CAT, GSH, GPx, and GST activities (65). Delphinidin, an anthocyanin, improves kidney health by lowering serum creatinine and urea, increasing creatinine clearance, enhancing renal antioxidants, and suppressing pro-inflammatory cytokines like TNF-α, IL-6, and IL-1β (96).

7.16 *Ananas comosus* (L.) Merr.

Ananas comosus (pineapple), a tropical Bromeliaceae fruit, shows promise in managing diabetes and nephropathy. Ethanol extracts improve kidney function by lowering serum creatinine, urea, uric acid, and serum protein levels, indicators of renal health. Pineapple extracts also decrease blood glucose and HbA1c, indicating blood sugar regulation (97). Compounds such as chlorogenic and ferulic acids enhance antioxidant and anti-inflammatory effects, reducing oxidative stress and TNF-α production, thereby protecting kidney tissue and supporting renal health (94, 98).

7.17 *Andrographis paniculata* (Burm.f.) Wall.

Andrographis paniculata, an herb of the Acanthaceae family, is known for its medicinal properties, especially its antioxidant, antidiabetic, and renoprotective effects. Chloroform root extracts lower blood glucose and kidney biomarkers like protein, urinary albumin, and creatinine. *In vivo* studies show *A. paniculata* inhibits albuminuria, proteinuria, and uremia, improving renal function (99). The active compound andrographolide reduces fasting blood glucose by decreasing ROS and pro-inflammatory cytokines through inhibition of the NF-κB pathway, thereby protecting the kidneys (100).

7.18 *Annona squamosa* L.

Annona squamosa (custard apple), a tropical fruit in the family Annonaceae, is traditionally used to treat ailments such as inflammation and diabetes. It improves glucose metabolism and kidney function. Studies show hot water extracts lower blood glucose and HbA1c, boost glucose uptake and insulin secretion, and help manage post-meal glucose spikes (101). Its antioxidants (SOD, GSH, and CAT) protect the kidneys by reducing serum creatinine, urea, uric acid, and BUN in ethanolic extracts (102). Compounds such as rutin, quercetin, and linalool contribute to its antidiabetic and kidney-protective effects, thereby enhancing glucose tolerance and reducing oxidative stress (53, 59, 73).

7.19 *Apium graveolens* L.

Apium graveolens (celery) is a biennial herb from the Apiaceae family used traditionally to treat hypertension, gout, diabetes, and nephropathy. Its aqueous extract lowers serum creatinine, urea, BUN, and uric acid levels and improves kidney health by regulating electrolytes such as sodium, calcium, and potassium (103). Compounds such as luteolin and ferulic acid help control blood glucose and protect renal tissue by reducing oxidative stress and inflammation (57, 94).

7.20 *Arachis hypogaea* L.

Arachis hypogaea (peanut) is a legume with high nutritional and therapeutic value, used to manage diabetes and kidney damage. Peanut seed oils have anti-diabetic effects by improving insulin sensitivity, increasing liver glycogen, and lowering blood glucose (104). Ethanolic extracts can reduce serum creatinine and improve kidney histology, enhancing kidney function. Peanut extracts also increase insulin levels and lower serum lipids, including cholesterol, TG, and LDL, thereby aiding the management of metabolic disorders (105).

7.21 *Artemisia absinthium* L.

Artemisia absinthium (wormwood), a perennial herb of the Asteraceae family, is used to treat parasitic infections and digestive ulcers. It has notable antidiabetic and nephroprotective effects. Its ethanol extract lowers blood glucose, creatinine, and urea levels, and urinary protein levels, increases insulin levels, and reduces cholesterol levels (106). Active compounds such as β-caryophyllene protect the kidneys by reducing serum creatinine and urea levels, boosting SOD, CAT, GSH, and GPx activities, and lowering renal MDA (107). Linalool reduces blood glucose by enhancing glucose uptake and decreases inflammatory markers, as well as urea, creatinine, and BUN (53).

7.22 *Artocarpus heterophyllus* Lam.

Artocarpus heterophyllus (jackfruit), a large tree in the Moraceae family, has nutritional and medicinal uses, especially its antidiabetic and antioxidant effects. Acetone extracts reduce fasting blood glucose, scavenging DPPH radicals, decreasing oxidative stress by inhibiting glucose absorption, increasing liver glycogen, and boosting antioxidant enzymes (108). The compound silymarin lowers serum creatinine, urea, uric acid, and BUN, and prevents ROS- and inflammation-mediated effects, thereby reducing renal MDA (109).

β-carotene also helps lower creatinine, urea, and uric acid, and normalize cytokines, decreasing renal oxidative stress (110).

7.23 *Asparagus racemosus* Willd.

Asparagus racemosus (shatavari), a plant in the family Asparagaceae, is known for its antidiabetic and kidney-protective effects. Ethanol extracts lower fasting glucose, TGs, and cholesterol, improve kidney function by reducing creatinine, urea, and MDA, and boost antioxidant enzymes GSH, SOD, and CAT (111, 112). Compounds like rutin and quercetin decrease blood and urinary glucose, increase renal GSH, SOD, and CAT, reduce MDA and LPO, and lower serum creatinine and urea (59, 73).

7.24 *Azadirachta indica* A. Juss.

Azadirachta indica, or neem, from the Meliaceae family, is used in Ayurveda for its therapeutic effects, including antidiabetic and kidney benefits. Ethanolic extracts lower blood glucose, while methanolic extracts increase antioxidant levels, including SOD, GSH, GST, and CAT, and reduce MDA, NO, creatinine, urea, and uric acid (113, 114). Rutin improves kidney function by decreasing blood and urinary glucose and renal MDA (73). Quercetin reduces blood glucose levels and enhances renal antioxidant activity, thereby protecting the kidneys from oxidative damage. kidneys (59).

7.25 *Berberis aristata* DC.

Berberis aristata, or Indian Barberry, in the Berberidaceae family, is used in Ayurveda for its antidiabetic and kidney-protective effects. Ethanolic extracts lower blood glucose and stimulate insulin release, showing hypoglycemic activity (115). Aqueous extracts have anti-urolithiatic and antioxidant effects by inhibiting calcium oxalate crystal formation, reducing kidney inflammation, and oxidative stress (116). Active compounds like berberine decrease glycemic levels, serum creatinine, urinary albumin, and suppress renal oxidative stress and NF-κB pathway (117). Palmatine enhances kidney function by lowering MDA, BUN, and creatinine clearance, while increasing GSH activity and sodium and potassium excretion, and reducing oxidative stress (118).

7.26 *Berberis vulgaris* L.

Berberis vulgaris, or European Barberry, of the Berberidaceae family, is used in herbal medicine for its antidiabetic, antioxidant, anti-inflammatory, and kidney-protective effects. Its aqueous extracts lower blood glucose, cholesterol, and TG levels, thereby improving lipid profiles. They also boost kidney health by reducing serum creatinine, urea, and uric acid levels and increasing antioxidant enzymes such as SOD, GSH, and CAT (119). Histology shows less glomerular damage and hypercellularity (120). Berberine, a key compound, helps lower blood sugar, serum creatinine, and urinary albumin, and reduces oxidative stress and inflammation in the kidneys (117).

7.27 *Biophytum sensitivum* (L.) DC.

Biophytum sensitivum, a member of the Oxalidaceae family, is traditionally used for various ailments and is valued for its antidiabetic and kidney-protective properties. Studies show that aqueous extracts reduce blood glucose, urinary albumin, and total protein, while improving lipid profiles by lowering cholesterol, TGs, and LDL, and increasing HDL (121). The compound orientin improves kidney function by reducing creatinine and urea levels, increasing creatinine clearance, and enhancing antioxidant enzymes such as SOD, CAT, and GSH. It also suppresses pro-inflammatory cytokines (TNF-α, IL-6, IL-1β) and balances oxidative stress markers (122). Epicatechin decreases glycemic levels, serum creatinine, and urinary albumin, improves creatinine clearance, and lowers renal MDA levels (123).

7.28 *Bixa orellana* L.

Bixa orellana, or Annatto, from the Bixaceae family, is known for its kidney-protective and antidiabetic effects. Aqueous extracts improve kidney function by lowering blood glucose, serum creatinine, urea, and uric acid levels (124). Bixin, a key compound, boosts antioxidant enzymes such as SOD, CAT, and GPx, reduces renal MDA levels, and suppresses inflammatory cytokines, thereby shielding renal tissues from damage (125).

7.29 *Bombax ceiba* L.

Bombax ceiba, or Silk Tree, from the Malvaceae family, has extracts that lower blood glucose and HbA1c and improve tolerance. They enhance lipid profiles by reducing cholesterol and TGs, increasing HDL, and boosting antioxidant activity via increased SOD and decreased MDA (126). These extracts also protect the kidneys by reducing oxidative stress and urinary albumin, creatinine, and BUN, with mangiferin further lowering serum creatinine, uric acid, and BUN, and reducing inflammation (127, 128). Lupeol benefits include reduced blood glucose, creatinine, urea, uric acid, and improved insulin sensitivity and electrolyte balance (129).

7.30 *Brassica juncea* (L.) Czern.

Brassica juncea, or Indian mustard, from the Brassicaceae family, shows notable antidiabetic and renoprotective effects. Aqueous extracts boost insulin secretion and lower blood glucose (130). The methanolic extract improves biochemical profiles by reducing glucose, TGs, and cholesterol levels, while increasing the antioxidant enzymes SOD and GSH. It also decreases renal MDA and NO levels, indicating renoprotection (131). The compounds cinnamic acid and kaempferol in *B. juncea* exert antihyperglycemic, antioxidant, and renoprotective effects by lowering blood sugar levels, reducing oxidative stress, and enhancing antioxidant activity in the kidneys (60, 132).

7.31 *Brassica oleracea* L.

Brassica oleracea, or cabbage, in the Brassicaceae family, has been traditionally used for its medicinal benefits, particularly for

glucose metabolism and kidney health. The ethanol extract improved blood glucose tolerance, insulin secretion, and lowered HbA1c, while increasing hemoglobin (133). The aqueous extract reversed diabetes-related weight loss, improved kidney function by reducing serum creatinine and urea levels, and increased antioxidant enzyme activity, including SOD and CAT (134). Gallic and chlorogenic acids in the extract showed antioxidant and anti-inflammatory effects, decreasing kidney inflammation markers and oxidative stress (76, 98).

7.32 *Caesalpinia decapetala* (Roth) Alston

Caesalpinia decapetala, a member of the Fabaceae family, exhibits antidiabetic and kidney-protective effects. Its hydromethanolic extract lowered blood glucose, cholesterol, TGs, and serum creatinine (135). Resveratrol and quercetin reduced oxidative stress by decreasing MDA and boosting antioxidants such as SOD, CAT, and GSH, and by lowering LPO and inflammation, thus offering renal protection (59, 136).

7.33 *Camellia sinensis* (L.) Kuntze

Camellia sinensis, or tea, belongs to the Theaceae family and is traditionally used for its antidiabetic and renoprotective properties. Its hot-water extract increases insulin secretion, delays starch digestion, inhibits dipeptidyl peptidase-IV (DPP-IV), and improves glucose uptake and tolerance, lowering blood sugar levels (137). Aqueous extracts boost antioxidant enzyme activity, reduce serum creatinine and urea, and protect renal tissues against oxidative stress (138). Bioactive compounds like epicatechin, rutin, and gallic acid lower glycemic levels, reduce oxidative stress, suppress inflammatory cytokines, and enhance renal antioxidant enzymes (73, 76, 123).

7.34 *Carica papaya* L.

Carica papaya, known as papaya, is a member of the family Caricaceae and has notable antidiabetic and kidney-protective effects. Its aqueous extracts lower blood glucose, cholesterol, and TGs, while raising HDL and NO levels, improving vascular and metabolic health (139). Chlorogenic acid and quercetin reduce renal oxidative stress, suppress inflammation, and enhance renal function by lowering serum creatinine, BUN, and MDA. These compounds boost antioxidants and safeguard renal tissue (59, 98).

7.35 *Catharanthus roseus* (L.) G.Don

Catharanthus roseus, or Madagascar periwinkle, in the family Apocynaceae, has antidiabetic and antioxidant effects. Extracts with dichloromethane and methanol improve glucose metabolism, increase GSH, reduce oxidative stress, lower blood glucose, and raise liver glycogen (140). Ethanolic extracts exhibit DPPH radical-scavenging activity, improve blood parameters, and reduce renal oxidative stress by lowering MDA and increasing SOD in the kidneys (141). Bioactive compounds such as vindoline enhance glucose tolerance, lipid profiles, and kidney function, lowering serum creatinine and urea levels while boosting SOD and CAT (142). Chlorogenic acid decreases renal oxidative stress, suppresses inflammation, and improves kidney function (98).

7.36 *Centella asiatica* (L.) Urb.

Centella asiatica, or Gotu kola, from the Apiaceae family, has notable antidiabetic and renoprotective effects. Aqueous extracts lower blood glucose and improve lipids by decreasing cholesterol, TGs, and LDL, while raising HDL (143). Ethanolic extracts boost antioxidant enzymes, reduce urea and creatinine, and enhance kidney function (144). Compounds like naringin bolster renal antioxidant levels and reduce oxidative stress and inflammation (145). Asiaticoside inhibits inflammatory biomarkers such as IL-6, interleukin-8 (IL-8), and TNF-α, lowers serum creatinine and BUN, and reduces urinary protein (146).

7.37 *Cinnamomum zeylanicum* Blume

Cinnamomum zeylanicum, or cinnamon, from the Lauraceae family, has strong antidiabetic and kidney-protective effects. Its methanolic, ethanolic, and acetone extracts lower blood glucose, oxidative stress, and LDL, and improve lipid profiles (147). Ethanolic extracts boost SOD, GSH, and CAT activity, reduce serum urea and creatinine, and protect kidney tissue from damage (148). Cinnamaldehyde enhances antioxidant defenses and reduces oxidative stress (149). Eugenol lowers serum creatinine, urea, and renal MDA, while increasing SOD and GSH activity (54).

7.38 *Coccinia grandis* (L.) Voigt

Coccinia grandis, or Ivy gourd, in the family Cucurbitaceae, has notable effects on glucose regulation and kidney protection. Aqueous extracts lower blood glucose and HbA1c and increase insulin secretion (150). Ethanolic extracts decrease oxidative stress, improve kidney function, and boost antioxidant activity in the liver and kidneys (151). Quercetin enhances renal antioxidant enzyme activity, reduces MDA levels, and improves renal function (59). Rutin decreases blood and urinary glucose levels, serum creatinine and urea, and renal LPO (73). Kaempferol improves glucose metabolism, reduces renal oxidative stress, and inhibits inflammatory markers (60).

7.39 *Coccinia indica* Wight & Arn.

In some studies, the species *Coccinia indica* (a synonym of *Coccinia grandis*) was mentioned instead of *Coccinia grandis*. In Ayurveda, it is used to treat diabetes, fever, and skin issues. It has significant antidiabetic and renoprotective effects. Ethanolic extracts lower blood sugar, improve glucose tolerance, and reduce cholesterol, TG, creatinine, and BUN levels (152). Constituents such as lupeol enhance insulin sensitivity, reduce serum urea and uric acid, and

balance renal electrolytes (129). β-carotene also lowers creatinine, uric acid, reduces oxidative stress, and modulates cytokines (110).

7.40 *Coriandrum sativum* L.

Coriandrum sativum, or coriander, from the Apiaceae family and native to the Mediterranean, has been traditionally used for diabetes, digestion, and as a diuretic. It exhibits strong antidiabetic and renoprotective effects. Aqueous extracts reduce hyperglycemia by boosting insulin, glucose uptake, and oxidation (153). Ethanolic extracts enhance kidney function by increasing antioxidants and lowering urea and creatinine (154). Compounds such as linalool improve glucose uptake, reduce inflammation, and protect the kidneys (53). Ferulic acid lowers BUN, creatinine, and oxidative stress, while rutin decreases glucose, urea, and kidney damage markers (73, 94).

7.41 *Curcuma longa* L.

Curcuma longa, or turmeric, from South Asia, is part of the Zingiberaceae family. Used in Ayurveda and Chinese medicine, it has anti-inflammatory, antioxidant, and wound-healing effects. Turmeric shows strong antidiabetic and kidney-protective properties. Extracts from *C. longa* lower blood glucose, food and water intake, and body weight in diabetic rats (155). Methanolic extracts reduce BUN, uric acid, and creatinine levels, thereby improving kidney tissue (156). Curcumin boosts antioxidant enzymes like SOD and GPx, decreases urinary albumin and renal MDA, and suppresses inflammation (157).

7.42 *Emblica officinalis* Gaertn.

Emblica officinalis, or Indian gooseberry, belongs to the family Phyllanthaceae and is native to Southeast Asia, mainly India. Traditionally, it has been used in Ayurveda to manage diabetes, enhance immunity, and improve kidney health. Hydroethanolic extracts increase renal antioxidant levels, including CAT, SOD, and GPx, while lowering serum creatinine and BUN levels (158). Compounds such as gallic acid improve renal function by reducing serum creatinine, urea, and uric acid levels, as well as oxidative stress and inflammation (76). Ellagic acid lowers blood sugar, increases serum insulin, and protects the kidneys by reducing ROS and cytokines TNF-α and IL-6 (159).

7.43 *Eriobotrya japonica* (Thunb.) Lindl.

Eriobotrya japonica (Loquat), from southeastern China and Japan, is valued for its medicinal uses, including managing diabetes and reducing oxidative stress and inflammation. Ethanolic extracts lower blood glucose, serum creatinine, urea, urinary proteins, and albumin (160). Active constituents such as epicatechin enhance creatinine clearance, reduce urinary albumin excretion, and reduce renal oxidative stress (123). Chlorogenic acid lowers serum creatinine and BUN levels, reduces renal stress, and reduces inflammation (98). Quercetin boosts renal antioxidants, decreases LPO, and reduces serum creatinine and urea (59).

7.44 *Euphorbia hirta* L.

The asthma plant (*Euphorbia hirta*), part of the Euphorbiaceae family, grows in tropical and subtropical regions. Traditionally, it treats respiratory issues, hyperglycemia, and kidney problems. Ethanolic extracts lower blood glucose, cholesterol, LDL, and VLDL and raise HDL (161). Compounds like kaempferol reduce oxidative stress and inflammation and boost renal antioxidant levels (60). Gallic acid helps kidney function by decreasing creatinine, urea, and uric acid, and reducing oxidative stress and inflammation (76).

7.45 *Ficus benghalensis* L.

Ficus benghalensis, or banyan tree, from the Indian subcontinent, belongs to the Moraceae family and has traditional Ayurvedic uses for diabetes, kidney health, and metabolic disorders. Its ethanolic extracts lower blood glucose, increase insulin levels, and enhance carbohydrate and lipid metabolism (162). They also decrease cholesterol, TGs, LDL, urea, and creatinine, while increasing HDL (163). Active compounds such as rutin reduce blood and urinary glucose levels, serum creatinine and urea, and renal MDA (73). Gallic acid improves kidney function by lowering serum creatinine, urea, and uric acid levels, boosting antioxidant activity, and decreasing oxidative stress and inflammation (76). Lupeol lowers blood glucose, urea, uric acid, creatinine, and serum sodium and potassium levels, and increases insulin resistance (129).

7.46 *Ficus racemosa* Willd.

The cluster fig, *Ficus racemosa*, in the family Moraceae, native to India and Southeast Asia, is used for its anti-inflammatory, antidiabetic, and nephroprotective effects. Ethanolic extracts lower blood glucose, serum TGs, and LDL, while increasing HDL and reducing serum creatinine, urinary albumin, sodium, and BUN (164). Lupeol enhances insulin sensitivity, decreases blood glucose, urea, uric acid, and creatinine, and lowers serum sodium and potassium (129). Ursolic acid reduces fasting glucose, serum creatinine, and BUN, boosts renal SOD activity, decreases MDA, and suppresses renal TNF-α and IL-1β (165). Rutin lowers blood and urinary glucose, serum creatinine and urea, and renal MDA levels (73).

7.47 *Fragaria ananassa* Duchesne

Fragaria ananassa (strawberry) is a widely cultivated fruit in the Rosaceae family, native to Europe. In traditional medicine, strawberries are valued for their health benefits, especially for improving metabolic and kidney health. Research shows that aqueous extracts of *F. ananassa* lower blood glucose, serum cholesterol, LDL, VLDL, TGs, and serum creatinine levels. They boost renal CAT activity and reduce MDA levels (166). Active compounds such as ellagic acid lower glycemic levels, increase serum insulin, suppress ROS and LPO, and improve renal SOD and GSH activity, while reducing MDA and inflammatory markers (159). Kaempferol and quercetin lower blood glucose, increase renal SOD, CAT, and GSH, and

reduce oxidative stress, inflammation, LPO, serum creatinine, and urea (59, 60).

7.48 *Garcinia kola* Heckel

Garcinia kola, or bitter kola, is a West African tropical plant used traditionally for its antidiabetic and renoprotective effects. Studies show methanolic extracts lower LPO, glucose, and oxidative stress, while increasing GSH (167). Kolaviron reduces serum creatinine, urea, renal MDA, and boosts SOD, CAT, GSH, GPx, and GST activity in the kidneys (168).

7.49 *Glycine max* (L.) Merr.

Glycine max (Soybean), from East Asia, is a key legume cultivated globally. Known for nutrition and medicinal uses, especially in managing diabetes and supporting kidney health, *G. max* extracts boost kidney antioxidant enzymes, lower blood glucose, improve glucose tolerance, and decrease serum BUN, creatinine, urea, uric acid, and urinary electrolytes and protein (169, 170). Compounds like β-sitosterol and kaempferol improve glucose uptake, lower cholesterol, TGs, urea, and creatinine, enhance renal SOD, CAT, and GSH activities, reduce MDA levels, and inhibit oxidative stress and inflammation (60, 171).

7.50 *Glycyrrhiza glabra* L.

Glycyrrhiza glabra, or licorice, has notable antidiabetic and kidney-protective effects. Its aqueous extracts improve biochemical and kidney functions, raising RBC count and boosting SOD and CAT activity, while decreasing serum creatinine and urea (172). Glycyrrhetinic acid reduces serum creatinine, urea, and uric acid, lowers LPO, NO, and suppresses pro-inflammatory cytokines in the kidneys (173).

7.51 *Gundelia tournefortii* L.

Gundelia tournefortii, known as Jerusalem artichoke, is an herb native to Europe and Asia in the family Fabaceae, traditionally used for blood glucose and kidney health. Aqueous extracts lowered blood glucose, serum creatinine, and urea (174). Compounds such as linolenic and ursolic acids reduced fasting glucose, serum creatinine, and BUN, improved renal SOD activity, decreased MDA levels, and suppressed inflammatory cytokines (165, 175). β-sitosterol increased muscle glucose uptake, improved insulin sensitivity, lowered cholesterol, TGs, urea, and creatinine, and boosted antioxidant activity in the kidneys (171).

7.52 *Heritiera fomes* Buch. -Ham.

Heritiera fomes, or Sundari, is a Malvaceae plant native to the Indian coastal regions, noted for its medicinal properties, especially antidiabetic and kidney-protective effects. Hot water extracts lower blood glucose, increase insulin, and impair starch digestion, DPP-IV activity, and glucose absorption (176). Ethanol extracts improve glucose tolerance, reduce ROS, and protect the kidneys, decreasing creatinine, uric acid, and BUN, while improving the lipid profile by lowering cholesterol, TG, and LDL and raising HDL (177). Compounds such as quercetin and epicatechin have antioxidant and anti-inflammatory benefits; quercetin increases renal SOD, CAT, and GSH, and reduces MDA, LPO, creatinine, and urea (59). Epicatechin enhances glycemic control and kidney health by lowering serum creatinine, urinary albumin, increasing clearance, and reducing renal MDA (123).

7.53 *Hibiscus rosa-sinensis* L.

Hibiscus rosa-sinensis, or hibiscus, is a Malvaceae native to Asia's tropical and subtropical regions. Traditionally used for their antidiabetic and nephroprotective effects, hydromethanolic extracts lower blood glucose, increase insulin levels, and improve kidney function. They also reduce serum creatinine, uric acid, urinary protein levels, and cholesterol, while increasing SOD and CAT activities and lowering oxidative stress (178). Active compounds like niacin help by improving kidney function, lowering serum creatinine, urea, cholesterol, TGs, and LDL, and raising HDL, reducing oxidative stress and kidney inflammation (179).

7.54 *Jatropha curcas* L.

Jatropha curcas, or physic nut, is a Euphorbiaceae native to the Americas and cultivated worldwide. It is known for medicinal uses such as managing blood glucose levels and supporting kidney health. Methanolic extracts decrease blood glucose, boost insulin, improve hematopoietic activity, and reduce TGs, cholesterol, and LDL while increasing HDL (180). Ethanolic extracts enhance kidney function by increasing CAT, SOD, and GSH activities and reducing MDA, urea, creatinine, sodium, and potassium (181). The compound iso-orientin reduces serum creatinine and BUN, improves renal SOD, GSH, and inhibits inflammatory cytokines and LPO (182).

7.55 *Linum usitatissimum* L.

Linum usitatissimum, or flaxseed, from the Linaceae family, shows notable effects on glycemic control and kidney health. Studies reveal that aqueous extracts lower fasting blood glucose and HbA1c levels, as well as kidney function markers such as serum creatinine and urea. It also decreases oxidative stress markers such as LPO, and increases antioxidant enzymes (SOD, CAT) (183). Active compounds such as linolenic acid reduce blood glucose, serum creatinine, urinary albumin, and urinary protein levels, and suppress renal inflammation (175). Ferulic acid also lowers BUN, creatinine, and MDA, counters oxidative stress, and reduces kidney inflammation (94).

7.56 *Mangifera indica* L.

Mangifera indica, or mango, is a tropical tree in the Anacardiaceae family with various therapeutic benefits. Ethanol extracts lower blood glucose, inhibit α-amylase, slow starch digestion, increase glucose uptake, and reduce oxidative stress and NO production (184). *In vivo*, *M. indica* enhances kidney function by increasing SOD, CAT, and GSH while lowering serum creatinine

and urea (185). Compounds such as mangiferin lower serum creatinine, uric acid, and BUN, increase antioxidant enzyme activity, and reduce inflammation (128). Catechin and quercetin help regulate blood glucose, reduce markers of kidney damage, and improve antioxidant levels (59, 186).

7.57 *Momordica charantia* L.

Momordica charantia (bitter melon) is a tropical plant in the Cucurbitaceae family, known for its medicinal uses, especially in managing diabetes and kidney function. Aqueous extracts reduce blood glucose, renal MDA, LPO, and oxidative stress by inhibiting free radicals. Serum creatinine and urea decrease, indicating improved kidney function (187). Active compounds such as rosmarinic acid lower serum creatinine and BUN, suppress free radicals and cytokines, and inhibit LPO by reducing MDA levels (75). Oleanolic acid lowers urinary albumin, urea, and creatinine and improves GFR (70). Diosgenin improves serum glucose, creatinine, urea, and protein while increasing kidney antioxidants (SOD, CAT, GPx, GSH) and decreasing LPO and myeloperoxidase (MPO) (188).

7.58 *Nigella sativa* L.

Nigella sativa, a member of the Ranunculaceae family, has various therapeutic effects, particularly in blood glucose regulation, insulin secretion, and kidney health. Studies show aqueous extracts lower blood glucose, increase serum insulin, and reduce urea, creatinine, urinary albumin, total cholesterol, TGs, and LDL, indicating a role in lipid balance (189). Active compounds thymol and thymoquinone decrease serum creatinine and urea, boost SOD, CAT, and GPx activities, lower renal MDA, and reduce kidney inflammatory cytokines (190, 191).

7.59 *Phaseolus vulgaris* L.

Phaseolus vulgaris (common bean), a member of the family Fabaceae, can modulate blood glucose and lipid levels. Its aqueous extracts lower blood glucose, urinary protein, albumin, cholesterol, and TG, aiding glycemic and lipid control (192). Compounds such as genistein and gallic acid reduce serum creatinine and urea levels, increase creatinine clearance, SOD, CAT, and GPx activities, lower MDA levels, and suppress inflammatory biomarkers, thereby reducing renal oxidative stress (76, 193). Hesperidin decreases BUN, creatinine, and urinary albumin, reduces oxidative stress and cytokines, and enhances GSH in renal tissue (194).

7.60 *Punica granatum* L.

Punica granatum (pomegranate) in the family Lythraceae is known for its therapeutic properties, particularly for regulating blood glucose, lipid profiles, and kidney function. Aqueous extracts lower fasting blood glucose, total cholesterol, TGs, LDL, and VLDL, and reduce tissue lipid peroxidase levels while increasing antioxidant enzymes such as SOD, CAT, GSH, and GPx (195). Hydromethanolic extracts improve glucose tolerance, lipid profiles, and reduce urinary albumin, protein, serum creatinine, and BUN (196). Compounds such as punicalagin, chlorogenic acid, and linolenic acid reduce serum creatinine, MDA, NO, MPO activity, and renal inflammation (98, 175, 197). Catechin further reduces blood glucose, plasma creatinine, albuminuria, serum TBARS, and LPO (186).

7.61 *Scoparia dulcis* L.

Scoparia dulcis, or sweet broomweed, from the family Scrophulariaceae, is native to the tropical Americas and parts of Asia. Used in folk medicine, it treats fever, cough, digestive issues, manages diabetes, and supports kidney health. Ethanol extracts lower fasting blood glucose, improve glucose tolerance, inhibit free radicals, and reduce oxidative stress (198). *In vivo* studies show that these extracts lower serum creatinine and BUN levels and LPO, and improve creatinine clearance (199). Compounds like luteolin lower blood glucose, enhance insulin sensitivity, inhibit renal apoptosis, and reduce serum creatinine, BUN, cholesterol, and TG (57). Apigenin and *p*-coumaric acid decrease BUN, boost SOD, CAT, GPx, and glutathione reductase (GR) activities, reduce LPO and NO, and suppress renal inflammation, apoptosis, and MDA levels (67, 200).

7.62 *Terminalia chebula* Retz.

Terminalia chebula (Haritaki), from Southeast Asia and the Combretaceae family, is used in Ayurveda and traditional Chinese medicine for digestion, liver health, and blood purification. It may help regulate blood sugar and support the kidneys. Methanol extracts reduce blood glucose, urinary albumin, serum creatinine, and urinary proteins, while decreasing serum TGs, LDL, and VLDL and increasing HDL (201). Compounds like chebulagic acid and gallic acid lower blood glucose, boost plasma insulin, and improve renal antioxidant activity (76, 202). Ellagic acid reduces glycemia, boosts insulin, suppresses ROS and LPO, decreases renal MDA, and lowers inflammatory biomarkers (159).

7.63 *Urtica dioica* L.

Urtica dioica, or nettle, native to Europe, Asia, and North America, belongs to the Urticaceae family. It's traditionally used for its anti-inflammatory, diuretic, kidney, and blood glucose benefits. Ethanol extracts of *U. dioica* lower BUN and serum creatinine, while boosting GSH activity (203). Compounds such as naringin and rutin reduce BUN, creatinine, uric acid, NO, and TNF-α levels, decrease renal MDA levels, and improve antioxidant activity (SOD, GPx, GST, GSH) (73, 145). Ellagic acid lowers blood sugar, raises serum insulin, suppresses ROS and LPO, and reduces kidney inflammation (159).

7.64 *Zingiber officinale* Roscoe

Zingiber officinale, or ginger, from Southeast Asia, has been used in traditional medicine for centuries. It treats digestive issues, inflammation, nausea, blood glucose regulation, and supports the kidneys. Ethanol extracts lower blood glucose, HbA1c, BUN, serum creatinine, urea, and urinary albumin, while improving lipids,

reducing oxidative stress, and decreasing pro-inflammatory cytokines (204). Compounds like 6-gingerol and 6-shogaol reduce serum creatinine, urea, and BUN, enhance urea clearance, lower renal MDA, and boost renal SOD, CAT, and GSH activities (80, 81).

8 Bioactive phytochemicals and DN

The growing popularity of herbal treatments in both developed and developing nations can be attributed to their inexpensiveness and broad accessibility, although their clinical safety requires rigorous evaluation. Kidney-related complications might be managed more effectively by identifying and isolating potentially active phytochemicals from medicinal plants. The sources of numerous herbal compounds are now being investigated to assess their potential applications in kidney dysfunction. Preclinical studies have extensively documented the renoprotective mechanisms of these plants. By helping to normalize key clinical markers such as BUN, uric acid, and serum creatinine, phytochemicals show potential in delaying disease progression and might help reduce the long-term economic burden of dialysis (259). With the use of modern technology, many biologically active phytochemicals that have been extracted and isolated to develop primary or adjuvant therapy are now being utilized to treat DN (Figure 2).

Following oral ingestion, phytochemicals need to dissolve in the gastrointestinal tract before absorption. Many compounds, like polyphenols, are directly absorbed through the intestinal walls. Once in the bloodstream, they are transported to the liver, where they undergo metabolic processes such as glucuronidation, sulfation, or methylation. Some phytochemicals, however, are poorly absorbed and continue further into the digestive system. In the lower gut, the gut microbiota break down these unabsorbed compounds into smaller, active metabolites that can enter the bloodstream. As a result, the therapeutic effects in patients may come from the original compound, its metabolites, or both. Whether directly absorbed or processed by gut bacteria, these active molecules are ultimately delivered to the kidneys, where they exert their biological and protective effects (260, 261).

There is high variability in the bioavailability of phytochemicals. For instance, quercetin is absorbed in the intestines and primarily circulates as conjugated metabolites, whereas curcumin is poorly absorbed systemically due to instability, limited bioavailability, and extensive metabolism (260–262). Similarly, berberine has low bioavailability due to low intestinal permeability, efflux mediated by P-glycoprotein, and intestinal and hepatic first-pass effects (263). After systemic absorption, the parent compounds and/or metabolites are delivered to peripheral tissues, including the kidney, where oxidative, inflammatory, and fibrotic pathways involved in DN may be affected.

For this reason, the nephroprotective activities observed in animal models cannot be attributed solely to the experimental dose. Bioavailability may have been significantly altered by absorption, metabolism, distribution within the body, and even formulation. Enhancing bioavailability through correct formulation may thus be important in making phytochemicals applicable in treatment (263).

However, since DN patients often receive complex polypharmacy, integrating these bioactive compounds requires careful assessment of potential pharmacokinetic and pharmacodynamic interactions with standard drugs, such as RAAS and SGLT2 inhibitors.

Camellia sinensis, widely known as green tea, offers numerous health benefits and has long been used to prevent and manage diabetes-related complications. *C. sinensis* is rich in polyphenols such as catechin, epicatechin, epicatechin gallate, epigallocatechin, and epigallocatechin gallate. Experiments suggested that green tea can protect from DN and alleviate kidney damage. It has been reported that catechin inhibits the development of AGEs and reduces pro-inflammatory cytokine levels, making it a promising natural remedy for the prevention of complications associated with diabetes (186).

Flavonoids such as quercetin and kaempferol have been experimentally proven to improve the conditions of DN due to their ability to reduce inflammation, oxidative stress, and free radical scavenging activity. Results from experimental diabetic rats given an oral dosage of 10 mg/kg quercetin suggested that the supplement functions as an antioxidant and may be used to cure or prevent DN (59). Similar outcomes were recorded with the same dose of kaempferol. The anti-inflammatory and antioxidant properties of kaempferol greatly mitigated the effects of kidney damage (60). Other phytochemicals, such as galactomannan, hesperidin,

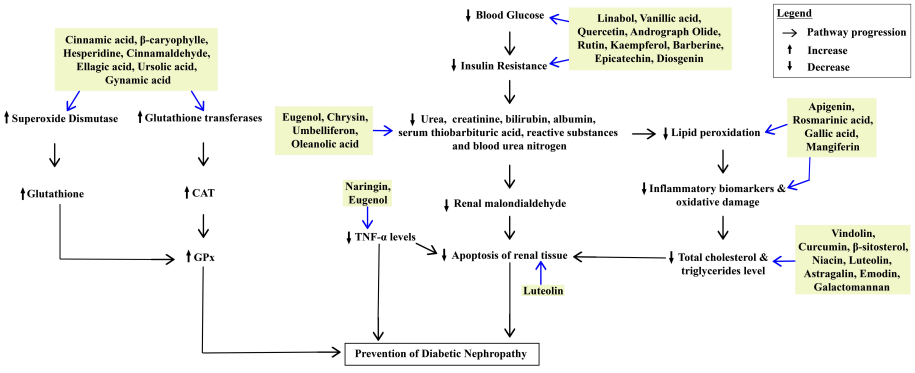


FIGURE 2
The mechanism of phytoconstituents in the management of DN.

apigenin, naringenin, berberine, mangiferin, resveratrol, gingerol, luteolin, chlorogenic acid, ferulic acid, eugenol, and many more from various medicinal plants, also showed potential renoprotective effects for the management of DN (Table 2). Supplementary Figure S1 represents all possible structures of bioactive phytoconstituents against DM and DN.

9 Clinical trials of alternative therapies against DN

Historically, medicinal plants have been widely used in various medical systems around the world to manage diabetes and related complications. Despite their long-standing role in primary healthcare, these plants have not yet been widely adopted as official treatments for diabetes and its associated conditions. This is due to several challenges, including insufficient knowledge of their chemical makeup, proper preparation methods, biologically active compounds, potential side effects, and the most effective dosage forms and routes of administration (285). Recently, the therapeutic potential of these plants has been investigated in several randomized controlled trials focusing on DN patients (Table 3). These studies include short-term treatments lasting 4 to 12 weeks as well as long-term multi-year trials, with sample sizes ranging from small pilot groups to large multicenter studies with over 200 participants.

Many randomized controlled trials have examined the effectiveness of individual botanical extracts and phytochemicals. *Ginkgo biloba* is particularly well-studied; a short-term 4-week trial showed initial reductions in urinary albumin excretion and blood lipids alongside increased creatinine clearance (286). Validating these early renal and metabolic benefits, a comprehensive 3-year double-blind trial involving 210 patients verified its long-term ability to slow the progression of the albumin-to-creatinine ratio (287). Similarly, a large 24-week multicenter trial involving 239 patients showed that *Abelmoschus manihot* (Huangkui capsule) effectively lowered the albumin-to-creatinine ratio and stabilized the estimated GFR, establishing its clinical efficacy alongside standard therapies (288). Other single-plant interventions displayed specific but varying metabolic and renal benefits. For instance, while both *Camellia sinensis* (12 weeks) and *Nigella sativa* (12 weeks) successfully lowered fasting blood glucose and improved GFR, *C. sinensis* additionally reduced serum TG and increased soluble receptor for AGE (sRAGE) levels (289, 290). Conversely, an 8-week trial of *Punica granatum* focused strictly on renal markers, consistently decreasing both urinary albumin and the albumin-to-creatinine ratio without reporting widespread lipid alterations (291).

Beyond single-plant extracts, traditional polyherbal formulations, especially from Traditional Chinese Medicine (TCM), have shown profound clinical efficacy. These intricate decoctions are typically studied in larger groups, ranging from 150 to over 200 patients, over extended periods to evaluate their comprehensive systemic effects. Formulations like the Tangshen formula (180 subjects, 24 weeks) have been reported to lower 24-hour urinary protein levels while improving GFR (292). Other complex mixtures suggested broader metabolic control; both the Zicuiyin decoction (88 subjects, 8 weeks) and the Qidan Tangshen granule (219 patients, 12 months) were reported to not only help maintain GFR stability and reduce urinary protein, but also aid in reducing blood glucose levels, as evidenced by lowered HbA1c (293, 294).

A major advancement in recent clinical trials is the measurement of molecular biomarkers, bridging preclinical mechanisms and human efficacy. Numerous trials have demonstrated how plant-based therapies modulate the innate immune system to translate into clinical outcomes. For instance, Total Glucosides of Paeony (TGP) and the Shenqi Dihuang decoction notably suppressed innate immune activation by reducing circulating pro-inflammatory cytokines such as TNF-α, IL-6, and monocyte chemoattractant protein-1 (MCP-1), as well as decreasing high-sensitivity C-reactive protein (295, 296). The clinical validation of oxidative stress modulation was demonstrated by the Qidan Tangshen granule, which elevated SOD levels and reduced markers of oxidative DNA damage (8-hydroxy-2′-deoxyguanosine and 3-nitrotyrosine) (294). Furthermore, studies with *Piper crocatum* and *Ginkgo biloba* leaf extracts demonstrated direct structural and tubular protection by decreasing early renal injury markers like neutrophil gelatinase-associated lipocalin (NGAL), MMP-9, N-acetyl-beta-glucosaminidase (NAG), and α1-microglobulin (297, 298).

Despite these encouraging findings, common methodological limitations call for cautious interpretation of the reported efficacy. First, many trials were open-label or unblinded, or did not clearly describe random sequence generation and allocation concealment. These shortcomings introduce performance, detection, and potential reporting biases, which may lead to overestimates of therapeutic benefits compared with double-blind, placebo-controlled designs. Second, most studies were conducted at a single center and involved specific regional cohorts, while few large multicenter trials were included (e.g., Huangkui capsule, Tangshen formula). Third, small sample sizes and limited reporting of pre-trial statistical power calculations increase the risk of Type II errors. Most importantly, interventions were predominantly short-term (typically 4 to 12 weeks). As a result, studies mainly assessed transient changes in renal biomarkers rather than hard, long-term clinical endpoints, such as progression to ESRD and overall renal survival.

Comparing these diverse trials indicates that both targeted single-extracts and synergistic complex formulations may offer renoprotective benefits. Among the reported clinical trials, while a few of the short-term 4- to 12-week studies have reported initial anti-proteinuric effects and early metabolic control, these observations remain preliminary. Meanwhile, a few of the long-term studies evaluating specific inflammatory biomarkers and structural renal parameters are encouraging. However, these long-term outcomes were predominantly not derived from single plants. As the number of studies is limited to date, we still lack sufficient evidence to determine the optimal treatment duration for specific medicinal plants. Further studies targeting individual medicinal plants are needed to verify these effects and establish optimal treatment durations for phytotherapy as a targeted adjuvant strategy for DN.

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TABLE 2 Medicinal plants and their phytoconstituents with promising effects against diabetes and DN.

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
Abelmoschus esculentus L. Moench	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content; protected the kidneys from oxidative damage	(53)
Abrus precatorius L.	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
Acacia arabica (Lam.) Willd.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Acacia catechu (L.f.) Willd.	Naringenin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Urea, uric acid, creatinine, bilirubin, urinary albumin, and serum TBARS levels, ↓ renal MDA levels, enhanced SOD, CAT, GSH, GPx, and GST activity in kidneys	(65)
Achillea millefolium L.	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
Aerva javanica Juss.	Vanillic acid	25, 50, and 100 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity	↓ Glycemic levels, ↓ serum creatinine, urea, uric acid, and urinary microproteinuria levels, ↑ urinary creatinine clearance, and suppressed production of inflammatory biomarkers	(77)
Ageratum conyzoides L.	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Albizia adianthifolia (Schumach.) W.Wight	Mangiferin	10, 20, and 40 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, uric acid, and BUN content, enhanced SOD, CAT, GSH, GST, and GPx activity, and ↓ inflammatory biomarkers	(128)
Allium cepa L.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Alstonia scholaris (L.) R. Br.	Vallesamine	5 mg/kg body weight	Transcriptomic analysis, RT-qPCR, Western blotting, molecular docking and molecular dynamics	Alleviated diabetic kidney injury through PPAR-δ regulation; modulated lipid-metabolism genes and proteins, and renal inflammatory/podocyte-associated markers, including IL-6, BCL-2 and podocin	(205)
Amaranthus tricolor L.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Vanillic acid	25, 50, and 100 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity	↓ Glycemic levels, ↓ serum creatinine, urea, uric acid, and urinary microproteinuria levels, ↑ urinary creatinine clearance, and suppressed production of inflammatory biomarkers	(77)
Anacardium occidentale L.	Naringenin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Urea, uric acid, creatinine, bilirubin, urinary albumin, and serum TBARS levels, ↓ renal MDA levels, enhanced SOD, CAT, GSH, GPx, and GST activity in kidneys	(65)
Andrographis paniculata (Burm.f.) Wall.	Andrographolide	2 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Fasting blood glucose levels, ↓ serum creatinine, BUN, and 24-h albuminuria, attenuated ROS production and pro-inflammatory cytokines	(100)
Anemarrhena asphodeloides Bunge	Mangiferin	10, 20, and 40 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, uric acid, and BUN content, enhanced SOD, CAT, GSH, GST, and GPx activity, and ↓ inflammatory biomarkers	(128)
Annona squamosa L.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content; protected the kidneys from oxidative damage	(53)
Apium graveolens L.	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
Arachis hypogaea L.	Resveratrol	5 and 10 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity, LPO	↓ Blood glucose levels, ↓ serum creatinine, urinary albumin and urea content, ↑ creatinine clearance, ↓ renal MDA levels, AGE accumulation, and oxidative stress	(136)
	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)

(Continued)

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Artemisia absinthium L.	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content; protected the kidneys from oxidative damage	(53)
Asparagus racemosus Willd.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Kaempferol	5, 10, and 50 µM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Azadirachta indica A. Juss.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Berberis aristata DC.	Berberine	100, 200 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↓ serum creatinine and urinary albumin levels, ↓ renal oxidative stress, and suppressed generation of inflammatory biomarkers	(117)
Berberis vulgaris L.	Berberine	100, 200 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↓ serum creatinine and urinary albumin levels, ↓ renal oxidative stress, and suppressed generation of inflammatory biomarkers	(117)
Biophytum sensitivum (L.) DC.	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
Bombax ceiba L.	Mangiferin	10, 20, and 40 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, uric acid, and BUN content, enhanced SOD, CAT, GSH, GST, and GPx activity; ↓ inflammatory biomarkers	(128)
	Lupeol	25 mg/kg body weight	Biochemical and kidney function parameters, serum electrolytes	↓ Blood glucose levels, improved insulin resistance, ↓ urea, uric acid, and creatinine levels, ↓ serum sodium and potassium ions	(129)
Brassica juncea (L.) Czern.	Kaempferol	5, 10, and 50 µM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Bunium persicum B.Fedtsch.	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content, protected the kidneys from oxidative damage	(53)
	Kaempferol	5, 10, and 50 µM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Caesalpinia decapetala (Roth) Alston	Resveratrol	5 and 10 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity, LPO	↓ Blood glucose levels, ↓ serum creatinine, urinary albumin and urea content, ↑ creatinine clearance, ↓ renal MDA levels, AGE accumulation, and oxidative stress	(136)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Lupeol	25 mg/kg body weight	Biochemical and kidney function parameters, serum electrolytes	↓ Blood glucose levels, improved insulin resistance, ↓ urea, uric acid, and creatinine levels, ↓ serum sodium and potassium ions	(129)
Camellia sinensis (L.) Kuntze	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
Carica papaya L.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Cassia auriculata L.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Catharanthus roseus (L.) G.Don	Vindoline	20 mg/kg body weight	Kidney function and biochemical parameters, lipid profile, LPO, OGTT, antioxidant activity	↓ Blood glucose levels, improved glucose tolerance, ↓ serum creatinine and urea content, ↓ renal MDA levels, ↓ total cholesterol, TGs, VLDL and LDL levels, ↑ HDL levels, enhanced SOD and CAT activity in kidneys	(142)
Centella asiatica (L.) Urb.	Asiaticoside	10 mg/kg body weight	Kidney function parameters, anti-inflammatory activity	↓ Serum creatinine, BUN content, and 24-hour urine protein; suppressed inflammatory biomarkers (IL-6, IL-8, TNF-α)	(146)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
Coccinia grandis (L.) Voigt	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Coccinia indica Wight & Arn.	Lupeol	25 mg/kg body weight	Biochemical and kidney function parameters, serum electrolytes	↓ Blood glucose levels, improved insulin resistance, ↓ urea, uric acid, and creatinine levels, ↓ serum sodium and potassium ions	(129)
Cola nitida (Vent.) Schott & Endl.	Naringenin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Urea, uric acid, creatinine, bilirubin, urinary albumin, and serum TBARS levels, ↓ renal MDA levels, enhanced SOD, CAT, GSH, GPx, and GST activity in kidneys	(65)
	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
Coriandrum sativum L.	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content; protected the kidneys from oxidative damage	(53)
Cornus officinalis Siebold & Zucc.	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in the kidneys	(159)
	Ursolic acid	35 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Fasting blood glucose levels, ↓ serum creatinine and BUN content, enhanced renal SOD activity, ↓ renal MDA levels, and suppressed overexpression of inflammatory cytokines in the kidneys	(165)
Curcuma longa L.	Curcumin	100 mg/kg body weight	Biochemical and kidney function parameters, OGTT, antioxidant activity, LPO, and lipid profile	↓ Blood glucose levels, ↑ insulin levels, improved glucose tolerance, ↓ total cholesterol and TG levels, ↓ urinary albumin excretion and serum creatinine content, enhanced SOD activity in kidneys, ↓ renal MDA levels	(157)
Emblica officinalis Gaertn.	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in the kidneys	(159)
Eriobotrya japonica (Thunb.) Lindl.	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Euphorbia hirta L.	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress and inflammatory biomarkers in kidneys	(60)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Ficus benghalensis L.	Lupeol	25 mg/kg body weight	Biochemical and kidney function parameters, serum electrolytes	↓ Blood glucose levels, improved insulin resistance, ↓ urea, uric acid, and creatinine levels, ↓ serum sodium and potassium ions	(129)
Ficus racemosa Willd.	Lupeol	25 mg/kg body weight	Biochemical and kidney function parameters, serum electrolytes	↓ Blood glucose levels, improved insulin resistance, ↓ urea, uric acid, and creatinine levels, ↓ serum sodium and potassium ions	(129)
	Ursolic acid	35 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Fasting blood glucose levels, ↓ serum creatinine and BUN content, enhanced renal SOD activity, ↓ renal MDA levels, and suppressed overexpression of inflammatory cytokines in kidneys	(165)
Forsythia suspensa	Phillygenin	50 mg/kg body weight	RNA sequencing, ELISA, Western blotting, immunofluorescence and immunohistochemistry	↓ Urinary albumin-to-creatinine ratio, inflammatory cytokines and podocyte apoptosis; inhibited TLR4/MyD88/NF-κB signaling and enhanced phosphorylation associated with the PI3K/AKT/GSK3β pathway	(264)
Fragaria ananassa Duchesne	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in kidneys	(159)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress and inflammatory biomarkers in kidneys	(60)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
Glycine max (L.) Merr.	β-Sitosterol	20 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, enhanced glucose uptake into skeletal muscles, ↓ total cholesterol and TGs, ↓ urea and creatinine content in plasma, enhanced SOD, CAT, and GSH activity in kidneys	(171)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress and inflammatory biomarkers in kidneys	(60)
Gundelia tournefortii L.	Linolenic acid	10 or 100 μM	Biochemical and kidney function parameters, anti-inflammatory activity	↓ Blood glucose levels, ↓ creatinine, urinary albumin, and total urinary protein levels; suppressed renal inflammatory biomarkers	(175)
	Ursolic acid	35 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Fasting blood glucose levels, ↓ serum creatinine and BUN content, enhanced renal SOD activity, ↓ renal MDA levels, and suppressed overexpression of inflammatory cytokines in the kidneys	(165)
	β-Sitosterol	20 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, enhanced glucose uptake into skeletal muscles, ↓ total cholesterol and TGs, ↓ urea and creatinine content in plasma, enhanced SOD, CAT, and GSH activity in kidneys	(171)
Heritiera fomes Buch.-Ham.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
	β-Sitosterol	20 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, enhanced glucose uptake into skeletal muscles, ↓ total cholesterol and TGs, ↓ urea and creatinine content in plasma, enhanced SOD, CAT, and GSH activity in the kidneys	(171)
Hibiscus rosa-sinensis L.	β-Sitosterol	20 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, enhanced glucose uptake into skeletal muscles, ↓ total cholesterol and TGs, ↓ urea and creatinine content in plasma, enhanced SOD, CAT, and GSH activity in kidneys	(171)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Linum usitatissimum L.	Linolenic acid	10 or 100 μM	Biochemical and kidney function parameters, anti-inflammatory activity	↓ Blood glucose levels, ↓ creatinine, urinary albumin, and total urinary protein levels; suppressed renal inflammatory biomarkers	(175)
Mangifera indica L.	Mangiferin	10, 20, and 40 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, uric acid, and BUN content, enhanced SOD, CAT, GSH, GST, and GPx activity, and ↓ inflammatory biomarkers	(128)
	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Mentha x piperita L.	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
Momordica charantia L.	Diosgenin	5, 10, and 20 mg/kg body weight	Biochemical and kidney function parameters, LPO, lipid profile, antioxidant and anti-inflammatory activity	↓ Blood glucose levels, ↓ serum creatinine, urea, and urinary protein excretion; enhanced SOD, CAT, GPx, and GSH activity in the kidneys; ↓ LPO, and MPO level	(188)
Moringa oleifera Lam.	*p*-Coumaric acid	20 mg/kg body weight	Biochemical and kidney function parameters, LPO, and antioxidant activity	↓ Blood glucose levels, serum urea, creatinine, and uric acid; enhanced SOD, CAT, and GST activity in the kidneys, and ↓ renal MDA levels	(200)
Murraya koenigii (L.) Spreng.	Linalool	25 mg/kg body weight	Biochemical parameters, kidney function, lipid profile, and antioxidant activity	↓ Glycemic levels, improved glucose uptake, ↓ inflammatory biomarkers, ↓ urea, creatinine, and BUN content, protected the kidneys from oxidative damage	(53)
	Myrcene	50 mg/kg body weight	Biochemical parameters, OGTT, antioxidant and anti-inflammatory activity, and histopathological analysis	↓ Blood glucose levels, improved glucose tolerance and glucose metabolism, inhibited inflammatory cytokine and pro-inflammatory signaling, ↓ glucose oxidative stress, enhanced renal SOD, CAT and GPx activity, ↓ renal MDA levels, decreased the matrix accumulation and fibrosis	(265)
Musa sapientum L	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)
Nasturtium officinale R.Br.	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
Ocimum sanctum L.	β-Sitosterol	20 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, enhanced glucose uptake into skeletal muscles, ↓ total cholesterol and TGs, ↓ urea and creatinine content in plasma, enhanced SOD, CAT, and GSH activity in kidneys	(171)
Olea europaea subsp. europaea	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
Panax ginseng f. *trifoliolatus* Makino	Ginsenoside Rb1	10 μM	Biochemical and kidney function parameters, antioxidant activity	↓ Glycemic levels, decreased creatinine, BUN, and proteinuria, inhibited ROS generation and the activity of aldose reductase, and prevented high glucose-induced cell apoptosis	(266)
Phyllanthus amarus Schumach. & Thonn.	Astragalin	5 and 10 mg/kg body weight	Biochemical and kidney function parameters, lipid profile, antioxidant activity	↓ Blood glucose levels, ↓ serum and urine creatinine, total cholesterol, TGs and LDL levels, enhanced renal SOD and GSH activity, ↓ MDA levels	(267)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
Plantago ovata Phil.	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress, and inflammatory biomarkers in the kidneys	(60)
Pterocarpus marsupium Roxb.	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
Punica granatum L.	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)
	Linolenic acid	10 or 100 μM	Biochemical and kidney function parameters, anti-inflammatory activity	↓ Blood glucose levels, ↓ creatinine, urinary albumin, and total urinary protein levels; suppressed renal inflammatory biomarkers	(175)
Raphanus sativus var. *sativus* (L.) Schmalh.	Syringaresinol	25 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity, and histopathological analysis	↓ Urine creatinine and albuminuria ameliorated renal hypertrophy, fibrosis, mesangial expansion, glomerular basement membrane thickening, upregulated antioxidant genes, and suppressed ROS production in the kidneys	(268)
Rehmannia glutinosa var. *makinoi* Matsuda	Catalpol	0, 60, 120 mg/kg body weight	Biochemical and kidney function parameters, histopathological analysis	↓ Blood glucose levels, ↓ glycated serum protein, 24 h urinary protein excretion, serum creatinine, and BUN attenuated kidney hypertrophy, inhibited the accumulation of extracellular matrix	(269)
Scoparia dulcis L.	Luteolin	80 mg/kg body weight	Biochemical parameters, kidney function, lipid profile	↓ Blood glucose levels, improved insulin resistance, inhibited apoptosis of renal tissue, ↓ serum creatinine and BUN content, ↓ total cholesterol, and TG levels	(57)
	p-Coumaric acid	20 mg/kg body weight	Biochemical and kidney function parameters, LPO, and antioxidant activity	↓ Blood glucose levels, serum urea, creatinine, and uric acid; enhanced SOD, CAT, and GST activity in kidneys, and ↓ renal MDA levels	(200)
Swietenia macrophylla King	Swietenine	25 and 50 mg/kg body weight (*in vivo*); 1, 5, and 10 μM (*in vitro*)	Kidney function tests, iron content, ROS, MDA, and GSH assays, TEM, Western blotting	Enhanced renal function, mitigated oxidative stress, and inhibited ferroptosis via activation of the Akt/GSK-3β/Nrf2 signaling pathway	(270)
Syzygium cumini (L.) Skeels	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress and inflammatory biomarkers in kidneys	(60)
Tamarindus indica L.	Naringenin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Urea, uric acid, creatinine, bilirubin, urinary albumin, and serum TBARS levels, ↓ renal MDA levels, enhanced SOD, CAT, GSH, GPx, and GST activity in kidneys	(65)
	Epicatechin	35 mg/dL body weight	Biochemical and kidney function parameters, LPO	↓ Glycemic levels, ↓ serum creatinine and urinary albumin excretion, ↑ creatinine clearance, and ↓ renal MDA levels	(123)
Terminalia chebula Retz.	Chebulagic acid	50 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity	↓ Blood glucose levels, ↑ plasma insulin concentration, ↓ serum creatinine and urea content, enhanced SOD, CAT, GPx, and GST activity in kidneys	(202)
	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in kidneys	(159)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Renoprotective effect models associated with DN					
Tinospora cordifolia (Willd.) Miers ex Hook.f. & Thomson	Magnoflorine	100 mg/kg body weight	Kidney function parameters, anti-inflammatory activity, and histopathological analysis	↓ Serum creatinine and BUN, ↓ proliferation of mesangial cells, inhibited inflammation and fibrosis in mesangial cells	(271)
	Berberine	100, 200 mg/kg body weight	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↓ serum creatinine and urinary albumin levels, ↓ renal oxidative stress, and suppressed generation of inflammatory biomarkers	(117)
Tribulus terrestris L.	Ruscogenin	0.3, 1.0, or 3.0 mg/kg body weight	Biochemical and kidney function parameters, anti-inflammatory activity, and histopathological analysis	↓ BUN and proteinuria, ↑ creatinine clearance, improved histological architecture in the diabetic kidney, and suppressed production of inflammatory cytokines in the kidneys	(272)
	Diosgenin	5, 10, and 20 mg/kg body weight	Biochemical and kidney function parameters, LPO, lipid profile, antioxidant and anti-inflammatory activity	↓ Blood glucose levels, ↓ serum creatinine, urea, and urinary protein excretion, enhanced SOD, CAT, GPx, and GSH activity in the kidneys, ↓ LPO, and MPO level	(188)
Trigonella foenum-graecum L.	Galactomannan	500 mg/kg body weight	Biochemical and kidney function parameters, LPO, lipid profile, antioxidant and anti-inflammatory activity	↓ Fasting blood glucose levels, ↓ serum creatinine, urinary albumin and urea content, ↓ levels of total cholesterol, TGs and LDL, enhanced SOD and GSH activity, suppressed ROS and inflammatory cytokines generation in kidneys	(273)
	Diosgenin	5, 10, and 20 mg/kg body weight	Biochemical and kidney function parameters, LPO, lipid profile, antioxidant and anti-inflammatory activity	↓ Blood glucose levels, ↓ serum creatinine, urea, and urinary protein excretion, enhanced SOD, CAT, GPx, and GSH activity in the kidneys, ↓ LPO, and MPO level	(188)
Tripterygium wilfordii Hook.f. L.	Triptolide	100 μg/kg	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Urine albumin, serum creatinine, and BUN, ↓ fasting blood glucose, enhanced renal SOD and GSH activity, ↓ MDA levels, suppressed production of free radical species, and pro-inflammatory cytokines	(274)
	Celastrol	50 and 100 μg/kg	Biochemical and kidney function parameters, antioxidant and anti-inflammatory activity	↓ Blood glucose level, improved insulin resistance, ↓ serum creatinine, BUN, and 24-hr urinary protein, suppressed production of free radical species and pro-inflammatory cytokines, and ↓ oxidative stress	(275)
Urtica dioica L.	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in kidneys	(159)
Vaccinium macrocarpon Aiton	Ellagic acid	100 mg/kg body weight	Kidney function and biochemical parameters, antioxidant and anti-inflammatory activity	↓ Glycemic levels, ↑ serum insulin levels, suppressed ROS generation and LPO, ↓ renal MDA levels, enhanced renal SOD and GSH activity, suppressed inflammatory biomarkers production in kidneys	(159)
	Quercetin	10 mg/kg body weight	Antioxidant activity, kidney function, and biochemical parameters,	↓ Blood glucose levels, ↑ renal SOD, CAT, and GSH levels, ↓ renal MDA and LPO, ↓ serum creatinine, urea, and BUN	(59)
	p-Coumaric acid	20 mg/kg body weight	Biochemical and kidney function parameters, LPO, and antioxidant activity	↓ Blood glucose levels, serum urea, creatinine, and uric acid; enhanced SOD, CAT, and GST activity in kidneys, and ↓ renal MDA levels	(200)
Vaccinium myrtillus L.	Delphinidin-3-*O*-glucoside	54, 162, and 486 mg/kg	Biochemical parameters, anti-inflammatory activity	↓ GSP, improved lipid metabolism, ↓ inflammatory cytokines by modulating the NF-κB signaling pathway	(242)
	Cyanidin-3-*O*-glucoside	54, 162, and 486 mg/kg	Biochemical parameters, anti-inflammatory activity	↓ GSP, improved lipid metabolism, ↓ inflammatory cytokines by modulating the NF-κB signaling pathway	(242)
Vigna angularis (Willd.) Ohwi & H.Ohashi	Kaempferol	5, 10, and 50 μM	Biochemical parameters, antioxidant activity	↓ Blood glucose levels, enhanced renal SOD and GSH activity, ↓ renal MDA, ↓ oxidative stress and inflammatory biomarkers in kidneys	(60)
	Catechin	5 mg/kg body weight	Biochemical and kidney function parameters, LPO	↓ Blood glucose levels, ↓ plasma creatinine content, albuminuria, ↓ serum TBARS, and LPO	(186)
Other renoprotective effect models					
Abelmoschus esculentus **L.** Moench	Eugenol	100 mg/kg body weight	Antioxidant activity, kidney function parameters	↓ Serum urea, creatinine, and LPO, ↓ renal MDA and TNF-α levels, ↑ renal SOD and GSH levels	(54)
Acacia catechu (L.f.) Willd.	Chrysin	25 and 50 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea, enhanced renal SOD, CAT, and GPx, and ↓ pro-inflammatory cytokines	(63)
	Umbelliferone	30 mg/kg	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and BUN content, enhanced renal GSH, SOD, and Nrf-2, suppressed MDA levels, and ↓ LPO	(64)
Achillea millefolium L.	Apigenin	5, 10, and 20 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity, LPO	↓ BUN and creatinine content, enhanced SOD, CAT, GPx, and GR activity in kidneys, ↓ levels of lipid peroxide and NO, suppressed renal inflammation and apoptosis by decreasing levels of inflammatory biomarkers	(67)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Other renoprotective effect models					
Achyranthes aspera L.	Oleanolic acid	40, 60, and 80 mg/kg body weight	Kidney function parameters, histopathological analysis	↓ Urinary albumin, urea, and creatinine levels; improved GFR	(70)
Aegle marmelos (L.) Corrêa	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
	Eugenol	100 mg/kg body weight	Antioxidant activity, kidney function parameters	↓ Serum urea, creatinine, and LPO, ↓ renal MDA and TNF-α levels, ↑ renal SOD and GSH levels	(54)
Aerva javanica Juss.	Rosmarinic acid	1, 2, and 5 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, suppressed production of free radicals and pro-inflammatory cytokines, inhibited LPO by reducing MDA levels	(75)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, and ↓ oxidative stress in the kidneys	(76)
Aframomum melegueta K.Schum.	6-Gingerol	12.5, 25, 50 mg/kg body weight	Kidney function parameters, LPO and antioxidant activity	↓ Serum creatinine, urea, and BUN content, ↑ urea clearance, ↓ renal MDA levels, and enhanced renal SOD, CAT, and GSH activity	(80)
	6-Shogaol	20 mg/kg body weight	Kidney function parameters, antioxidant activity	↓ Serum creatinine and BUN content, ↓ renal MDA levels, enhanced SOD, CAT, and GSH activity	(81)
Albizia adianthifolia (Schumach.) W.Wight	β-Caryophyllene	200 and 400 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Creatinine and urea content, enhanced SOD, CAT, GSH, and GPx activity, and ↓ renal MDA level	(107)
Allium cepa L.	Allicin	50 mg/kg	Kidney function parameters, LPO, and antioxidant activity	↓ Serum creatinine, BUN, urinary protein and albumin, ↓ renal MDA levels, enhanced renal GSH and SOD activity	(86)
Allium sativum L.	Allicin	50 mg/kg	Kidney function parameters, LPO, and antioxidant activity	↓ Serum creatinine, BUN, urinary protein and albumin, ↓ renal MDA levels, enhanced renal GSH and SOD activity	(86)
	Diallyl sulfide	50 mg/kg body weight	Kidney function parameters, antioxidant activity	↓ Serum creatinine, BUN, and urinary protein excretion; inhibited ROS generation and ↓ oxidative stress in the kidneys	(89)
Aloe vera Burm.f.	Emodin	10 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity	↓ Urinary albumin, creatinine, uric acid, BUN content, ↓ total cholesterol, and TGs; enhanced renal SOD and GSH activity	(92)
Amaranthus tricolor L.	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidneys	(94)
Anacardium occidentale L.	Delphinidin	25 mg/kg body weight	Kidney function parameters, LPO, anti-inflammatory and antioxidant activity	↓ Serum creatinine and urea levels, ↑ creatinine clearance, enhanced renal SOD, GR, GSH, GPx, and ↓ MDA levels, and suppressed production of pro-inflammatory cytokines	(96)
Ananas comosus (L.) Merr.	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidneys	(94)
Annona squamosa L.	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Apium graveolens L.	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidney	(94)
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Artemisia absinthium L.	β-Caryophyllene	200 and 400 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Creatinine and urea content, enhanced SOD, CAT, GSH, and GPx activity, and ↓ renal MDA levels	(107)
Artocarpus heterophyllus Lam.	Silymarin	80 mg/kg body weight	Kidney function parameters, antioxidant activity	↓ Serum creatinine, urea, uric acid, and BUN content; prevented the generation of ROS, inflammatory biomarkers, and ↓ renal MDA levels	(109)
	β-Carotene	10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of creatinine, urea, uric acid, ↓ renal MDA levels, and normalized levels of pro-inflammatory cytokines	(110)
Asparagus racemosus Willd.	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Other renoprotective effect models					
Azadirachta indica A. Juss.	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Berberis aristata DC.	Palmatine	50 and 100 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity, LPO	↓ Urine flow rate, creatinine clearance, and renal MDA levels, ↓ BUN and creatinine content, enhanced GSH activity in kidneys, ↑ fractional excretion of Na and K, inhibited renal oxidative stress	(118)
Biophytum sensitivum (L.) DC.	Orientin	40 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory, LPO	↓ Creatinine and urea levels and ↑ creatinine clearance, enhanced renal SOD, CAT, GSH, GR, and GPx activity, suppressed production of pro-inflammatory cytokines in the kidneys, ↓ hydrogen peroxide, and TBARS to normal levels	(122)
Bixa orellana L.	Bixin	100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory, LPO	↓ Serum levels of creatinine, urea, uric acid, ↓ MDA level in kidneys, ↑ the activation of SOD, CAT, and GPx, ameliorated renal inflammation by suppressing the production of pro-inflammatory cytokines	(125)
Brassica juncea (L.) Czern.	Cinnamic acid	20 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ Uric acid, urea, and creatinine levels, ↓ nitrate, and MDA levels returned to normalcy, enhanced renal GSH activity, and suppressed oxidative stress in the kidneys	(132)
Brassica oleracea L.	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, and ↓ oxidative stress in the kidneys	(76)
	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
Bunium persicum B.Fedtsch.	β-Caryophyllene	200 and 400 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Creatinine and urea content, enhanced SOD, CAT, GSH, and GPx activity, and ↓ renal MDA levels	(107)
Camellia sinensis (L.) Kuntze	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, and ↓ oxidative stress in the kidneys	(76)
Carica papaya L.	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Cassia auriculata L.	Hesperidin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ BUN, creatinine, urinary albumin levels, ↓ renal tissue oxidative stress, and inflammatory cytokines; enhanced GSH activity	(194)
Catharanthus roseus (L.) G.Don	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
Centella asiatica (L.) Urb.	Naringin	25, 50, and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ BUN, creatinine, uric acid, NO, and TNF-α, ↓ renal MDA levels, enhanced GSH, GST, GPx, and SOD activity in kidneys	(145)
Cinnamomum zeylanicum Blume	Cinnamaldehyde	20 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Serum creatinine and urea content, ↓ renal MDA levels, enhanced SOD, CAT, and GSH activity in kidneys	(149)
	Eugenol	100 mg/kg body weight	Antioxidant activity, kidney function parameters	↓ Serum urea, creatinine, and LPO, ↓ renal MDA and TNF-α levels, ↑ renal SOD and GSH levels	(54)
Coccinia grandis (L.) Voigt	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Coccinia indica Wight & Arn.	β-Carotene	10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of creatinine, urea, uric acid, ↓ renal MDA levels, and normalized levels of pro-inflammatory cytokines	(110)
Cola nitida (Vent.) Schott & Endl.	Apigenin	5, 10, and 20 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity, LPO	↓ BUN and creatinine content, enhanced SOD, CAT, GPx, and GR activity in kidneys, ↓ levels of lipid peroxide and NO, suppressed renal inflammation and apoptosis by decreasing levels of inflammatory biomarkers	(67)
Coriandrum sativum L.	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidneys	(94)

(Continued)

10.3389/fimmu.2026.1964602

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TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Other renoprotective effect models					
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Cornus officinalis Siebold & Zucc.	Gymnemic acid	20, 40, and 60 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea content, enhanced renal GSH, SOD, and CAT activity, suppressed production of renal inflammatory biomarkers, and ↓ renal MDA levels	(276)
Emblica officinalis Gaertn.	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Eriobotrya japonica (Thunb.) Lindl.	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
Euphorbia hirta L.	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
Ficus benghalensis L.	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
Ficus racemosa Willd.	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, and ↓ renal MDA levels	(73)
Garcinia kola Heckel	Kolaviron	100 and 200 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of serum creatinine and urea, ↓ renal MDA levels, enhanced SOD, CAT, GSH, GPx, and GST activity in kidneys	(168)
Glycyrrhiza glabra L.	Glycyrrhetinic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, ↓ LPO and NO levels, and suppressed production of pro-inflammatory cytokines in the kidneys	(173)
Hibiscus rosa-sinensis L.	Niacin	50 mg/kg body weight	Kidney function parameters, lipid profile, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea content, ↑ creatinine clearance, ↓ total cholesterol, TGs, and LDL and ↑ HDL levels, ↓ renal MDA levels, enhanced CAT and GR activity, suppressed ROS and inflammatory cytokines generation in the kidneys	(179)
Jatropha curcas L.	Iso-orientin	50 mg/kg body weight	Kidney function parameters, LPO, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN, enhanced renal SOD and GSH activity, ↓ MDA levels and LPO, suppressed production of pro-inflammatory cytokines	(182)
Linum usitatissimum L.	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidneys	(94)
Lycopersicon esculentum Mill.	Cinnamic acid	20 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ Uric acid, urea, and creatinine levels, ↓ nitrate, and MDA levels returned to normalcy, enhanced renal GSH activity, and suppressed oxidative stress in the kidneys	(132)
	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
Mentha x piperita L.	Hesperidin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ BUN, creatinine, urinary albumin levels, ↓ renal tissue oxidative stress, and inflammatory cytokines; enhanced GSH activity	(194)
	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
Mimosa pudica L.	Syringic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Creatinine and BUN, ↑ creatinine clearance rate, enhanced renal GSH activity, suppressed production of ROS, and pro-inflammatory cytokines in the kidneys	(277)

(Continued)

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Other renoprotective effect models					
Momordica charantia L.	Rosmarinic acid	1, 2, and 5 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, suppressed production of free radicals and pro-inflammatory cytokines, inhibited LPO by reducing MDA levels	(75)
	Oleanolic acid	40, 60, and 80 mg/kg body weight	Kidney function parameters, histopathological analysis	↓ Urinary albumin, urea, and creatinine levels; improved GFR	(70)
Momordica dioica Roxb. ex Willd.	Oleanolic acid	40, 60, and 80 mg/kg body weight	Kidney function parameters, histopathological analysis	↓ Urinary albumin, urea, and creatinine levels; improved GFR	(70)
	Niacin	50 mg/kg body weight	Kidney function parameters, lipid profile, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea content, ↑ creatinine clearance, ↓ total cholesterol, TGs, and LDL and ↑ HDL levels, ↓ renal MDA levels, enhanced CAT and GR activity, suppressed ROS and inflammatory cytokines generation in the kidneys	(179)
Moringa oleifera Lam.	Vanillin	150 mg/kg body weight	Kidney function parameters, lipid profile, antioxidant activity	↓ Plasma creatinine, urea, and uric acid levels, enhanced SOD, CAT, GSH, GPx activity in kidneys, ↓ MDA levels, and suppressed renal oxidative stress	(278)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
Musa sapientum L	Syringic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Creatinine and BUN, ↑ creatinine clearance rate, enhanced renal GSH activity, suppressed production of ROS, and pro-inflammatory cytokines in the kidneys	(277)
	β-Carotene	10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of creatinine, urea, uric acid, ↓ renal MDA levels, and normalized levels of pro-inflammatory cytokines	(110)
Nasturtium officinale R.Br.	Lutein	1mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ BUN and creatinine content, enhanced total GSH, total oxidant system, and total antioxidant system levels in the kidneys, and ↓ renal MDA levels	(279)
	β-Carotene	10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of creatinine, urea, uric acid, ↓ renal MDA levels, and normalized levels of pro-inflammatory cytokines	(110)
Nigella sativa L.	Thymol	100 mg/kg body weight	Biochemical and kidney function parameters, LPO, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea levels, enhanced renal SOD, CAT, and GPx activity, ↓ renal MDA levels, and suppressed production of pro-inflammatory cytokines in the kidneys	(190)
	Thymoquinone	50 mg/kg body weight	Antioxidant and anti-inflammatory activity, LPO, and histopathological analysis	↑ Renal SOD, CAT, and GPx activity, ↓ renal MDA levels, ameliorated hydropic degeneration and hypertrophy of epithelia of the proximal tubules and tubular and glomerular swelling, suppressed production of pro-inflammatory cytokines in the kidneys	(191)
Ocimum sanctum L.	Eugenol	100 mg/kg body weight	Antioxidant activity, kidney function parameters	↓ Serum urea, creatinine, and LPO, ↓ renal MDA and TNF-α levels, ↑ renal SOD and GSH levels	(54)
	Apigenin	5, 10, and 20 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity, LPO	↓ BUN and creatinine content, enhanced SOD, CAT, GPx, and GR activity in kidneys, ↓ levels of lipid peroxide and NO, suppressed renal inflammation and apoptosis by decreasing levels of inflammatory biomarkers	(67)
Olea europaea subsp. europaea	Oleuropein	50 mg/kg body weight	Kidney function parameters, LPO, antioxidant and anti-inflammatory	↓ Levels of serum creatinine and urea, enhanced renal GSH activity, ↓ MDA and NO levels, and suppressed renal inflammatory biomarkers	(280)
	Oleanolic acid	40, 60, and 80 mg/kg body weight	Kidney function parameters, histopathological analysis	↓ Urinary albumin, urea, and creatinine levels; improved GFR	(70)
Phaseolus vulgaris L.	Genistein	15 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and urea, ↑ creatinine clearance, enhanced renal SOD, CAT, and GPx activity, ↓ MDA levels, suppressed production of inflammatory biomarkers, and ↓ oxidative stress in the kidneys	(193)
	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
	Hesperidin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ BUN, creatinine, urinary albumin levels, ↓ renal tissue oxidative stress, and inflammatory cytokines; enhanced GSH activity	(194)
Phyllanthus amarus Schumach. & Thonn	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, ↓ renal MDA levels	(73)
Piper betle L.	Eugenol	100 mg/kg body weight	Antioxidant activity, kidney function parameters	↓ Serum urea, creatinine, and LPO, ↓ renal MDA and TNF-α levels, ↑ renal SOD and GSH levels	(54)
	β-Carotene	10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Levels of creatinine, urea, uric acid, ↓ renal MDA levels, and normalized levels of pro-inflammatory cytokines	(110)

(Continued)

10.3389/fimmu.2026.1964602

TABLE 2 Continued

Species	Compounds	Dose	Assay	Antidiabetic and renoprotective effects	Ref.
Other renoprotective effect models					
Punica granatum L.	Punicalagin	50 mg/kg body weight	Kidney function parameters, LPO, antioxidant and anti-inflammatory activity	↓ Serum creatinine levels, ↓ renal MDA, NO, and MPO activity, and suppressed renal inflammatory biomarkers	(197)
	Chlorogenic acid	3, 10, and 30 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, ↓ renal oxidative stress, and suppressed production of inflammatory cytokines in the kidneys	(98)
Rheum ribes L.	Emodin	10 mg/kg body weight	Biochemical and kidney function parameters, antioxidant activity	↓ Urinary albumin, creatinine, uric acid, BUN content, ↓ total cholesterol, and TGs; enhanced renal SOD and GSH activity	(92)
	Hesperidin	50 mg/kg and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ BUN, creatinine, urinary albumin levels, ↓ renal tissue oxidative stress, and inflammatory cytokines, enhanced GSH activity	(194)
Salvia miltiorrhiza var. miltiorrhiza	Tanshinone IIA	12.5 and 25 mg/kg body weight	Biochemical and kidney function parameters, anti-inflammatory activity, and histopathological analysis	↓ Serum creatinine level, ↓ renal necrosis, promoted proliferation of tubular epithelial cells, improved renal arterial perfusion, and ameliorated renal inflammation	(281)
	Rosmarinic acid	1, 2, and 5 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine and BUN content, suppressed production of free radicals and pro-inflammatory cytokines, inhibited LPO by reducing MDA level	(75)
	Ferulic acid	50 mg/kg body weight	Kidney function parameters, LPO, and antioxidant activity	↓ BUN and creatinine content, ↓ renal MDA levels, prevented oxidative stress and inflammatory biomarkers in the kidneys	(94)
Scoparia dulcis L.	Apigenin	5, 10, and 20 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity, LPO	↓ BUN and creatinine content, enhanced SOD, CAT, GPx, and GR activity in kidneys, ↓ levels of lipid peroxide and NO, suppressed renal inflammation and apoptosis by decreasing levels of inflammatory biomarkers	(67)
Syzygium cumini (L.) Skeels	Betulinic acid	30 mg/kg body weight	Kidney function parameters	↓ Serum creatinine, BUN, urea, and uric acid levels; ameliorated diabetes-induced renal fibrosis	(282)
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, ↓ renal MDA levels	(73)
Tamarindus indica L.	Taxifolin	50 mg/kg body weight	Kidney function parameters, LPO, antioxidant and anti-inflammatory activity	↓ Serum urea and creatinine levels, enhanced renal SOD, CAT, and GSH levels, ↓ MDA levels, and suppressed production of pro-inflammatory cytokines in the kidneys	(283)
Tanacetum parthenium (L.) Sch.Bip.	Parthenolide	5 and 10 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ Serum creatinine and urea levels, enhanced renal SOD and CAT activity, ↓ MDA levels, and suppressed ROS production	(284)
Terminalia chebula Retz.	Gallic acid	50 and 100 mg/kg body weight	Kidney function parameters, antioxidant and anti-inflammatory activity	↓ Serum creatinine, urea, and uric acid levels, enhanced renal SOD, CAT, GSH, and GPx activity, suppressed production of inflammatory biomarkers, ↓ oxidative stress in the kidneys	(76)
Urtica dioica L.	Naringin	25, 50, and 100 mg/kg body weight	Kidney function parameters, antioxidant activity, LPO	↓ BUN, creatinine, uric acid, NO, and TNF-α, ↓ renal MDA levels, enhanced GSH, GST, GPx, and SOD activity in kidneys	(145)
	Rutin	100, 500, and 1000 mg/kg body weight	Kidney function and biochemical parameters, LPO assay	↓ Blood glucose and urinary glucose excretion, ↓ serum creatinine and urea levels, ↓ renal MDA levels	(73)
Zingiber officinale Roscoe	6-Gingerol	12.5, 25, 50 mg/kg body weight	Kidney function parameters, LPO and antioxidant activity	↓ Serum creatinine, urea, and BUN content, ↑ urea clearance, ↓ renal MDA levels, and enhanced renal SOD, CAT, and GSH activity	(80)
	6-Shogaol	20 mg/kg body weight	Kidney function parameters, antioxidant activity	↓ Serum creatinine and BUN content, ↓ renal MDA levels, enhanced SOD, CAT, and GSH activity	(81)

↓, decrease; ↑, increase.

10.3389/fimmu.2026.1964602

10.3389/fimmu.2026.1964602

TABLE 3 Clinical trial findings of alternative therapies against DN.

Medicinal plants/ formulation	Health condition and no. of subjects	Study design	Dosage regimen	Study duration	Results	Ref.
Abelmoschus manihot alcohol extract	T2DM and DN, 68	Randomized controlled trial	Oral extract + conventional therapy	8 weeks	↓ Urinary microalbumin ↓ Proteinuria ↓ NAG	(299)
Abelmoschus manihot (Huangkui capsule)	T2DM and DN, 239	Multicenter, randomized, open-label, parallel-controlled trial	2.15 g three times a day	24 weeks	↓ Albumin-to-creatinine ratio	(288)
Acanthopanax senticosus	Early DN, 47	Randomized controlled trial	Injection therapy	8 weeks	↓ Urinary albumin excretion ↓ Urinary endothelin	(300)
Astragalus injection (Milkvetch)	Early DN, 69	Randomized controlled trial	Intravenous injection + captopril	3 months	↓ Urinary albumin excretion rate ↓ BUN ↓ Serum creatinine	(301)
Camellia sinensis (L.) Kuntze	T2DM and DN, 39	Double blind, randomized, placebo-controlled trial	400 mg, leaves extract, twice daily	12 weeks	↓ Fasting blood glucose ↓ TGs ↑ Expression of sRAGE ↑ GFR ↑ Renal function	(289)
Compound *Fructus arctii* mixture (*Fructus arctii* & *Astragalus membranaceus*)	DN, 54	Controlled clinical trial	Oral mixture	3 months	↓ 24 h urinary protein ↓ Urinary albumin	(302)
Curcuma longa L.	T2DM and DN, 100	Randomized controlled trial	500 mg, capsule, daily with antidiabetic medication	12 weeks	↓ Serum creatinine ↓ Blood urea ↓ Total urinary protein, urinary albumin, and globulin	(303)
Elevaté® Body Balance (Phytoformula)	T2DM with albuminuria, 20	Controlled clinical trial	1.5 g/day, oral	90 days	↓ HbA1c ↓ Albuminuria	(304)
Ginkgo biloba L.	DN, 60	Controlled trial	20 mL injection, once daily	4 weeks	↓ Urinary albumin excretion ↓ Blood lipids ↑ Creatinine clearance ↑ Renal function	(286)
Ginkgo biloba leaf extract	Early DN, 68	Randomized controlled trial	9.6 mg thrice daily + conventional therapy	3 months	↓ Urinary microalbumin ↓ α1-microglobulin ↓ NAG	(298)
Ginkgo biloba L.	T2DM (early DN), 210	Randomized, double-blind, placebo-controlled trial	24 mg thrice daily	3 years	↓ Albumin-to-creatinine ratio progression	(287)
Hibiscus sabdariffa	DN, 41	Double-blind randomized clinical trial	425 mg tablets twice daily	8 weeks	↓ Systolic blood pressure ↓ Urinary albumin	(305)
Hibiscus sabdariffa L.	T2DM with DN and hypertension, 70	Randomized clinical trial	500 mg extract twice daily, added to valsartan	3 months	Improved kidney function Improved lipid levels	(306)
Liuwei Dihuang pill	Early DN, 72	Randomized controlled trial	Oral pill + conventional therapy	3 months	↓ Urinary albumin excretion rate ↓ β2-microglobulin ↓ Erythrocyte aldose reductase activity	(307)
Modified Shenqi pill	DN, 150	Randomized controlled trial	Oral pills combined with conventional western medicine	3 months	↓ Netrin-1 ↓ 24 h urine protein ↓ BUN ↓ Serum creatinine	(308)
Nigella sativa L.	DN	Randomized controlled trial	2.5 mL oil, orally, once daily	12 weeks	↓ Fasting blood glucose ↓ Serum creatinine ↓ Blood urea ↓ 24 h urinary protein ↑ GFR ↑ Hemoglobin level	(290)

(Continued)

TABLE 3 Continued

Medicinal plants/ formulation	Health condition and no. of subjects	Study design	Dosage regimen	Study duration	Results	Ref.
Paeonia lactiflora Pall (Total glucosides of paeony [TGP])	T2DM and DKD, 76	Open-label, randomized, parallel-group trial	1800 mg/day TGP + losartan	6 months	↓ Urinary albumin excretion rate ↓ High-sensitivity C-reactive protein ↓ MCP-1 ↓ TNF-α	(295)
Panax notoginseng	Early DN, 58	Randomized controlled trial	8 mL intravenous drip daily	Not specified	↓ Urinary albumin ↓ β2-microglobulin ↓ α1-microglobulin	(309)
Piper crocatum Ruiz & Pav.	Early renal dysfunction, 387	Double-blind, randomized, placebo-controlled trial	*Piper crocatum*-enriched functional foods	12 weeks	↓ MMP-9 ↓ NGAL ↓ MMP-9-to-NGAL ratio	(297)
Punica granatum L. (pomegranate juice powder)	DN, 46	Double-blind randomized clinical trial	4 capsules (500 mg each) daily	8 weeks	↓ Urinary albumin ↓ Urinary albumin-to-creatinine ratio	(291)
Qidan Tangshen granule	DKD, 219	Randomized, controlled clinical trial	Oral granule	12 months	↓ HbA1c ↓ Urinary albumin-to-creatinine ratio ↑ GFR ↑ SOD ↓ 8-hydroxy-2′-deoxyguanosine ↓ 3-nitrotyrosine	(294)
Safflower yellow (from *Carthamus tinctorius*)	DN, 76	Randomized controlled trial	150 mg/day intravenous drip	15 days (two courses)	↓ Urinary albumin excretion rate ↓ β2-microglobulin ↓ α1-microglobulin	(310)
Shenqi Dihuang decoction	Early DN, 205	Controlled clinical trial	Oral decoction + conventional therapy	Not specified	↓ Inflammatory factors (IL-6, IL-8, TNF-α, C-reactive protein) ↓ Urine microalbuminuria ↓ β2-microglobulin ↓ cystatin-C	(296)
Tangshen formula	DKD, 180	Double blind, randomized, placebo-controlled trial	8 g granules dissolved in warm water, orally, twice daily	24 weeks	↓ 24 h urinary protein ↑ GFR	(292)
Tongxinluo capsule	DN, 63	Randomized controlled trial	2 capsules thrice daily	8 weeks	↑ Creatinine clearance ↓ Urinary albumin excretion rate ↓ β2-microglobulin ↓ Plasma endothelin-1 ↑ Renal function	(311)
Tripterygium wilfordii Hook.f. L.	T2DM and DN, 65	Randomized, controlled clinical trial	40 mg tablet, three times daily for 12 weeks, then 20 mg tablet, three times daily for 12 weeks	24 weeks	↓ 24 h urinary protein ↓ Serum creatinine	(312)
Tripterygium wilfordii polyglycoside (TWP)	DN, 49	Randomized controlled trial	30–90 mg/day, added to ARB therapy	48 weeks	↓ Proteinuria Preserved estimated GFR/renal function ↓ Risk of ESRD	(313)
Zicuiyin decoction	DKD, 88	Randomized controlled trial	Crude drug amount 75 g, 150 mL, twice daily	8 weeks	↓ 24 h urinary protein ↓ Urinary albumin excretion ↓ Serum creatinine ↓ HbA1c ↓ Urinary albumin-creatinine ratio ↑ GFR	(293)

10 Renoprotective mechanisms of medicinal plants

While bioactive phytochemicals from medicinal plants offer new therapeutic approaches, their pharmacological effects and potential clinical risks remain incompletely understood (17). Before implementing these botanical therapies in clinical settings, it is essential to carefully consider potential toxicity, dose consistency, extract standardization, and possible herb–drug interactions. Medicinal plants are reported to exert renoprotective effects through various mechanisms, including reducing oxidative stress, inhibiting inflammation, preventing the formation of AGEs, and

attenuating cell death. These actions are believed to contribute to protection from kidney damage, particularly in conditions like DN (Figure 3).

One of the primary factors influencing the onset and progression of DN is oxidative stress (314). The glomerular and renal tubular cells exhibit structural and functional changes due to oxidative stress–induced renal damage in DN. Renal cells are often protected from ROS by a self-defense mechanism involving renal antioxidants. In hyperglycemic conditions, ROS production can exceed the capacity of the kidney's antioxidant defenses, leading to oxidative stress (315, 316). This imbalance between ROS and antioxidants disrupts cellular redox signaling and negatively impacts cellular metabolism (317). Moreover, it activates several pathways involved in DN development. As a result, managing oxidative stress may be an effective strategy for preserving kidney function and delaying or preventing the progression of DN (318).

Medicinal plants, acting as natural antioxidants, play a vital role in this process. They prevent the formation of ROS, neutralize and eliminate existing ROS, enhance the activity of antioxidant enzymes, and stimulate the production of proteins involved in antioxidant pathways, thereby helping to reduce oxidative stress linked to DN (319). Excessive ROS production in pancreatic β-cells can lead to the activation of stress signaling pathways, resulting in the stimulation of transcription factors like NF-κB, which are associated with inflammation and apoptosis. This process ultimately results in β-cell death and reduced insulin production. Moreover, evidence indicates that increased ROS production triggers NF-κB activation in other kidney cells, including renal tubular, mesangial, and glomerular endothelial cells, as well as dendritic cells (320–322). The activation of NF-κB promotes the transcription of proinflammatory genes encoding chemokines and cytokines. In addition, this response is implicated in diabetes-related atherosclerosis, vascular dysfunction, and the production of profibrotic and proinflammatory mediators. Thus, suppressing NF-κB–mediated inflammation may represent a good therapeutic approach for DN (323).

Building on their ability to modulate major signaling cascades such as the NLRP3 inflammasome, emerging evidence indicates that medicinal plants and their bioactive compounds extend beyond general anti-inflammatory effects by directly targeting key upstream components of the innate immune system. Certain phytocons tituents specifically interfere with the TLR4/MyD88 signaling pathway and prevent NLRP3 inflammasome formation, effectively stopping the inflammatory process at its origin. For example, bixin has been found to reduce renal inflammation and fibrosis by directly modulating the TLR4/MyD88 pathway (125). Likewise, triptolide defends against diabetic podocyte injury by directly inhibiting the NLRP3 inflammasome pathway (268). By suppressing these specific innate immune sensors, medicinal plants efficiently block the maturation of IL-1β and IL-18, thereby decreasing macrophage-driven renal damage and providing precise innate immune regulation in DN.

Experimental studies suggest that extracts from certain medicinal plants may reduce inflammatory mediators in kidney cells, thereby potentially improving conditions associated with DN (Figure 3). Accumulation of AGEs at the site of microvascular injury is also considered a critical step in the pathogenesis of DN (324). The cytotoxic effects of AGEs on renal cells are enhanced by their ability to modify lipids and proteins, leading to oxidative stress and inflammation (325). RAGEs are expressed on many cell types in

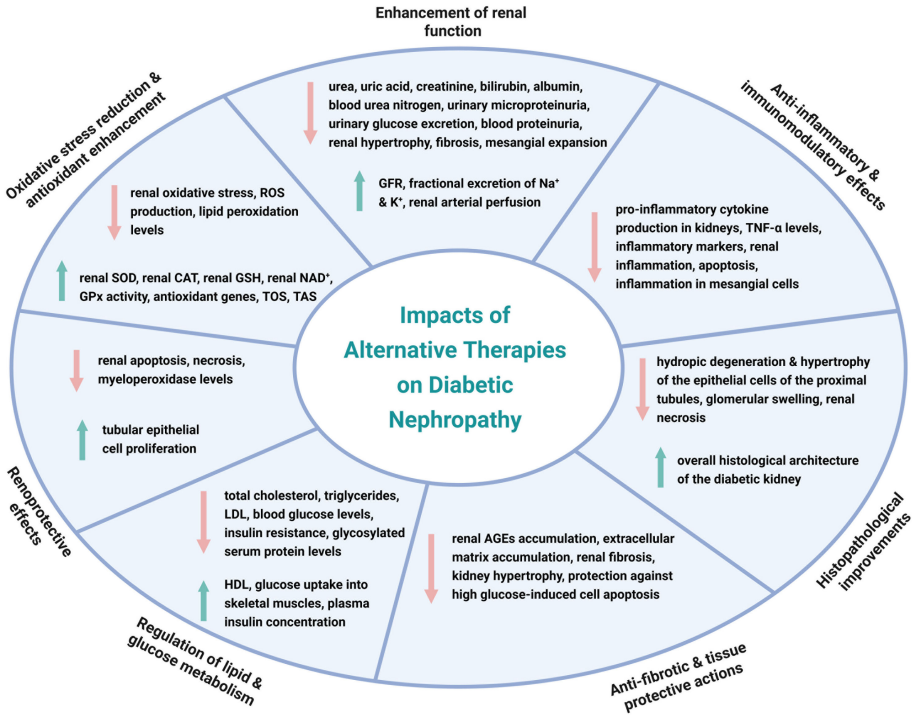


FIGURE 3
Mechanistic pathways underlying the therapeutic actions of various medicinal plants in the management of DN.

the kidney, including podocytes, mesangial cells, and proximal tubular epithelial cells, allowing AGEs to activate inflammatory pathways through the expression of various inflammatory mediators (326). Therefore, the use of medicinal plants, due to their anti-inflammatory and antioxidant properties, may inhibit AGE formation and may subsequently assist in managing DN by reducing AGE accumulation (Figure 3). Furthermore, DN is fundamentally driven by ECM remodeling, leading to mesangial expansion and tubulointerstitial fibrosis (327). Emerging evidence suggests that combining phytochemicals might yield unique, non-additive effects on ECM turnover by simultaneously targeting complementary nodes within profibrotic signaling pathways. For example, in *Salvia miltiorrhiza*, a species known for its renoprotective effects, combining salvianolic acid B with tanshinone IIA has provided greater renoprotection than either compound alone by synergistically modulating TGF-β1/Smads, PI3K/Akt/NF-κB, and MMP pathways (328). Likewise, certain alkaloid-flavonoid combinations have demonstrated strong anti-fibrotic synergy via these same mechanisms (329).

Within the molecular pathways underlying DN, NF-κB and NLRP3 have emerged as the most prominent targets of multiple phytochemicals, linking hyperglycemia-induced oxidative and metabolic stress to innate inflammatory signaling. For instance, andrographolide acts through the Akt/NF-κB pathway to suppress renal oxidative stress (100), whereas triptolide inhibits NLRP3 inflammasome activation, thereby enhancing Nrf2/HO-1 signaling (274). Similarly, other phytoconstituents, including galactomannan and mangiferin, modulate NF-κB–associated inflammatory and oxidative stress pathways (128, 273), while the TGF-β1/Wnt/β-catenin axis is a key pathway involved in renal fibrogenesis (228). Altogether, these findings indicate the potential of multi-target phytochemicals to simultaneously address metabolic stress, inflammation, oxidative defense, and fibrosis. Nonetheless, further validation is required to establish the precise contribution of these interconnected nodes.

11 Future strategies to prevent DN

Increasing the availability of affordable treatment options for the avoidance and management of kidney disease risk factors, as well as raising public awareness of the significance of the detection and treatment of risk factors of DN, are crucial steps toward minimizing the burden of DN. Personalized screening procedures, including blood pressure monitoring, urine analysis, and plasma glucose assessment, should be part of these approaches. By identifying those who are at risk of developing DN and diagnosing the condition early, such screening will enable the implementation of relevant and successful therapies (330). In both developing and developed nations worldwide, patients who have a higher chance of developing CKD due to DN are often referred to nephrologists at a later stage in life (331). Early referral to a nephrologist can help prevent the progression of CKD by enabling proactive steps such as developing vascular access for hemodialysis and optimizing metabolic, nutritional, endocrine, hematologic, and hemodynamic functions (332). Preventing DN can be achieved through education and self-management, which promote adherence to treatment and lifestyle modifications to regulate blood circulation and glucose levels (333). Effective prevention requires collaborative efforts from healthcare professionals, government agencies, and pharmaceutical companies. The outlook for the prevention and control of DN is optimistic despite the existing challenges because of the rapid development of novel treatment approaches, combined management strategies, individualized care, and patient education.

Natural remedies may assist in the management of DN by potentially reducing pro-inflammatory cytokines through the inhibition of innate immune responses and pathways, oxidative stress, and kidney injury mechanisms. Medicinal plants have shown therapeutic potential in the treatment of DN due to the presence of diverse bioactive compounds with various therapeutic effects, as well as their association with enhanced kidney function, reduced protein levels in the urine, and slowed disease progression (322). However, more research is needed to confirm their efficacy in the prevention and mitigation of the illness. The antioxidant properties of medicinal plants are believed to help protect the kidneys from damage by counteracting oxidative stress and promoting the body's natural processes. Further research should be performed to identify and harness these components and their mechanisms in the treatment of DN, as well as their ability to support conventional drugs (334).

Translating the renoprotective mechanisms of medicinal plants from preclinical models into clinical efficacy reveals a substantial translational bottleneck. For a select few plants, preclinical and clinical studies have assessed similar parameters and reported consistently positive outcomes. For example, *Curcuma longa*, *Nigella sativa*, and *Tripterygium wilfordii* have each been associated with reductions in serum creatinine, blood urea, and urinary protein or albumin in both experimental animal models and human randomized controlled trials. Similarly, *Punica granatum* has been reported to reduce urinary albumin levels in both *in vivo* models and clinical evaluations. However, the parameters assessed often differ substantially: animal studies primarily measure localized tissue-level markers (e.g., SOD, CAT, GSH, MDA) and histopathological improvements, while clinical trials of plants such as *Camellia sinensis* and *Ginkgo biloba* rely on systemic surrogate markers like eGFR, urinary albumin-to-creatinine ratio, and α1-microglobulin. As a result, it remains unclear whether the substantial structural and antioxidant benefits observed in animal kidneys translate directly to human tissues.

Although dozens of species show promising preliminary effects, most remain at the preclinical stage, and very few medicinal plants have advanced to human clinical trials. This lack of progress highlights a key limitation in phytotherapy research: insufficient evidence to determine whether the renoprotective parameters and pharmacological doses established in animal models are viable, safe, or effective for people with DN.

To ensure safe integration of these therapies, future research should shift from broad observational profiling of medicinal plants to frameworks that analyze individual constituents. These

frameworks should prospectively chart herb–drug interactions within pharmacokinetic and pharmacodynamic areas (335). Such frameworks should outline the effects of phytochemicals on CYP enzymes, transporters, renal elimination, and phase II metabolic pathways. They should also integrate these data with common pharmacodynamic targets relevant to DN, such as RAAS signaling, natriuresis, and glycemic regulation (336). This interaction mapping is especially important for patients on polypharmacy involving RAAS inhibitors, SGLT2 inhibitors, metformin, and sulfonylureas, as the risks of interactions vary significantly by drug class and kidney function (337). To accurately identify true phytochemical synergy and distinguish it from mere additivity or antagonism, future preclinical research should employ multi-arm, fixed-ratio experimental designs that evaluate compound combinations against direct ECM endpoints such as collagen, fibronectin, and α-SMA, rather than relying solely on overall indices such as serum creatinine (338). By integrating *in vitro* assays, model-based pharmacokinetic predictions, and real-world pharmacovigilance, researchers can better distinguish theoretical interactions from actual clinical risks, potentially facilitating the safe use of natural remedies in preventing and managing DN.

Although current evidence suggests potential renoprotective effects of phytoconstituents such as curcumin, quercetin, mangiferin, oleuropein, and triptolide, patient-specific multi-omics frameworks remain largely unexplored for delineating the specific efficacy of these phytochemicals in DN. Proteomic and metabolomic profiling in human DKD cohorts have shown the potential to distinguish DKD stages, facilitating molecular stratification (339). Consequently, clinical trials involving curcumin have indicated possible therapeutic benefits of the phytoconstituent in DN (340). In contrast, multi-omics studies involving compounds such as mangiferin remain in their preclinical stage, limited to experimental DN models (341). Thus, the integration of patient-derived multi-omics studies may offer valuable insights into precision phytotherapy in DN, potentially guiding the targeted selection of individual phytoconstituents based on specific molecular signatures.

12 Conclusion

DN continues to be a significant complication of diabetes, with increasing prevalence and a substantial number of cases progressing to ESRD worldwide. While contemporary pharmacotherapies provide essential disease-modifying benefits, the clinical complexity of DN leaves room for adjunctive and complementary strategies. Medicinal plants and their isolated phytochemicals may offer a promising complementary approach, as preclinical studies indicate their potential to support and enhance renal function. However, existing evidence does not conclusively prove that these botanical interventions are fundamentally safer or more cost-efficient than standard treatments. Before medicinal plants can be safely translated into clinical practice, extensive research is required to address essential pharmacological variables, including safety profiling, extract standardization, dose variability, and overall bioavailability. Furthermore, potential toxicities and herb–drug interactions with existing DN regimens must be thoroughly assessed. Ultimately, while these natural alternatives have promising therapeutic potential, extensive clinical trials are necessary to confirm their effectiveness and safety as complementary options for enhancing patient outcomes and reducing disease burden.

Author contributions

SA: Data curation, Formal analysis, Methodology, Writing – original draft. FA: Data curation, Formal analysis, Visualization, Writing – review and editing. TRS: Data curation, Formal analysis, Visualization, Writing – original draft. SAA: Formal analysis, Visualization, Writing – review and editing. MAA: Formal analysis, Writing – review and editing. MSA: Formal analysis, Visualization, Writing – review and editing. KUB: Formal analysis, Visualization, Writing – review and editing. MJU: Conceptualization, Supervision, Project administration, Formal analysis, Writing – review and editing.

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Supplementary material

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References

1. Sun H, Saeedi P, Karuranga S, Pinkepank M, Ogurtsova K, Duncan BB, et al. IDF Diabetes Atlas: global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res Clin Pract*. (2022) 183:109119. doi: 10.1016/j.diabres.2021.109119

2. Ansari P, Akther S, Khan JT, Islam SS, Masud M, Rahman A, et al. Hyperglycaemia-linked diabetic foot complications and their management using conventional and alternative therapies. *Appl Sci*. (2022) 12:11777. doi: 10.3390/app122211777

3. Katsarou A, Gudbjörnsdottir S, Rawshani A, Dabelea D, Bonifacio E, Anderson BJ, et al. Type 1 diabetes mellitus. *Nat Rev Dis Primers*. (2017) 3:17016. doi: 10.1038/nrdp.2017.16

4. Martínez-Castelao A, Navarro-González J, Górriz J, De Alvaro F. The concept and the epidemiology of diabetic nephropathy have changed in recent years. *J Clin Med*. (2015) 4:1207–16. doi: 10.3390/jcm4061207

5. Van Buren PN, Toto R. Hypertension in diabetic nephropathy: epidemiology, mechanisms, and management. *Adv Chronic Kidney Dis*. (2011) 18:28–41. doi: 10.1053/j.ackd.2010.10.003

6. Schwartz M. Renal pathology patterns in type II diabetes mellitus: relationship with retinopathy. *Nephrol Dial Transplant*. (1998) 13:2547–52. doi: 10.1093/ndt/13.10.2547

7. ElSayed NA, McCoy RG, Aleppo G, Balapattabi K, Beverly EA, Briggs Early K, et al. Chronic kidney disease and risk management: standards of care in diabetes—2025. *Diabetes Care*. (2025) 48:S239–51. doi: 10.2337/dc25-S011

8. De Boer IH, Caramori ML, Chan JCN, Heerspink HJL, Khunti K, Liew A, et al. GLP-1 receptor agonists and other incretin mimetics for diabetes and chronic kidney disease—a KDIGO commentary. *Kidney Int*. (2025) 107:767–71. doi: 10.1016/j.kint.2024.12.019

9. Awdishu L, Maxson R, Gratt C, Rubenzik T, Battistella M. KDIGO 2024 clinical practice guideline on evaluation and management of chronic kidney disease: a primer on what pharmacists need to know. *Am J Health Syst Pharm*. (2025) 82:660–71. doi: 10.1093/ajhp/zxaf044

10. Zhang H, Chen J, Wang N, Chen W, Xu Y, Gui D. Advancements in pharmacotherapy for diabetic kidney disease: a systematic review. *Diabetes Metab Res Rev*. (2025) 41:e70077. doi: 10.1002/dmrr.70077

11. Ahmed M, Kishore G, Khader H, Kasturirangan M. Risk factors and management of diabetic nephropathy. *Saudi J Kidney Dis Transpl*. (2013) 24:1242. doi: 10.4103/1319-2442.121310

12. Abhirami BL, Krishna AA, Kumaran A, Chiu CH. Targeting NF-κB in diabetic nephropathy: exploring the therapeutic potential of phytoconstituents. *Arch Pharmacal Res*. (2025) 48:577–637. doi: 10.1007/s12272-025-01555-z

13. Efiong EE, Effa E, Peters E, Erukainure OL, Amadi PU, Ikebuiro JO, et al. Multi-target modulation of polyphenols in diabetic kidney disease therapeutics: a comprehensive review. *Food Sci Nutr*. (2025) 13:e70882. doi: 10.1002/fsn3.70882

14. Zeng Q, Wang W, Li Y, Wang M, Liu Z, Yan L, et al. Beneficial effects of food-medicine homologous herbs for patients with diabetic kidney disease. *Front Immunol*. (2025) 16:1698240. doi: 10.3389/fimmu.2025.1698240

15. Kim HK, Kim SJ, Gil WJ, Yang CS. Exploring the therapeutic potential of phytochemicals: challenges and strategies for clinical translation. *Phytomedicine*. (2025) 145:157090. doi: 10.1016/j.phymed.2025.157090

16. Nazar CMJ. Mechanism of hypertension in diabetic nephropathy. *J Nephropharmacol*. (2014) 3:49–55.

17. Tziomalos K, Athyros VG. Diabetic nephropathy: new risk factors and improvements in diagnosis. *Rev Diabetes Stud*. (2015) 12:110–8. doi: 10.1900/RDS.2015.12.110

18. Kawanami D, Matoba K, Utsunomiya K. Dyslipidemia in diabetic nephropathy. *Ren Replace Ther*. (2016) 2:16. doi: 10.1186/s41100-016-0028-0

19. Ayodele OE, Alebiosu CO, Salako BL. Diabetic nephropathy: a review of the natural history, burden, risk factors and treatment. *J Natl Med Assoc*. (2004) 96:1445–54.

20. Epstein M, Sowers JR. Diabetes mellitus and hypertension. *Hypertension*. (1992) 19:403–18. doi: 10.1161/01.HYP.19.5.403

21. Rask-Madsen C, King GL. Kidney complications: factors that protect the diabetic vasculature. *Nat Med*. (2010) 16:40–1. doi: 10.1038/nm0110-40

22. Rutledge JC, Ng KF, Aung HH, Wilson DW. Role of triglyceride-rich lipoproteins in diabetic nephropathy. *Nat Rev Nephrol*. (2010) 6:361–70. doi: 10.1038/nrneph.2010.59

23. Chen HM, Shen WW, Ge YC, Zhang YD, Xie HL, Liu ZH. The relationship between obesity and diabetic nephropathy in China. *BMC Nephrol*. (2013) 14:69. doi: 10.1186/1471-2369-14-69

24. Su S, Wang W, Sun T, Ma F, Wang Y, Li J, et al. Smoking as a risk factor for diabetic nephropathy: a meta-analysis. *Int Urol Nephrol*. (2017) 49:1801–7. doi: 10.1007/s11255-017-1638-3

25. Jeng CJ, Hsieh YT, Yang CM, Yang CH, Lin CL, Wang IJ. Diabetic retinopathy in patients with diabetic nephropathy: development and progression. *PloS One*. (2016) 11:e0161897. doi: 10.1371/journal.pone.0161897

26. Navarro-González JF, Mora-Fernández C. The role of inflammatory cytokines in diabetic nephropathy. *J Am Soc Nephrol*. (2008) 19:433–42. doi: 10.1681/ASN.2007091048

27. Thrailkill KM, Clay Bunn R, Fowlkes JL. Matrix metalloproteinases: their potential role in the pathogenesis of diabetic nephropathy. *Endocrine*. (2009) 35:1–10. doi: 10.1007/s12020-008-9114-6

28. Badal SS, Danesh FR. New insights into molecular mechanisms of diabetic kidney disease. *Am J Kidney Dis*. (2014) 63:S63–83. doi: 10.1053/j.ajkd.2013.10.047

29. Herman-Edelstein M, Doi SQ. "Pathophysiology of diabetic nephropathy". In: Blaine J, editor. *Proteinuria: basic mechanisms, pathophysiology and clinical relevance*. Cham: Springer International Publishing (2016) p. 41–65. doi: 10.1007/978-3-319-43359-2_4

30. Tuttle KR. Back to the future: glomerular hyperfiltration and the diabetic kidney. *Diabetes*. (2017) 66:14–6. doi: 10.2337/dbi16-0056

31. Ruggenenti P, Cravedi P, Remuzzi G. The RAAS in the pathogenesis and treatment of diabetic nephropathy. *Nat Rev Nephrol*. (2010) 6:319–30. doi: 10.1038/nrneph.2010.58

32. Flier JS, Underhill LH, Brownlee M, Cerami A, Vlassara H. Advanced glycosylation end products in tissue and the biochemical basis of diabetic complications. *N Engl J Med*. (1988) 318:1315–21. doi: 10.1056/NEJM198805193182007

33. Abel M, Ritthaler U, Zhang Y, Deng Y, Schmidt AM, Greten J, et al. Expression of receptors for advanced glycosylated end-products in renal disease. *Nephrol Dial Transplant*. (1995) 10:1662–7. doi: 10.1093/ndt/10.9.1662

34. Tanji N, Markowitz GS, Fu C, Kislinger T, Taguchi A, Pischetsrieder M, et al. Expression of advanced glycation end products and their cellular receptor RAGE in diabetic nephropathy and nondiabetic renal disease. *J Am Soc Nephrol*. (2000) 11:1656–66. doi: 10.1681/ASN.V1191656

35. Rodríguez-Iturbe B, Pons H, Herrera-Acosta J, Johnson RJ. Role of immunocompetent cells in nonimmune renal diseases. *Kidney Int*. (2001) 59:1626–40. doi: 10.1046/j.1523-1755.2001.0590051626.x

36. Nishikawa T, Edelstein D, Du XL, Yamagishi S, Matsumura T, Kaneda Y, et al. Normalizing mitochondrial superoxide production blocks three pathways of hyperglycaemic damage. *Nature*. (2000) 404:787–90. doi: 10.1038/35008121

37. Ha H, Hwang I-A, Park JH, Lee HB. Role of reactive oxygen species in the pathogenesis of diabetic nephropathy. *Diabetes Res Clin Pract*. (2008) 82:S42–5. doi: 10.1016/j.diabres.2008.09.017

38. Wang N, Zhang C. Oxidative stress: a culprit in the progression of diabetic kidney disease. *Antioxidants*. (2024) 13:455. doi: 10.3390/antiox13040455

39. Tuttle KR. Linking metabolism and immunology: diabetic nephropathy is an inflammatory disease. *J Am Soc Nephrol*. (2005) 16:1537–8. doi: 10.1681/ASN.2005040393

40. Jin J, Zhang M. Exploring the role of NLRP3 inflammasome in diabetic nephropathy and the advancements in herbal therapeutics. *Front Endocrinol*. (2024) 15:1397301. doi: 10.3389/fendo.2024.1397301

41. Chen Y, Chen R, Ji X, Zeng Z, Guan C. NLRP3 inflammasome-mediated pyroptosis in diabetic nephropathy: pathogenic mechanisms and therapeutic targets. *J Inflammation Res*. (2025) 18:8399–418. doi: 10.2147/JIR.S524246

42. Imai E, Horio M, Watanabe T, Iseki K, Yamagata K, Hara S, et al. Prevalence of chronic kidney disease in the Japanese general population. *Clin Exp Nephrol*. (2009) 13:621–30. doi: 10.1007/s10157-009-0199-x

43. Yamagata K, Ishida K, Sairenchi T, Takahashi H, Ohba S, Shiigai T, et al. Risk factors for chronic kidney disease in a community-based population: a 10-year follow-up study. *Kidney Int*. (2007) 71:159–66. doi: 10.1038/sj.ki.5002017

44. Toyama T, Kitagawa K, Oshima M, Kitajima S, Hara A, Iwata Y, et al. Age differences in the relationships between risk factors and loss of kidney function: a general population cohort study. *BMC Nephrol*. (2020) 21:477. doi: 10.1186/s12882-020-02121-z

45. Perazella MA. Drug-induced nephropathy: an update. *Expert Opin Drug Saf*. (2005) 4:689–706. doi: 10.1517/14740338.4.4.689

46. Alsaad KO, Herzenberg AM. Distinguishing diabetic nephropathy from other causes of glomerulosclerosis: an update. *J Clin Pathol*. (2007) 60:18–26. doi: 10.1136/jcp.2005.035592

47. Ramesh P, Okigbo R. Effects of plants and medicinal plant combinations as anti-infectives. *Afr J Pharm Pharmacol*. (2008) 2:130–5. doi: 10.5897/AJPP.9000130

48. Shirazian S, Radhakrishnan J. Gastrointestinal disorders and renal failure: exploring the connection. *Nat Rev Nephrol*. (2010) 6:480–92. doi: 10.1038/nrneph.2010.84

49. Wagner H. Synergy research: approaching a new generation of phytopharmaceuticals. *Fitoterapia*. (2011) 82:34–7. doi: 10.1016/j.fitote.2010.11.016

50. Goyal S, Gupta N, Chatterjee S, Nimesh S. Natural plant extracts as potential therapeutic agents for the treatment of cancer. *Curr Top Med Chem*. (2017) 17:96–106. doi: 10.2174/1568026616666160530154407

51. Ahmed BT, Kumar SA. Antioxidant and antidiabetic properties of *Abelmoschus esculentus* extract–an *in vitro* assay. *Res J Biotechnol*. (2016) 11:34–41.

52. Wahyuningsih SPA, Savira NII, Anggraini DW, Winarni D, Suhargo L, Kusuma BWA, et al. Antioxidant and nephroprotective effects of okra pods extract (*Abelmoschus esculentus* L.) against lead acetate-induced toxicity in mice. *Scientifica (Cairo)*. (2020) 2020:1–10. doi: 10.1155/2020/4237205

53. Deepa B, Anuradha CV. Linalool, a plant derived monoterpene alcohol, rescues kidney from diabetes-induced nephropathic changes via blood glucose reduction. *Diabetol Croat*. (2011) 40:121–38.

54. Barhoma RAE. The role of eugenol in the prevention of chromium-induced acute kidney injury in male albino rats. *Alex J Med*. (2018) 54:711–5. doi: 10.1016/j.ajme.2018.05.006

55. Umamahesh B, Veeresham C. Antihyperglycemic and insulin secretagogue activities of *Abrus precatorius* leaf extract. *Pharmacognosy Res*. (2016) 8:303. doi: 10.4103/0974-8490.188881

56. Reddy SK, Begum A. Evaluation of hepato-nephro protective activity of ethanolic extract of *Abrus precatorius* Linn. root against paracetamol induced toxicity in male Sprague Dawley rats. *World J Pharm Res*. (2015) 4:2500–13.

57. Xiong C, Wu Q, Fang M, Li H, Chen B, Chi T. Protective effects of luteolin on nephrotoxicity induced by long-term hyperglycaemia in rats. *J Int Med Res*. (2020) 48:300060520903642. doi: 10.1177/0300060520903642

58. Ansari P, Flatt PR, Harriott P, Hannan JMA, Abdel-Wahab YHA. Identification of multiple pancreatic and extra-pancreatic pathways underlying the glucose-lowering actions of *Acacia arabica* bark in type 2 diabetes and isolation of active phytoconstituents. *Plants (Basel)*. (2021) 10:1190. doi: 10.3390/plants10061190

59. Anjaneyulu M, Chopra K. Quercetin, an anti-oxidant bioflavonoid, attenuates diabetic nephropathy in rats. *Clin Exp Pharmacol Physiol*. (2004) 31:244–8. doi: 10.1111/j.1440-1681.2004.03982.x

60. Sharma D, Gondaliya P, Tiwari V, Kalia K. Kaempferol attenuates diabetic nephropathy by inhibiting RhoA/Rho-kinase mediated inflammatory signalling. *BioMed Pharmacother*. (2019) 109:1610–9. doi: 10.1016/j.biopha.2018.10.195

61. Lakshmi T, Ramasamy R, Thirumalaikumaran R. Preliminary Phytochemical analysis and *in vitro* antioxidant, FTIR spectroscopy, anti-diabetic activity of *Acacia catechu* ethanolic seed extract. *Pharmacogn J*. (2015) 7:356–62. doi: 10.5530/pj.2015.6.7

62. D'souza P, Holla R, Swamy G. Amelioration of diabetic nephropathy in streptozotocin-induced diabetic rats by *Acacia catechu* leaves extract. *J Health Allied Sci NU*. (2019) 9:116–20. doi: 10.1055/s-0039-3402084

63. Kandemir F, Kucukler S, Eldutar E, Caglayan C, Gülçin I. Chrysin protects rat kidney from paracetamol-induced oxidative stress, inflammation, apoptosis, and autophagy: a multi-biomarker approach. *Sci Pharm*. (2017) 85:4. doi: 10.3390/scipharm85010004

64. Hassanein EHM, Mohamed WR, Shalkami A-G, Khalaf MM, Hemeida RAM. Renoprotective effects of umbelliferone on methotrexate-induced renal injury through regulation of Nrf-2/Keap-1, P38MAPK/NF-κB, and apoptosis signaling pathways. *Food Chem Toxicol*. (2018) 116:152–60. doi: 10.1016/j.fct.2018.03.041

65. Rajappa R, B. Magesh S, Sarvajayakesavulu S, Rao V, Madhunapantula S. Nephroprotective effect of naringenin against multiple low dose streptozotocin (MLDSTZ) induced renal damage in mice. *BioMed Pharmacol J*. (2017) 10:583–93. doi: 10.13005/bpj/1145

66. Zangeneh MM. Hematoprotective and nephroprotective effects of *Achillea millefolium* aqueous extract in diabetic mice. *Res J Pharmacogn*. (2018) 5:57–68. doi: 10.22127/rjp.2018.64878

67. Hussein MM, Althagafi HA, Alharthi F, Albrakati A, Alsharif KF, Theyab A, et al. Apigenin attenuates molecular, biochemical, and histopathological changes associated with renal impairments induced by gentamicin exposure in rats. *Environ Sci pollut Res Int*. (2022) 29:65276–88. doi: 10.1007/s11356-022-20235-9

68. Priyamvada PM, Sha A, Mohapatra AK. Evaluation of antidiabetic and antioxidant activities of *Achyranthes aspera* leaf extracts: an *in vitro* study. *J Pharmacogn Phytochem*. (2021) 10:103–10.

69. Jayakumar T, Sridhar MP, Bharathprasad TR, Ilayaraja M, Govindasamy S, Balasubramanian MP. Experimental studies of *Achyranthes aspera* (L) preventing nephrotoxicity induced by lead in albino rats. *J Health Sci*. (2009) 55:701–8. doi: 10.1248/jhs.55.701

70. Patil CR, Jadhav RB, Singh PK, Mundada S, Patil PR. Protective effect of oleanolic acid on gentamicin induced nephrotoxicity in rats. *Phytother Res*. (2010) 24:33–7. doi: 10.1002/ptr.2861

71. Kesari AN, Gupta RK, Singh SK, Diwakar S, Watal G. Hypoglycemic and antihyperglycemic activity of *Aegle marmelos* seed extract in normal and diabetic rats. *J Ethnopharmacol*. (2006) 107:374–9. doi: 10.1016/j.jep.2006.03.042

72. Dwivedi J, Singh M, Sharma S, Sharma S. Antioxidant and nephroprotective potential of *Aegle marmelos* leaves extract. *J Herbs Spices Med Plants*. (2017) 23:363–77. doi: 10.1080/10496475.2017.1345029

73. Sadeghnia HR, Yousefsani BS, Rashidfar M, Boroushaki MT, Asadpour E, Ghorbani A. Protective effect of rutin on hexachlorobutadiene-induced nephrotoxicity. *Ren Fail*. (2013) 35:1151–5. doi: 10.3109/0886022X.2013.815546

74. Movaliya V, Khamar D, Setty M. Nephroprotective activity of aqueous extract of *Aerva javanica* roots in cisplatin induced renal toxicity in rats. *Pharmacologyonline*. (2011) 1:68–74.

75. Domitrović R, Potočnjak I, Crnčević-Orlić Ž, Škoda M. Nephroprotective activities of rosmarinic acid against cisplatin-induced kidney injury in mice. *Food Chem Toxicol*. (2014) 66:321–8. doi: 10.1016/j.fct.2014.02.002

76. Moradi A, Abolfathi M, Javadian M, Heidarian E, Roshanmehr H, Khaledi M, et al. Gallic acid exerts nephroprotective, anti-oxidative stress, and anti-inflammatory effects against diclofenac-induced renal injury in male rats. *Arch Med Res*. (2021) 52:380–8. doi: 10.1016/j.arcmed.2020.12.005

77. Singh B, Kumar A, Singh H, Kaur S, Arora S, Singh B. Protective effect of vanillic acid against diabetes and diabetic nephropathy by attenuating oxidative stress and upregulation of NF-κB, TNF-α and COX-2 proteins in rats. *Phytother Res*. (2022) 36:1338–52. doi: 10.1002/ptr.7392

78. Mohammed A, Koorbanally NA, Islam MS. Phytochemistry, antioxidative activity and inhibition of key enzymes linked to type 2 diabetes by various parts of *Aframomum melegueta in vitro*. *Acta Pol Pharm*. (2016) 73:403–17.

79. Abdou RM, El-Maadawy WH, Hassan M, El-Dine RS, Aboushousha T, El-Tanbouly ND, et al. Nephroprotective activity of *Aframomum melegueta* seeds extract against diclofenac-induced acute kidney injury: a mechanistic study. *J Ethnopharmacol*. (2021) 273:113939. doi: 10.1016/j.jep.2021.113939

80. Kuhad A, Tirkey N, Pilkhwal S, Chopra K. 6-Gingerol prevents cisplatin-induced acute renal failure in rats. *Biofactors*. (2006) 26:189–200. doi: 10.1002/biof.5520260304

81. Gwon MG, Gu H, Leem J, Park KK. Protective effects of 6-shogaol, an active compound of ginger, in a murine model of cisplatin-induced acute kidney injury. *Molecules*. (2021) 26:5931. doi: 10.3390/molecules26195931

82. Nyunaï N, Njikam N, Abdenneb E, Mbafor J, Lamnaouer D. Hypoglycaemic and antihyperglycaemic activity of *Ageratum conyzoides* L. in rats. *Afr J Tradit Complement Altern Med*. (2010) 6:123–30. doi: 10.4314/ajtcam.v6i2.57083

83. Ola-Davies O, Akinrinde A. Acute sodium arsenite-induced hematological and biochemical changes in wistar rats: protective effects of ethanol extract of *Ageratum conyzoides*. *Pharmacognosy Res*. (2016) 8:26. doi: 10.4103/0974-8490.178645

84. Ozougwu JC. Anti-diabetic effects of *Allium cepa* (onions) aqueous extracts on alloxan-induced diabetic *Rattus novergicus*. *J Med Plants Res*. (2011) 5:1134–9.

85. Elsani MM, Chinnala KM, Achanta P, Vangala VL. Evaluation for nephroprotective activity of ethanolic extract of *Allium cepa* Linn. in gentamicin-induced nephrotoxicity in rats. *Asian J Pharm Clin Res*. (2017) 10:356. doi: 10.22159/ajpcr.2017.v10i3.16271

86. El-Kashef D, El-Kenawi A, Suddek G, Salem H. Allicin ameliorates kidney function and urinary bladder sensitivity in cyclosporine A-treated rats. *Hum Exp Toxicol*. (2017) 36:681–91. doi: 10.1177/0960327116660864

87. Ozougwu J, Eyo J. Studies on the anti-diabetic activity of *Allium sativum* (garlic) aqueous extracts on alloxan-induced diabetic albino rat. *Pharmacologyonline*. (2010) 2:1079–88.

88. Anusuya N, Durgadevi P, Dhinek A, Mythily S. Nephroprotective effect of ethanolic extract of garlic (*Allium sativum* L.) on cisplatin induced nephrotoxicity in male wistar rats. *Asian J Pharm Clin Res*. (2013) 6:97–100.

89. Pedraza-Chaverrí J, Maldonado PD, Barrera D, Cerón A, Medina-Campos ON, Hernández Pando R. Protective effect of diallyl sulfide on oxidative stress and nephrotoxicity induced by gentamicin in rats. *Mol Cell Biochem*. (2003) 254:125–30. doi: 10.1023/A:1027372102135

90. Muñiz-Ramirez A, Perez RM, Garcia E, Garcia FE. Antidiabetic activity of *Aloe vera* leaves. *Evid Based Complement Alternat Med*. (2020) 2020:6371201. doi: 10.1155/2020/6371201

91. Iftkhar A, Hasan IJ, Sarfraz M, Jafri L, Ashraf MA. Nephroprotective effect of the leaves of *Aloe barbadensis* (*Aloe vera*) against toxicity induced by diclofenac sodium in albino rabbits. *West Indian Med J*. (2015) 64:462–7. doi: 10.7727/wimj.2016.052

92. Ali BH, Al-Salam S, Al Husseini IS, Al-Lawati I, Waly M, Yasin J, et al. Abrogation of cisplatin-induced nephrotoxicity by emodin in rats. *Fundam Clin Pharmacol*. (2013) 27:192–200. doi: 10.1111/j.1472-8206.2011.01003.x

93. Clemente A, Desai P. Evaluation of the hematological, hypoglycemic, hypolipidemic and antioxidant properties of *Amaranthus tricolor* leaf extract in rat. *Trop J Pharm Res*. (2011) 10:595–602. doi: 10.4314/tjpr.v10i5.8

94. Bami E, Ozakpınar OB, Ozdemir-Kumral ZN, Köroglu K, Ercan F, Cirakli Z, et al. Protective effect of ferulic acid on cisplatin induced nephrotoxicity in rats. *Environ Toxicol Pharmacol*. (2017) 54:105–11. doi: 10.1016/j.etap.2017.06.026

95. Olatunji LA, Okwusidi JI, Soladoye AO. Antidiabetic effect of *Anacardium occidentale* stem-bark in fructose-diabetic rats. *Pharm Biol*. (2005) 43:589–93. doi: 10.1080/13880200500301712

96. Alvi K, Ijaz MU, Hamza A, Ashraf A, Arooj J, Ehsan N. Nephroprotective effects of delphinidin against bisphenol A induced kidney damage in rats. *Pak Vet J*. (2022) 43:189–93.

97. Adeel M, Mansha MA, Asghar A, Rahman A, Saeed MA. Ameliorating hyperglycemia by *Ananas comosus* leaves extract. *Int J Biol Res*. (2020) 3:33–49.

98. Domitrović R, Cvijanović O, Šušnić V, Katalinić N. Renoprotective mechanisms of chlorogenic acid in cisplatin-induced kidney injury. *Toxicology*. (2014) 324:98–107. doi: 10.1016/j.tox.2014.07.004

99. Nalamolu KR. Anti-hyperglycemic and renal protective activities of *Andrographis paniculata* roots chloroform extract. *Iran J Pharmacol Ther*. (2006) 5:47–50.

100. Ji X, Li C, Ou Y, Li N, Yuan K, Yang G, et al. Andrographolide ameliorates diabetic nephropathy by attenuating hyperglycemia-mediated renal oxidative stress and inflammation via Akt/NF-κB pathway. *Mol Cell Endocrinol*. (2016) 437:268–79. doi: 10.1016/j.mce.2016.06.029

101. Ansari P, Flatt PR, Harriott P, Abdel-Wahab YHA. Evaluation of the antidiabetic and insulin releasing effects of *A. squamosa*, including isolation and characterization of active phytochemicals. *Plants (Basel)*. (2020) 9:1348. doi: 10.3390/plants9101348

102. Neelima S, Dwarakanadha Reddy P, Kothapalli Bannoth CS. Nephroprotective activity of *Annona squamosa* leaves against paracetamol-induced nephrotoxicity in rats: *in vitro* and *in vivo* experiments. *Future J Pharm Sci*. (2020) 6:131. doi: 10.1186/s43094-020-00149-4

103. Naushad M, Urooj M, Ahmad T, Husain GM, Kazmi MH, Zakir M. Nephroprotective effect of *Apium graveolens* L. against cisplatin-induced nephrotoxicity. *J Ayurveda Integr Med*. (2021) 12:607–15. doi: 10.1016/j.jaim.2021.06.005

104. Kumar M. Antihyperglycemic and antioxidant potential of oil from *Arachis hypogaea* L. in streptozotocin-nicotinamide induced diabetic rats. *Afr J Pharm Pharmacol*. (2013) 7:2374–81. doi: 10.5897/AJPP2013.3609

105. Ibrahim AK, AL-Azawi AH. Nephro-protective effect of (*Arachis hypogaea* L.) peanut skin extracts on CCl4 induced kidney damage in mice. *Exp Anim*. (2019) 900:15.

106. Daradka HM, Abas MM, Mohammad MAM, Jaffar MM. Antidiabetic effect of *Artemisia absinthium* extracts on alloxan-induced diabetic rats. *Comp Clin Pathol*. (2014) 23:1733–42. doi: 10.1007/s00580-014-1963-1

107. Refaat B, El-Boshy M. Protective antioxidative and anti-inflammatory actions of β-caryophyllene against sulfasalazine-induced nephrotoxicity in rat. *Exp Biol Med (Maywood)*. (2022) 247:691–9. doi: 10.1177/15353702211073804

108. Ajiboye BO, Ojo OA, Oyinloye BE, Okesola MA, Oluwatosin A, Boligon AA, et al. Investigation of the *in vitro* antioxidant potential of polyphenolic-rich extract of *Artocarpus heterophyllus* Lam stem bark and its antidiabetic activity in streptozotocin-induced diabetic rats. *J Evid Based Complementary Altern Med*. (2020) 25:2515690X2091612. doi: 10.1177/2515690X20916123

109. Mashayekhi M. Renoprotective effect of silymarin on gentamicin-induced nephropathy. *Afr J Pharm Pharmacol*. (2012) 6:2241–46. doi: 10.5897/AJPP12.628

110. Akkara PJ, Sabina EP. Pre-treatment with beta carotene gives protection against nephrotoxicity induced by bromobenzene via modulation of antioxidant system, pro-inflammatory cytokines and pro-apoptotic factors. *Appl Biochem Biotechnol*. (2020) 190:616–33. doi: 10.1007/s12010-019-03111-0

111. Ramachandran V, Dipanjan M, Umasankar P, Dhanabal SP, Satishkumar M, Antony S, et al. Hypoglycemic, antioxidant and hypolipidemic activity of *Asparagus racemosus* on streptozotocin-induced diabetic in rats. *Adv Appl Sci Res*. (2011) 2:179–85.

112. Roy S, Pradhan S, Mandal S, Das K, Nandi DK. Nephroprotective efficacy of *Asparagus racemosus* root extract on acetaminophen-induced renal injury in rats. *Acta Biol Szeged*. (2018) 62:17–23. doi: 10.14232/abs.2018.1.17-23

113. Patil P, Patil S, Mane A, Verma S. Antidiabetic activity of alcoholic extract of neem (*Azadirachta indica*) root bark. *Natl J Physiol Pharm Pharmacol*. (2013) 3:142. doi: 10.5455/njppp.2013.3.134-138

114. Abdel Moneim AE, Othman MS, Aref AM. *Azadirachta indica* attenuates cisplatin-induced nephrotoxicity and oxidative stress. *BioMed Res Int*. (2014) 2014:1–11. doi: 10.1155/2014/647131

115. Pareek A, Suthar M. Antidiabetic activity of extract of *Berberis aristata* root in streptozotocin induced diabetic rats. *Pharmacologyonline*. (2010) 2:179–85.

116. Ambwani S, Goswami R, Fuloria N, Tandon R, Ambwani TK. Assessment of *in vitro* anti-urolithiasis potential of *Berberis aristata* DC. *Explor Anim Med Res*. (2023) 09:340–8. doi: 10.31783/elsr.2023.92340348

117. Yu J, Zong G, Wu H, Zhang K. Podoplanin mediates the renoprotective effect of berberine on diabetic kidney disease in mice. *Acta Pharmacol Sin*. (2019) 40:1544–54. doi: 10.1038/s41401-019-0263-3

118. Khaksari M, Esmaili S, Abedloo R, Khastar H. Palmatine ameliorates nephrotoxicity and hepatotoxicity induced by gentamicin in rats. *Arch Physiol Biochem*. (2021) 127:273–8. doi: 10.1080/13813455.2019.1633354

119. Meliani N, Dib MEA, Allali H, Tabti B. Hypoglycaemic effect of *Berberis vulgaris* L. in normal and streptozotocin-induced diabetic rats. *Asian Pac J Trop BioMed*. (2011) 1:468–71. doi: 10.1016/S2221-1691(11)60102-0

120. Laamech J, Hilaly JE. Nephroprotective effects of *Berberis vulgaris* L. total extract on lead acetate-induced toxicity in mice. *Pharm Sci*. (2016) 78:326–33. doi: 10.4172/pharmaceutical-sciences.1000122

121. Babas S, Luka C, Istifanus G, Mayel M. Determination of the antidiabetic property of the aqueous extract of *Biophytum sensitivum* on streptozotocin induced diabetic rats. *Glob Sci J*. (2020) 8:2137–50.

122. Ijaz MU, Aziz S, Faheem M, Abbas K, Nasir S, Naz H, et al. Orientin attenuates cisplatin-induced renal toxicity by reducing oxidative stress and inflammation. *Pak Vet J*. (2021) 41:574–8. doi: 10.29261/pakvetj/2021.076

123. Chennasamudram SP, Kudugunti S, Boreddy PR, Moridani MY, Vasylyeva TL. Renoprotective effects of (+)-catechin in streptozotocin-induced diabetic rat model. *Nutr Res*. (2012) 32:347–56. doi: 10.1016/j.nutres.2012.03.015

124. Bhat VR, Chianeh YR, Udupa P, U A, Sheikh S. Hepatoprotective and renoprotective potentials of the aqueous extract of *Bixa orellana* on streptozotocin induced diabetic rats. *Biochem Pharmacol*. (2018) 7. doi: 10.4172/2167-0501.1000255

125. Ma J-Q, Zhang Y-J, Tian Z-K, Liu C-M. Bixin attenuates carbon tetrachloride induced oxidative stress, inflammation and fibrosis in kidney by regulating the Nrf2/TLR4/MyD88 and PPAR-γ/TGF-β1/Smad3 pathway. *Int Immunopharmacol*. (2021) 90:107117. doi: 10.1016/j.intimp.2020.107117

126. Xu GK, Qin XY, Wang GK, Xie GY, Li XS, Sun CY, et al. Antihyperglycemic, antihyperlipidemic and antioxidant effects of standard ethanol extract of *Bombax ceiba* leaves in high-fat-diet- and streptozotocin-induced type 2 diabetic rats. *Chin J Nat Med*. (2017) 15:168–77. doi: 10.1016/S1875-5364(17)30033-X

127. Xu GK, Sun CY, Qin XY, Han Y, Li Y, Xie GY, et al. Effects of ethanol extract of *Bombax ceiba* leaves and its main constituent mangiferin on diabetic nephropathy in mice. *Chin J Nat Med*. (2017) 15:597–605. doi: 10.1016/S1875-5364(17)30087-0

128. Pal PB, Sinha K, Sil PC. Mangiferin attenuates diabetic nephropathy by inhibiting oxidative stress mediated signaling cascade, TNFα related and mitochondrial dependent apoptotic pathways in streptozotocin-induced diabetic rats. *PloS One*. (2014) 9:e107220. doi: 10.1371/journal.pone.0107220

129. Dharsan R, Priya V, Ponnulakshmi R, Gayathri R, Madhan K, Shyamaladevi B, et al. Attenuation of diabetic nephropathy by a plant sterol lupeol: a biochemical analysis. *Drug Invent Today*. (2019) 12:1404–13.

130. Thirumalai T, Therasa SV, Elumalai E, David E. Hypoglycemic effect of *Brassica juncea* (seeds) on streptozotocin induced diabetic male albino rat. *Asian Pac J Trop BioMed*. (2011) 1:323–5. doi: 10.1016/S2221-1691(11)60052-X

131. Kumar A, Rana AK, Singh A, Singh A. Bioactivity of methanolic extract of *Brassica juncea* in animal model of diabetes mellitus. *Chin Herb Med*. (2019) 11:434–7. doi: 10.1016/j.chmed.2019.04.007

132. Tohamy AA, Aref AM, Abdel Moneim AE, Sayed RH. Cinnamic acid attenuates cisplatin-induced hepatotoxicity and nephrotoxicity. *J Basic Environ Sci*. (2016) 3:1–9. doi: 10.21608/jbes.2016.369594

133. Shah MA, Sarker MMR, Gousuddin M. Antidiabetic potential of *Brassica oleracea* var. *italica* in type 2 diabetic Sprague Dawley (SD) rats. *Int J Pharmacogn Phytochem Res*. (2016) 8:462–9. doi: 10.13140/RG.2.1.4509.3529

134. Kataya HAH, Hamza AA. Red cabbage (*Brassica oleracea*) ameliorates diabetic nephropathy in rats. *Evid Based Complement Alternat Med*. (2008) 5:281–7. doi: 10.1093/ecam/nem029

135. Hussain L, Qadir MI, Rehman S. Antihyperglycemic and hypolipidemic potential of *Caesalpinia decapetala* in alloxan-induced diabetic rabbits. *Bangladesh J Pharmacol*. (2014) 9:529–32. doi: 10.3329/bjp.v9i4.19961

136. Gowd V, Kang Q, Wang Q, Wang Q, Chen F, Cheng KW. Resveratrol: evidence for its nephroprotective effect in diabetic nephropathy. *Adv Nutr*. (2020) 11:1555–68. doi: 10.1093/advances/nmaa075

137. Ansari P, Flatt PR, Harriott P, Abdel-Wahab YHA. Anti-hyperglycaemic and insulin-releasing effects of *Camellia sinensis* leaves and isolation and characterisation of active compounds. *Br J Nutr*. (2021) 126:1149–63. doi: 10.1017/S0007114520005085

138. Ramya S, Prasanna G. Nephroprotective effect of *Camellia sinensis* L. on lead acetate induced male albino rats. *Int J PharmTech Res*. (2013) 5:511–5.

139. Juárez-Rojop IE, Díaz-Zagoya JC, Ble-Castillo JL, Miranda-Osorio PH, Castell-Rodríguez AE, Tovilla-Zárate CA, et al. Hypoglycemic effect of *Carica papaya* leaves in streptozotocin-induced diabetic rats. *BMC Complement Altern Med*. (2012) 12:236. doi: 10.1186/1472-6882-12-236

140. Singh SN, Vats P, Suri S, Shyam R, Kumria MML, Ranganathan S, et al. Effect of an antidiabetic extract of *Catharanthus roseus* on enzymic activities in streptozotocin induced diabetic rats. *J Ethnopharmacol*. (2001) 76:269–77. doi: 10.1016/S0378-8741(01)00254-9

141. Hashim M, Arif H, Tabassum B, Arif A, Rehman AA, Rehman S, et al. Protective effect of *Catharanthus roseus* extract on cadmium-induced toxicity in albino rats: a putative mechanism of detoxification. *Metabolites*. (2022) 12:1059. doi: 10.3390/metabo12111059

142. Oguntibeju O, Aboua Y, Goboza M. Vindoline-a natural product from *Catharanthus roseus* reduces hyperlipidemia and renal pathophysiology in experimental type 2 diabetes. *Biomedicines*. (2019) 7:59. doi: 10.3390/biomedicines7030059

143. Rahman S, Jamal MM, Parvin A, Mahfuz-Al-Mamun M, Islam MR. Antidiabetic activity of *Centella asiatica* (L.) urbana in alloxan induced type 1 diabetic model rats. *J Biosci*. (2012) 19:23–7. doi: 10.3329/jbs.v19i0.12996

144. Rao BV, Beera Heggade RH, Jagannath HH. Ethanolic leaf extract of *Centella asiatica* L. possesses nephroprotection in an animal model of nephrotoxicity. *Indian J Pharm Educ Res*. (2024) 58:S274–81. doi: 10.5530/ijper.58.1s.29

145. Amini N, Maleki M, Badavi M. Nephroprotective activity of naringin against chemical-induced toxicity and renal ischemia/reperfusion injury: a review. *Avicenna J Phytomed*. (2022) 12:357–70. doi: 10.22038/ajp.2022.19620

146. Zhuang LG, Zhang R, Jin GX, Pei XY, Wang Q, Ge XX. Asiaticoside improves diabetic nephropathy by reducing inflammation, oxidative stress, and fibrosis: an *in vitro* and *in vivo* study. *World J Diabetes*. (2024) 15:2111–22. doi: 10.4239/wjd.v15.i10.2111

147. Singh R, Parasuraman S, Kathiresan S. Antioxidant and antidiabetic activities of methanolic extract of bark of *Cinnamomum zeylanicum* in diabetic rats. *FRA*. (2020) 10:16–23. doi: 10.5530/fra.2020.1.4

148. Quyamuddin M, Kumar P, Kumar S, Haque MR. Nephroprotective activity of ethanolic extract of *Cinnamomum zeylanicum* bark against acetaminophen induced nephrotoxicity in albino rats. *J Drug Delivery Ther*. (2020) 10:80–6. doi: 10.22270/jddt.v10i4-s.4292

149. Sharma UK, Kumar R, Gupta A, Ganguly R, Pandey AK. Renoprotective effect of cinnamaldehyde in food color induced toxicity. *3 Biotech*. (2018) 8:212. doi: 10.1007/s13205-018-1241-z

150. Attanayake A, Jayatilaka K, Pathirana C, Mudduwa L. Antihyperglycemic activity of *Coccinia grandis* (L.) Voigt in streptozotocin induced diabetic rats. *Indian J Tradit Knowl*. (2015) 14:376–81.

151. Islam Z, Tahsin MR, Ullah FA, Islam TT, Tasnim NT, Sharif NT, et al. *In vivo* assessment of antidiabetic potential and mapping of pharmacological properties of ethanolic extract of leaves of *Coccinia grandis* on alloxan-induced diabetic rats. *Asian J Adv Res Rep*. (2019) 7:1–9. doi: 10.9734/ajarr/2019/v7i230169

152. Basavarajappa GM, Nanjundan PK, Alabdulsalam A, Asif AH, Shekharappa HT, Anwer MK, et al. Improved renoprotection in diabetes with combination therapy of *Coccinia indica* leaf extract and low-dose pioglitazone. *Separations*. (2020) 7:58. doi: 10.3390/separations7040058

153. Gray AM, Flatt PR. Insulin-releasing and insulin-like activity of the traditional anti-diabetic plant *Coriandrum sativum* (coriander). *Br J Nutr*. (1999) 81:203–9. doi: 10.1017/S0007114599000392

154. Singh DM, Puri D, Sawhney SK, Barman M, Bhardwaj S, Mishra R, et al. Nephroprotective screening of *Coriandrum sativum* L. leaves against gentamicin induced renaltoxicity in Wistar albino rats. *J Biol Act Prod Nat*. (2019) 9:465–83. doi: 10.1080/22311866.2019.1707709

155. Mohammed A, Wudil A, Alhassan A, Imam A, Muhammad I, Idi A. Hypoglycemic activity of *Curcuma longa* Linn root extracts on alloxan induced diabetic rats. *Saudi J Life Sci*. (2017) 2:43–9.

156. Khorsandi L, Orazizadeh M. Protective effect of *Curcuma longa* extract on acetaminophen induced nephrotoxicity in mice. *Daru*. (2008) 16:155–9.

157. Kim BH, Lee ES, Choi R, Nawaboot J, Lee MY, Lee EY, et al. Protective effects of curcumin on renal oxidative stress and lipid metabolism in a rat model of type 2 diabetic nephropathy. *Yonsei Med J*. (2016) 57:664. doi: 10.3349/ymj.2016.57.3.664

158. Purena R, Seth R, Bhatt R. Protective role of *Emblica officinalis* hydro-ethanolic leaf extract in cisplatin induced nephrotoxicity in rats. *Toxicol Rep*. (2018) 5:270–7. doi: 10.1016/j.toxrep.2018.01.008

159. ALTamimi JZ, AlFaris NA, Alshammari GM, Alagal RI, Aljabryn DH, Aldera H, et al. Ellagic acid protects against diabetic nephropathy in rats by regulating the transcription and activity of Nrf2. *J Funct Foods*. (2021) 79:104397. doi: 10.1016/j.jff.2021.104397

160. Tabassum N, Sabeeha S. Evaluation of renal and hypoglycaemic activities of ethanolic extract of *Eriobotrya japonica* fruits in Swiss albino mice. *World J Pharm Res*. (2015) 4:2028–39. doi: 10.13140/RG.2.1.1358.2963

161. Maurya AK, Tripathi S, Ahmed Z, Sahu RK. Antidiabetic and antihyperlipidemic effect of *Euphorbia hirta* in streptozotocin induced diabetic rats. *Der Pharmacia Lettre*. (2012) 4:703–7.

162. Kasireddy G, B S, Nadithe L, Chinnam P. Experimental evaluation of hypoglycemic effect of bark extract of *Ficus benghalensis* in streptozotocin-induced diabetic rats. *Natl J Physiol Pharm Pharmacol*. (2021) 11:1. doi: 10.5455/njppp.2021.11.12359202023012021

163. Bhowmik P, Shohan F, Baroi JA, Pranto TI, Ullah MR, Rupak M, et al. Evaluation of the effects of ethanolic extract of *Ficus benghalensis* on the lipid profile and kidney function in rat model. *Int Res J Gastroenterol Hepatol*. (2024) 7:22–8.

164. Joshi H, Vaishnav D, Sanghvi G, Rabadia S, Airao V, Sharma T, et al. *Ficus recemosa* bark extract attenuates diabetic complications and oxidative stress in STZ-induced diabetic rats. *Pharm Biol*. (2016) 54:1586–95. doi: 10.3109/13880209.2015.1110596

165. Xu H, Wang X, Cheng Y, Zhao J, Zhou Y, Yang J, et al. Ursolic acid improves diabetic nephropathy via suppression of oxidative stress and inflammation in streptozotocin-induced rats. *BioMed Pharmacother*. (2018) 105:915–21. doi: 10.1016/j.biopha.2018.06.055

166. Mandave P, Khadke S, Karandikar M, Pandit V, Ranjekar P, Kuvalekar A, et al. Antidiabetic, lipid normalizing, and nephroprotective actions of the strawberry: a potent supplementary fruit. *Int J Mol Sci*. (2017) 18:124. doi: 10.3390/ijms18010124

167. Ogbole FA, Crown OO, Olayeriju OS, Olaleye MT, Akindahunsi AA. Antidiabetic effect of methanolic extract of *Garcinia kola* leaves on streptozotocin-induced diabetic rats. *Glob Sci J*. (2019) 7:634–41.

168. Adedara IA, Daramola YM, Dagunduro JO, Aiyegbusi MA, Farombi EO. Renoprotection of kolaviron against benzo (A) pyrene-induced renal toxicity in rats. *Ren Fail*. (2015) 37:497–504. doi: 10.3109/0886022X.2015.1006085

169. Badole SL, Bodhankar SL. Investigation of antihyperglycemic activity of *Glycine max* (L.) Merr. on serum glucose level in diabetic mice. *J Complement Integr Med*. (2009) 6:4. doi: 10.2202/1553-3840.1207

170. Ramasamy A, Jothivel N, Das S, Swapna A, Albert AP, Barnwal P, et al. Evaluation of the protective role of *Glycine max* seed extract (soybean oil) in drug-induced nephrotoxicity in experimental rats. *J Diet Suppl*. (2018) 15:583–95. doi: 10.1080/19390211.2017.1358792

171. Rajendran S, Rajagopal P, G SB, Jayaraman S, Krithika C, Kasthuri R, et al. Nephro protective effects of beta sitosterol in high fat diet-induced diabetic nephropathy in experimental rats. *J Pharm Negat Results*. (2022) 13:1511–25.

172. Zangeneh MM, Zangeneh A, Tahvilian R, Moradi R. Evaluation of the nephroprotective effect of *Glycyrrhiza glabra* L aqueous extract on CCl4-induced nephrotoxicity in mice. *Comp Clin Pathol*. (2018) 27:1119–26. doi: 10.1007/s00580-018-2707-4

173. Abd El-Twab SM, Hozayen WG, Hussein OE, Mahmoud AM. 18 β-glycyrrhetinic acid protects against methotrexate-induced kidney injury by up-regulating the Nrf2/ARE/HO-1 pathway and endogenous antioxidants. *Ren Fail*. (2016) 38:1516–27. doi: 10.1080/0886022X.2016.1216722

174. Mohammadi G, Zangeneh MM, Rashidi K, Zangeneh A. Evaluation of nephroprotective and antidiabetic effects of *Gundelia tournefortii* aqueous extract on diabetic nephropathy in male mice. *Res J Pharmacogn*. (2018) 5:65–73. doi: 10.22127/rjp.2018.69223

175. Kim DH, Yoo TH, Lee SH, Kang HY, Nam BY, Kwak SJ, et al. Gamma linolenic acid exerts anti-inflammatory and anti-fibrotic effects in diabetic nephropathy. *Yonsei Med J*. (2012) 53:1165. doi: 10.3349/ymj.2012.53.6.1165

176. Ansari P, Flatt PR, Harriott P, Abdel-Wahab YHA. Insulin secretory and antidiabetic actions of *Heritiera fomes* bark together with isolation of active phytomolecules. *PloS One*. (2022) 17:e0264632. doi: 10.1371/journal.pone.0264632

177. Sultana MS, Golder M, Biswas B, Karmakar UK, Bokshi B, Alam MJ, et al. Antioxidative and antidiabetic potentials of the pneumatophores of *Heritiera fomes* Buch. Ham. *Dhaka Univ J Pharm Sci*. (2022) 20:283–91. doi: 10.3329/dujps.v20i3.59794

178. Lina HZ, Samy MM, Samir AEB, Fatma AM, Kawther MT, Abdelaaty AS. Hypoglycemic and antioxidant effects of *Hibiscus rosa-sinensis* L. leaves extract on liver and kidney damage in streptozotocin induced diabetic rats. *Afr J Pharm Pharmacol*. (2017) 11:161–9. doi: 10.5897/AJPP2017.4764

179. Cho K, Kim H, Rodriguez-Iturbe B, Vaziri ND. Niacin ameliorates oxidative stress, inflammation, proteinuria, and hypertension in rats with chronic renal failure. *Am J Physiol Renal Physiol*. (2009) 297:F106–13. doi: 10.1152/ajprenal.00126.2009

180. Johnson M, Olufunmilayo L, Adegboyega C, Adetayo O. Evaluation of antidiabetic and the effect of methanolic leaf extract of *Jatropha curcas* on some biochemical parameters in alloxan-induced diabetic male albino rats. *Eur J Med Plants*. (2014) 4:1501–12. doi: 10.9734/EJMP/2014/12500

181. Asuk AA, Atangwho IJ, Ugwu MN, Ati BU. *Jatropha curcas* leaf extract and fractions attenuate hyperglycemia, tissue oxidation, and kidney dysfunction in diabetic rats. *J Biochem Technol*. (2021) 12:84–90. doi: 10.51847/3ChBg8qAJJ

182. Fan X, Wei W, Huang J, Liu X, Ci X. Isoorientin attenuates cisplatin-induced nephrotoxicity through the inhibition of oxidative stress and apoptosis via activating the SIRT1/SIRT6/Nrf-2 pathway. *Front Pharmacol*. (2020) 11:264. doi: 10.3389/fphar.2020.00264

183. Srivastava V, Pattanayak C, Sarkar S, Das MC, Prakash A. Evaluation of antidiabetic, renoprotective, and antioxidant properties of aqueous extracts of *Linum usitatissimum* in streptozotocin-induced diabetic rats. *Med J Dr DY Patil Vidyapeeth*. (2024) 17:S308–14. doi: 10.4103/mjdrdypu.mjdrdypu_693_22

184. Ngo DH, Ngo DN, Vo TTN, Vo TS. Mechanism of action of *Mangifera indica* leaves for anti-diabetic activity. *Sci Pharm*. (2019) 87:13. doi: 10.3390/scipharm87020013

185. Bibu KJ, Jor AD. Protective effect of *Mangifera indica* Linn. on gentamicin induced nephrotoxicity in rats. *Indian J Anim Sci*. (2010) 80:965–68. doi: 10.56093/ijans.v80i10.1504

186. Hase M, Babazono T, Karibe S, Kinae N, Iwamoto Y. Renoprotective effects of tea catechin in streptozotocin- induced diabetic rats. *Int Urol Nephrol*. (2007) 38:693–9. doi: 10.1007/s11255-006-0051-0

187. Rezaeizadeh A, Zakaria Z, Abdollahi M, Meng GY, Mustapha NM, Hamid MB, et al. Antioxidant and antihyperglycaemic effects of an aqueous extract from *Momordica charantia* fruit in a type II diabetic rat model. *J Med Plants Res*. (2011) 5:2990–3001.

188. Kanchan DM, Somani GS, Peshattiwar VV, Kaikini AA, Sathaye S. Renoprotective effect of diosgenin in streptozotocin induced diabetic rats. *Pharmacol Rep*. (2016) 68:370–7. doi: 10.1016/j.pharep.2015.10.011

189. Bensiameur-Touati K, Kacimi G, Haffaf EM, Berdja S, Aouichat-Bouguerra S. *In vivo* subacute toxicity and antidiabetic effect of aqueous extract of *Nigella sativa*. *Evid Based Complement Alternat Med*. (2017) 2017:8427034. doi: 10.1155/2017/8427034

190. El-Sheikh SMA, Bahaa HM, Galal AAA, Metwally MMM, Said MA, Alattar RH, et al. Gastroprotective, hepatoprotective, and nephroprotective effects of thymol against the adverse effects of acetylsalicylic acid in rats: biochemical and histopathological studies. *Saudi J Biol Sci*. (2022) 29:103289. doi: 10.1016/j.sjbs.2022.103289

191. Erboga M, Kanter M, Aktas C, Sener U, Fidanol Erboga Z, Bozdemir Donmez Y, et al. Thymoquinone ameliorates cadmium-induced nephrotoxicity, apoptosis, and oxidative stress in rats is based on its anti-apoptotic and anti-oxidant properties. *Biol Trace Elem Res*. (2016) 170:165–72. doi: 10.1007/s12011-015-0453-x

192. Luka C, Olatunde A, Tijjani H, Olisa-Enewe I. Effect of aqueous extract of *Phaseolus vulgaris* L.(red kidney beans) on alloxan-induced diabetic Wistar rats. *Int J Sci Invent Today*. (2013) 2:292–301.

193. Gholampour F, Mohammadi Z, Karimi Z, Owji SM. Protective effect of genistein in a rat model of ischemic acute kidney injury. *Gene*. (2020) 753:144789. doi: 10.1016/j.gene.2020.144789

194. Kumar M, Dahiya V, Kasala ER, Bodduluru LN, Lahkar M. The renoprotective activity of hesperetin in cisplatin induced nephrotoxicity in rats: molecular and biochemical evidence. *BioMed Pharmacother*. (2017) 89:1207–15. doi: 10.1016/j.biopha.2017.03.008

195. Bagri P, Ali M, Aeri V, Bhowmik M, Sultana S. Antidiabetic effect of *Punica granatum* flowers: effect on hyperlipidemia, pancreatic cells lipid peroxidation and antioxidant enzymes in experimental diabetes. *Food Chem Toxicol*. (2009) 47:50–4. doi: 10.1016/j.fct.2008.09.058

196. Ankita P, Deepti B, Nilam M. Flavonoid rich fraction of *Punica granatum* improves early diabetic nephropathy by ameliorating proteinuria and disturbed glucose homeostasis in experimental animals. *Pharm Biol*. (2015) 53:61–71. doi: 10.3109/13880209.2014.910533

197. Fouad AA, Qutub HO, Al-Melhim WN. Nephroprotection of punicalagin in rat model of endotoxemic acute kidney injury. *Toxicol Mech Methods*. (2016) 26:538–43. doi: 10.1080/15376516.2016.1211207

198. Zulfiker AHM, Ripa FA, Rahman MM, Ullah MO, Hamid K, Khan MMR, et al. Antidiabetic and antioxidant effect of *Scoparia dulcis* in alloxan induced albino mice. *Int J PharmTech Res*. (2010) 2:2527–34.

199. Jose S, Adikay S. Effect of the ethanolic extract of *Scoparia dulcis* in cisplastin induced nephrotoxicity in Wistar rats. *Indian J Pharm Educ Res*. (2015) 49:S68–74. doi: 10.5530/ijper.49.4s.8

200. Mani A, Kushwaha K, Khurana N, Gupta J. p-Coumaric acid attenuates high-fat diet-induced oxidative stress and nephropathy in diabetic rats. *J Anim Physiol Anim Nutr*. (2022) 106:872–80. doi: 10.1111/jpn.13645

201. Singh A, Srivastav R, Pandey AK. Effect of the seeds of *Terminalia chebula* on blood serum, lipid profile and urine parameters in STZ induced diabetic rats. *J Pharmacogn Phytochem*. (2018) 7:1–5.

202. Ramalingam S, Muthu K, Almansour AI, Arumugam N, Perumal K. Chebulagic acid ameliorates oxidative stress and inflammation via regulation of insulin signaling pathway in high-fat diet/streptozotocin-induced diabetic rats. *Rev Bras Farmacogn*. (2024) 35:123–33. doi: 10.1007/s43450-024-00600-0

203. Khan MA, Kassianos AJ, Hoy WE, Alam AK, Healy HG, Gobe GC. Promoting plant-based therapies for chronic kidney disease. *J Evid Based Complementary Altern Med*. (2022) 27:2515690X221079688. doi: 10.1177/2515690X221079688

204. Al Hroob AM, Abukhalil MH, Alghonmeen RD, Mahmoud AM. Ginger alleviates hyperglycemia-induced oxidative stress, inflammation and apoptosis and protects rats against diabetic nephropathy. *BioMed Pharmacother*. (2018) 106:381–9. doi: 10.1016/j.biopha.2018.06.148

205. Qin S-S, Tian C-B, Qian Y-L, Wu X-R, Wen X, Zhao Y-L, et al. *Alstonia scholaris* (L.) R. Br. ameliorated diabetic nephropathy through PPAR-δ pathway. *J Ethnopharmacol*. (2025) 348:119839. doi: 10.1016/j.jep.2025.119839

206. Zhang N, Chen A, Dong Y, Dou D. *Fructus arctii* mitigates diabetic nephropathy via the Apoh/PPAR-γ pathway. *Mol Immunol*. (2025) 181:18–28. doi: 10.1016/j.molimm.2025.02.017

207. Bhattacharjee C, Dutta A. Nephro protective activity of bark of *Artocarpus heterophyllus* Lam. *Int J BioMed Adv Res*. (2017) 8:259–63.

208. Singh PK, Baxi D, Doshi A, A VR. Antihyperglycaemic and renoprotective effect of *Boerhaavia diffusa* L. in experimental diabetic rats. *J Complement Integr Med*. (2011) 8:20121020. doi: 10.2202/1553-3840.1533

209. Gheith I, El-Mahmoudy A. Novel and classical renal biomarkers as evidence for the nephroprotective effect of *Carica papaya* leaf extract. *Biosci Rep*. (2018) 38:BSR20181187. doi: 10.1042/BSR20181187

210. Cheng D, Liang B, Li Y. Antihyperglycemic effect of *Ginkgo biloba* extract in streptozotocin-induced diabetes in rats. *BioMed Res Int*. (2013) 2013:1–7. doi: 10.1155/2013/162724

211. Chen L-Z, Yang S, Ren D-M, Wang X-L, Shen T, Wang X-N. Ethnopharmacological relevance and anti-inflammatory mechanisms of *Lycopi herba* in diabetic nephropathy: integrated chemical profiling and pharmacological evaluation. *J Ethnopharmacol*. (2026) 364:121505. doi: 10.1016/j.jep.2026.121505

212. Najla O, Olfat A, Kholoud S, Enas N, Hanan S. Hypoglycemic and biochemical effects of *Matricaria chamomilla* leave extract in streptozotocin-induced diabetic rats. *J Health Sci*. (2012) 2:43–8.

213. Mahendran G, Rahman L. Ethnomedicinal, phytochemical and pharmacological updates on peppermint (Mentha × piperita L.) a review. *Phytother Res*. (2020) 34:2088–139. doi: 10.1002/ptr.6664

214. Lan PT, Trang TTH, Thom VT, Minh PH, Vui DT, Hang PTN, et al. Antidiabetic and renoprotective effects of ethyl acetate fraction of *Mimosa pudica* Linn. leaves extract in streptozotocin-induced diabetic mice. *Curr Bioact Compd*. (2022) 18:e070322201801. doi: 10.2174/1573407218666220307103828

215. Gupta R, Katariya P, Mathur M, Bajaj V, Yadav S, Kamal R, et al. Antidiabetic and renoprotective activity of *Momordica dioica* in diabetic rats. *Diabetol Croat*. (2011) 40:81–8.

216. Omodanisi E, Aboua Y, Oguntibeju O. Assessment of the anti-hyperglycaemic, anti-inflammatory and antioxidant activities of the methanol extract of *Moringa oleifera* in diabetes-induced nephrotoxic male Wistar rats. *Molecules*. (2017) 22:439. doi: 10.3390/molecules22040439

217. Lanjhiyana S, Garabadu D, Ahirwar D, Bigoniya P, Rana AC, Patra KC, et al. Hypoglycemic activity studies on root extracts of *Murraya koenigii* root in alloxan-induced diabetic rats. *J Nat Prod Plant Resour*. (2011) 1:91–104.

218. Dikshit P, Shukla K, Tyagi MK, Garg P, Gambhir JK, Shukla R. Antidiabetic and antihyperlipidemic effects of the stem of *Musa sapientum* Linn. in streptozotocin-induced diabetic rats. *J Diabetes*. (2012) 4:378–85. doi: 10.1111/j.1753-0407.2012.00198.x

219. Thadani S, Salman M, Tewari S, Singh S, Bhagchandani D, Ahmad A. Renoprotective effect of *Ocimum sanctum* in comparison with olmesartan medoxomil and pitavastatin in metformin treated diabetic rats. *Int J Pharm Sci Res*. (2015) 6:4433. doi: 10.13040/IJPSR.0975-8232.6(10).4433-41

220. Mansour HMM, Zeitoun AA, Abd-Rabou HS, El Enshasy HA, Dailin DJ, Zeitoun MAA, et al. Antioxidant and anti-diabetic properties of olive (*Olea europaea*) leaf extracts: *in vitro* and *in vivo* evaluation. *Antioxidants*. (2023) 12:1275. doi: 10.3390/antiox12061275

221. Attele AS, Zhou YP, Xie JT, Wu JA, Zhang L, Dey L, et al. Antidiabetic effects of *Panax ginseng* berry extract and the identification of an effective component. *Diabetes*. (2002) 51:1851–8. doi: 10.2337/diabetes.51.6.1851

222. Karuna R, Bharathi V, Reddy S, Ramesh B, Saralakumari D. Protective effects of *Phyllanthus amarus* aqueous extract against renal oxidative stress in streptozotocin-induced diabetic rats. *Indian J Pharmacol*. (2011) 43:414. doi: 10.4103/0253-7613.83112

223. Afolabi OB, Akinola AG, Raji AK, Obafemi OT, Olasehinde OR, Olorunlagba AO, et al. Renoprotective effect of polyphenolic-rich extract of *Phyllanthus niruri* in streptozotocin-induced diabetic nephropathy via moderation of oxidative stress and inhibition of kidney injury. *J Mol Histol*. (2026) 57:13. doi: 10.1007/s10735-025-10667-1

224. Santhakumari P, Prakasam A, Pugalendi KV. Antihyperglycemic activity of *Piper betle* leaf on streptozotocin-induced diabetic rats. *J Med Food*. (2006) 9:108–12. doi: 10.1089/jmf.2006.9.108

225. Hannan JMA, Ali L, Khaleque J, Akhter M, Flatt PR, Abdel-Wahab YHA. Aqueous extracts of husks of *Plantago ovata* reduce hyperglycaemia in type 1 and type 2 diabetes by inhibition of intestinal glucose absorption. *Br J Nutr*. (2006) 96:131–7. doi: 10.1079/BJN20061819

226. Maruthupandian A, Mohan V. Antidiabetic, antihyperlipidaemic and antioxidant activity of *Pterocarpus marsupium* Roxb. in alloxan induced diabetic rats. *Int J PharmTech Res*. (2011) 3:1681–7.

227. Zhang R, Zhou J, Jia Z, Zhang Y, Gu G. Hypoglycemic effect of *Rehmannia glutinosa* oligosaccharide in hyperglycemic and alloxan-induced diabetic rats and its mechanism. *J Ethnopharmacol*. (2004) 90:39–43. doi: 10.1016/j.jep.2003.09.018

228. Dai X, Su S, Cai H, Wei D, Yan H, Zheng T, et al. Protective effects of total glycoside from *Rehmannia glutinosa* leaves on diabetic nephropathy rats via regulating the metabolic profiling and modulating the TGF-β1 and Wnt/β-catenin signaling pathway. *Front Pharmacol*. (2018) 9:1012. doi: 10.3389/fphar.2018.01012

229. Hamzeh S, Farokhi F, Heydari R, Manaffar R. Renoprotective effect of hydroalcoholic extract of *Rheum ribes* root in diabetic female rats. *Avicenna J Phytomed*. (2014) 4:392–401.

230. Lee SH, Kim YS, Lee SJ, Lee BC. The protective effect of *Salvia miltiorrhiza* in an animal model of early experimentally induced diabetic nephropathy. *J Ethnopharmacol*. (2011) 137:1409–14. doi: 10.1016/j.jep.2011.08.007

231. Kulkarni CR, Joglekar MM, Patil SB, Arvindekar AU. Antihyperglycemic and antihyperlipidemic effect of *Santalum album* in streptozotocin induced diabetic rats. *Pharm Biol*. (2012) 50:360–5. doi: 10.3109/13880209.2011.604677

232. Krishnakumari S, Bhuvaneswari P. Nephroprotective effects of ethanolic extract of *Sesamum indicum* seeds (Linn.) in streptozotocin induced diabetic male albino rats. *Int J Green Pharm*. (2012) 6:330. doi: 10.4103/0973-8258.108249

233. Thomson H, Ojo O, Flatt P, AbdelWahab Y. Antidiabetic actions of aqueous bark extract of *Swertia chirayita* on insulin secretion, cellular glucose uptake and protein glycation. *J Exp Integr Med*. (2014) 4:268. doi: 10.5455/jeim.160814.or.110

234. Dey P, Singh J, Suluvoy JK, Dilip KJ, Nayak J. Utilization of *Swertia chirayita* plant extracts for management of diabetes and associated disorders: present status, future prospects and limitations. *Nat Prod Bioprospect*. (2020) 10:431–43. doi: 10.1007/s13659-020-00277-7

235. Bhuyan ZA, Rokeya B, Masum N, Hossain S, Mahmud I. Antidiabetic effect of *Syzygium cumini* L. seed on type 2 diabetic rats. *Dhaka Univ J Biol Sci*. (1970) 19:157–64. doi: 10.3329/dujbs.v19i2.8959

236. Koyagura N, Kumar V, Jamadar M, Huilgol S, Nayak N, Yendigeri S, et al. Antidiabetic and hepatoprotective activities of *Tamarindus indica* fruit pulp in alloxan induced diabetic rats. *Int J Pharmacol Clin Sci*. (2013) 2:33–40.

237. Navin S, Jha J, Abhinav D, Kumar DA, Ali M, Ghosh A, et al. Comparative hypoglycemic and nephroprotective effect of *Tinospora cordifolia* and metformin on alloxan induced diabetic mice. *World J Pharm Med Res*. (2017) 3:89–92.

238. Kishore L, Mazumder PM. Extraction and evaluation of *Trigonella foenum-graecum* L. seeds extract for attenuating the progression of nephropathy in diabetic rats. *Asian J Pharm Pharmacol*. (2019) 5:933–41. doi: 10.31024/ajpp.2019.5.5.12

239. Shivam, Gupta AK. Phytochemical profiling and renoprotective effects of methanolic extract of *Trigonella foenum-graecum* on diabetic nephropathy: an experimental study on rats. *Curr Indian Sci*. (2025) 4:e2210299X384759. doi: 10.2174/012210299X384759251117054626

240. Chang B, Chen W, Zhang Y, Yang P, Liu L. *Tripterygium wilfordii* mitigates hyperglycemia-induced upregulated Wnt/β-catenin expression and kidney injury in diabetic rats. *Exp Ther Med*. (2018) 15:3874–82. doi: 10.3892/etm.2018.5901

241. Shareef SM, Khaleel RA, Maryoosh TM. Nephroprotective effect of cranberry (*Vaccinium oxycoccos*) in streptozocin-induced diabetic nephropathy in mice. *Drug Metab Pers Ther*. (2024) 39:35–45. doi: 10.1515/dmpt-2023-0092

242. Cao X, Yao F, Liu W, Wang Y, Zhang Z, Zhang C, et al. *Vaccinium myrtillus* L. ameliorates diabetic nephropathy via modulating metabolites and gut microbiota in rats. *Front Pharmacol*. (2025) 16:1541947. doi: 10.3389/fphar.2025.1541947

243. Gopalakrishnan V, Sophia D, Rathi M, Meenakshi P, Suganya S, Raj C, et al. Amelioration of nitrobenzene-induced nephrotoxicity by the ethanol extract of the herb *Euphorbia hirta*. *Pharmacognosy Res*. (2011) 3:201. doi: 10.4103/0974-8490.85009

244. Swathi N, Sreedevi A, Bharathi K. Evaluation of nephroprotective activity of fruits of *Ficus hispida* on cisplatin-induced nephrotoxicity. *Pharmacogn J*. (2011) 3:62–8. doi: 10.5530/pj.2011.22.12

245. Yadav YC, Srivastava D, Saini V, Singhal S, Seth A, Kumar S, et al. Nephroprotective and curative activity of methanolic extract of *Ficus religiosa* L. latex in albino rats using cisplatin induced nephrotoxicity. *Pharmacologyonline*. (2011) 1:132–9.

246. Tebekeme O, Prosper AE. *Garcinia kola* extract reduced cisplatin-induced kidney dysfunction in rats. *Afr J Biochem Res*. (2007) 1:124–6.

247. Mansoor M, Brahmini CS, Rao S. Phytochemical and nephroprotective activity of *Ginkgo biloba* against gentamicin-induced nephrotoxicity in rats. *Int J Adv Pharm Med Bioallied Sci*. (2015) 3:98–101.

248. Kingsley UI. Effects of tomato extract (*Lycopersicon esculentum*) on carbimazole-induced alterations in kidney of albino rats. *Int J Res Rev*. (2018) 5:72–9.

249. Ullah N, Khan M, Khan T, Asif A, Ahmad W. *Mentha piperita* in nephrotoxicity a possible intervention to ameliorate renal derangements associated with gentamicin. *Indian J Pharmacol*. (2014) 46:166. doi: 10.4103/0253-7613.129309

250. Khan A, Simaiya P, Dubey B, Basedia D. Ethanolic extract of *Musa sapientum* spotted peel treatment for better ameliorates carbon tetrachloride nephrotoxicity. *World J Pharm Res*. (2022) 11:858–71.

251. Karami M, Mostafazadeh M, Sadeghi H, Sadeghi H, Mehraban F, Panahi KE, et al. Nephroprotective effect of *Nasturtium officinale* (watercress) ethanol extract and vitamin E on vancomycin-induced nephrotoxicity in rats. *Jundishapur J Nat Pharm Prod*. (2018) 13:e67178. doi: 10.5812/jjnpp.67178

252. Kalkan Y, Kapakin KAT, Kara A, Atabay T, Karadeniz A, Simsek N, et al. Protective effect of *Panax ginseng* against serum biochemical changes and apoptosis in kidney of rats treated with gentamicin sulphate. *J Mol Histol*. (2012) 43:603–13. doi: 10.1007/s10735-012-9412-4

253. Shehzadi I, Shah N, Khan M, Shuaib M, Shah M, Khan A, et al. *In vivo* antioxidant potential of *Raphanus sativus* seeds in rat kidney against CCl4-induced toxicity. *Pol J Environ Stud*. (2019) 29:277–84. doi: 10.15244/pjoes/94453

254. Mahalakshmi A, Prasanna G. Nephroprotective effect of *Syzygium cumini* Linn. against paracetamol induced toxicity in albino rats. *Adv Pharmacol Toxicol*. (2011) 12:25–32.

255. Ullah N, Azam Khan M, Khan T, Ahmad W. Protective potential of *Tamarindus indica* against gentamicin-induced nephrotoxicity. *Pharm Biol*. (2014) 52:428–34. doi: 10.3109/13880209.2013.840318

256. Mazani M, Mahmoodzadeh Y, Chinifroush AMM, Banaei S, Rezagholizadeh L, Mohammadnia A. Renoprotective effects of the methanolic extract of *Tanacetum parthenium* against carbon tetrachloride-induced renal injury in rats. *Avicenna J Phytomed*. (2018) 8:370–9.

257. Akhtar F, Azhar M, Aslam M, Javed K. Nephroprotective effect of khar-e-khasak khurd (*Tribulus terristris* Linn) on gentamicin-induced experimental nephrotoxicity in rats. *Asian J Res Nephrol*. (2020) 3:86–93.

258. Baracho NCDV, Monteiro NF, Borges MG, Arguelho RRDM. Effect of aqueous extract of the *Vigna angularis* in rats subjected to an experimental model of moderate chronic kidney disease. *Acta Cir Bras*. (2016) 31:527–32. doi: 10.1590/S0102-865020160080000005

259. Basist P, Parveen B, Zahiruddin S, Gautam G, Parveen R, Khan MA, et al. Potential nephroprotective phytochemicals: mechanism and future prospects. *J Ethnopharmacol*. (2022) 283:114743. doi: 10.1016/j.jep.2021.114743

260. Zhu X, Ding G, Ren S, Xi J, Liu K. The bioavailability, absorption, metabolism, and regulation of glucolipid metabolism disorders by quercetin and its important glycosides: a review. *Food Chem*. (2024) 458:140262. doi: 10.1016/j.foodchem.2024.140262

261. Almeida AF, Borge GIA, Piskula M, Tudose A, Tudoreanu L, Valentová K, et al. Bioavailability of quercetin in humans with a focus on interindividual variation. *Compr Rev Food Sci Food Saf*. (2018) 17:714–31. doi: 10.1111/1541-4337.12342

262. Jamwal R. Bioavailable curcumin formulations: a review of pharmacokinetic studies in healthy volunteers. *J Integr Med*. (2018) 16:367–74. doi: 10.1016/j.joim.2018.07.001

263. Cui Y, Zhou Q, Jin M, Jiang S, Shang P, Dong X, et al. Research progress on pharmacological effects and bioavailability of berberine. *Naunyn-Schmiedeberg's Arch Pharmacol*. (2024) 397:8485–514. doi: 10.1007/s00210-024-03199-0

264. Feng Q, Yu X, Xie J, Liu F, Zhang X, Li S, et al. Phillygenin improves diabetic nephropathy by inhibiting inflammation and apoptosis via regulating TLR4/MyD88/NF-κB and PI3K/AKT/GSK3β signaling pathways. *Phytomedicine*. (2025) 136:156314. doi: 10.1016/j.phymed.2024.156314

265. Yang L, Liao M. Influence of myrcene on inflammation, matrix accumulation in the kidney tissues of streptozotocin-induced diabetic rat. *Saudi J Biol Sci*. (2021) 28:5555–60. doi: 10.1016/j.sjbs.2020.11.090

266. He J, Hong Q, Chen B, Cui S, Liu R, Cai G, et al. Ginsenoside Rb1 alleviates diabetic kidney podocyte injury by inhibiting aldose reductase activity. *Acta Pharmacol Sin*. (2022) 43:342–53. doi: 10.1038/s41401-021-00788-0

267. Sun M, Ye H, Zheng C, Jin Z, Yuan Y, Weng H. Astragalin ameliorates renal injury in diabetic mice by modulating mitochondrial quality control via AMPK-dependent PGC1α pathway. *Acta Pharmacol Sin*. (2023) 44:1676–86. doi: 10.1038/s41401-023-01064-z

268. Li G, Liu C, Yang L, Feng L, Zhang S, An J, et al. Syringaresinol protects against diabetic nephropathy by inhibiting pyroptosis via NRF2-mediated antioxidant pathway. *Cell Biol Toxicol*. (2023) 39:621–39. doi: 10.1007/s10565-023-09790-0

269. Dong Z, Chen CX. Effect of catalpol on diabetic nephropathy in rats. *Phytomedicine*. (2013) 20:1023–9. doi: 10.1016/j.phymed.2013.04.007

270. Duan J, Pei F, Miao J, Liu S, Tan L, Lu M, et al. Swietenine improved the progression of diabetic nephropathy through inhibiting ferroptosis via activating akt/GSK-3β/Nrf2 signaling pathway. *J Ethnopharmacol*. (2025) 349:119981. doi: 10.1016/j.jep.2025.119981

271. Chang L, Wang Q, Ju J, Li Y, Cai Q, Hao L, et al. Magnoflorine ameliorates inflammation and fibrosis in rats with diabetic nephropathy by mediating the stability of lysine-specific demethylase 3A. *Front Physiol*. (2020) 11:580406. doi: 10.3389/fphys.2020.580406

272. Lu H-J, Tzeng T-F, Liou S-S, Da Lin S, Wu M-C, Liu I-M. Ruscogenin ameliorates diabetic nephropathy by its anti-inflammatory and anti-fibrotic effects in streptozotocin-induced diabetic rat. *BMC Complement Altern Med*. (2014) 14:110. doi: 10.1186/1472-6882-14-110

273. Alsuliam SM, Albadr NA, Almaiman SA, Al-Khalifah AS, Alkhaldy NS, Alshammari GM. Fenugreek seed galactomannan aqueous and extract protects against diabetic nephropathy and liver damage by targeting NF-κB and Keap1/Nrf2 axis. *Toxics*. (2022) 10:362. doi: 10.3390/toxics10070362

274. Lv C, Cheng T, Zhang B, Sun K, Lu K. Triptolide protects against podocyte injury in diabetic nephropathy by activating the Nrf2/HO-1 pathway and inhibiting the NLRP3 inflammasome pathway. *Ren Fail*. (2023) 45:2165103. doi: 10.1080/0886022X.2023.2165103

275. Zhang M, Chen Y, Yang M, Fan X, Xie H, Zhang L, et al. Celastrol attenuates renal injury in diabetic rats via MAPK/NF-κB pathway. *Phytother Res*. (2019) 33:1191–8. doi: 10.1002/ptr.6314

276. Gumbar S, Bhardwaj S, Mehan S, Khan Z, Narula AS, Kalfin R, et al. Renal mitochondrial restoration by gymnemic acid in gentamicin-mediated experimental nephrotoxicity: evidence from serum, kidney and histopathological alterations. *Front Pharmacol*. (2023) 14:1218506. doi: 10.3389/fphar.2023.1218506

277. Hardevinder Pal S, Thakur Gurjeet S, Randhir S. Evaluation of the renoprotective effect of syringic acid against nephrotoxicity induced by cisplatin in rats. *J Appl Pharm Sci*. (2021) 11:080–5. doi: 10.7324/JAPS.2021.11s109

278. Makni M, Chtourou Y, Garoui E, Boudawara T, Fetoui H. Carbon tetrachloride-induced nephrotoxicity and DNA damage in rats: protective role of vanillin. *Hum Exp Toxicol*. (2012) 31:844–52. doi: 10.1177/0960327111429140

279. Kocaturk H, Bedir F, Sener E, Aktas M, Akyuz S, Dabanlioglu B, et al. The effect of lutein on vancomycin-induced oxidative kidney damage in rats. *Ann Med Res*. (2021) 28:1643. doi: 10.5455/annalsmedres.2020.09.982

280. Yin M, Jiang N, Guo L, Ni Z, Al-Brakati AY, Othman MS, et al. Oleuropein suppresses oxidative, inflammatory, and apoptotic responses following glycerol-induced acute kidney injury in rats. *Life Sci*. (2019) 232:116634. doi: 10.1016/j.lfs.2019.116634

281. Dou J-Y, Zhang M, Cen H, Chen Y-Q, Wu Y-F, Lu F, et al. *Salvia miltiorrhiza* Bunge (Danshen) and bioactive compound tanshinone IIA alleviates cisplatin-induced acute kidney injury through regulating PXR/NF-κB signaling. *Front Pharmacol*. (2022) 13:860383. doi: 10.3389/fphar.2022.860383

282. Sharma A, Thakur R, Lingaraju MC, Kumar D, Mathesh K, Telang AG, et al. Betulinic acid attenuates renal fibrosis in rat chronic kidney disease model. *BioMed Pharmacother*. (2017) 89:796–804. doi: 10.1016/j.biopha.2017.01.181

283. Algefare AI. Renoprotective and oxidative stress-modulating effects of taxifolin against cadmium-induced nephrotoxicity in mice. *Life*. (2022) 12:1150. doi: 10.3390/life12081150

284. Albalawi RS, Binmahfouz LS, Hareeri RH, Shaik RA, Bagher AM. Parthenolide phytosomes attenuated gentamicin-induced nephrotoxicity in rats via activation of Sirt-1, Nrf2, OH-1, and NQO1 axis. *Molecules*. (2023) 28:2741. doi: 10.3390/molecules28062741

285. Naveen YP, Urooj A, Byrappa K. A review on medicinal plants evaluated for anti-diabetic potential in clinical trials: present status and future perspective. *J Herb Med*. (2021) 28:100436. doi: 10.1016/j.hermed.2021.100436

286. Jie L, Hai H. Clinical observation of *Gingko biloba* extract injection in treating early diabetic nephropathy. *Chin J Integr Med*. (2005) 11:226–8. doi: 10.1007/BF02836510

287. Wang Y, Sun Z, Yu J, Yang B, Yin H, Wang Y. Effects of *ginkgo biloba* extract on prevention of development of diabetic nephropathy in type 2 diabetic patients. *Diabetologia*. (2012) 55:456.

288. Yin J, Chen S, Wei L, Jia W. 11-LB: the efficacy and safety of *Abelmoschus manihot* for reduction of microalbuminuria in patients with type 2 diabetes and diabetic nephropathy—a multicenter, randomized, open-label, parallel-controlled clinical trial. *Diabetes*. (2023) 72:11–LB. doi: 10.2337/db23-11-LB

289. Barocio-Pantoja M, Quezada-Fernández P, Cardona-Müller D, Jiménez-Cázarez MB, Larios-Cárdenas M, González-Radillo OI, et al. Green tea extract increases soluble RAGE and improves renal function in patients with diabetic nephropathy. *J Med Food*. (2021) 24:1264–70. doi: 10.1089/jmf.2020.0212

290. Ansari Z, Nasiruddin M, Khan R, Haque S. Protective role of *Nigella sativa* in diabetic nephropathy: a randomized clinical trial. *Saudi J Kidney Dis Transpl*. (2017) 28:9. doi: 10.4103/1319-2442.198093

291. Faridi E, Faridi P, Vosoughi M, Ansari R, Dehghanian A, Roozbeh J, et al. The effectiveness and safety of pomegranate (*Punica granatum* L.) juice powder in patients with diabetic nephropathy, a double-blinded randomized clinical trial. *Tradit Integr Med*. (2026) 10:374–81. doi: 10.18502/tim.v10i4.20724

292. Li P, Chen Y, Liu J, Hong J, Deng Y, Yang F, et al. Efficacy and safety of Tangshen formula on patients with type 2 diabetic kidney disease: a multicenter double-blinded randomized placebo-controlled trial. *PloS One*. (2015) 10:e0126027. doi: 10.1371/journal.pone.0126027

293. Liu J, Gao L, Fu B, Yang H, Zhang L, Che S, et al. Efficacy and safety of Zicuiyin decoction on diabetic kidney disease: a multicenter, randomized controlled trial. *Phytomedicine*. (2022) 100:154079. doi: 10.1016/j.phymed.2022.154079

294. Yang H, Xia S, Cong Y, Yang X, Min J, Wu T. Effects of Qidan Tangshen granule on diabetic kidney disease in patients with type 2 diabetes. *Diabetes Res Clin Pract*. (2024) 209:111128. doi: 10.1016/j.diabres.2024.111128

295. Zhu Q, Qi X, Wu Y, Wang K. Clinical study of Total Glucosides of Paeony for the treatment of diabetic kidney disease in patients with diabetes mellitus. *Int Urol Nephrol*. (2016) 48:1873–80. doi: 10.1007/s11255-016-1345-5

296. Wang M-R, Yu L-H, Wang T-T, Wang Y-M, Han M-X. Effect of Shenqi Dihuang decoction on inflammatory factor, renal function and microcirculation in patients with early diabetic nephropathy. *China J Chin Mater Med*. (2018) 43:1276–81. doi: 10.19540/j.cnki.cjcmm.2018.0050

297. Setyawati A, Jafar N, Raihan M, Kristianto H. Randomized controlled trial on the regulation of MMP-9-NGAL ratio in diabetic patients with early renal dysfunction by *Piper crocatum* ruiz & pav. functional foods. *J Pharm Pharmacogn Res*. (2025) 13:1847–60. doi: 10.56499/jppres25.2499_13.6.1847

298. Zhu H, Shi Z, Chen Y. Effect of extract of *Ginkgo bilboa* leaf on early diabetic nephropathy. *Chin J Integr Tradit West Med*. (2005) 25:889–91.

299. Yu JY, Xiong NN, Guo HF. Clinical observation on diabetic nephropathy treated with alcohol of *Abelmoschus manihot*. *Chin J Integr Tradit West Med*. (1995) 15:263–5.

300. Ni HX, Luo SS, Shao GM. Effect of *Acanthopanax senticosus* injection on plasma and urinary endothelin in early stage of diabetic nephropathy. *Chin J Integr Tradit West Med*. (2001) 21:105–7.

301. Liu Y, Yang L, Liu J. Clinical observation on treatment of early diabetic nephropathy by milkvetch injection combined with captopril. *Chin J Integr Tradit West Med*. (2005) 25:993–5.

302. Wang H, Chen Y. Clinical observation on treatment of diabetic nephropathy with Compound *Fructus arctii* mixture. *Chin J Integr Tradit West Med*. (2004) 24:589–92.

303. Hendre A, Patil S, Sontakke A, Phatak R. Ameliorating effect of turmeric on kidney function in patients with type 2 diabetes mellitus. *Int J Pharm Sci Res*. (2022) 13:4019–24. doi: 10.13040/IJPSR.0975-8232.13(10).4019-24

304. Espinoza NAG, Tello HM, Córdova RS, Sarabia NM, Gaxiola JIO, Gingele JAP, et al. Microalbuminuria in patients with type 2 diabetes mellitus treated with a phytoformula as adjuvant. *Am Heart J*. (2024) 278:6–7. doi: 10.1016/j.ahj.2024.09.019

305. Najarzade A, Hemayati R, Zavar Reza J, Fallahzade H, Mozaffari-Khosravi H, Taghizadeh M, et al. Effects of *Hibiscus subdariffa* on albuminuria and hypertension in patients with diabetic nephropathy. *TB*. (2016) 14:107–18.

306. Mirzaei H, Enayati A, Kiani M, Ghorbani S, Zengin G, Amirkhanlou S. Comparison of *Hibiscus sabdariffa* L. extract and hydrochlorothiazide as adjuncts to valsartan in managing hypertension in type 2 diabetic nephropathy: a randomized clinical trial. *Avicenna J Phytomed*. (2025) 16:330–41. doi: 10.22038/ajp.2025.26575

307. Song X-Y, Chen Q, Qi X-Y. Effect of Liuwei Dihuang pill on erythrocyte aldose reductase activity in early diabetic nephropathy patients. *Chin J Integr Tradit West Med*. (2004) 24:1087–90.

308. Zhang X, Gao C, Yin X. Effect of modified Shenqi pill combined with conventional western medicine on diabetic nephropathy and its influence on netrin-1 level and renal function. *Afr Health Sci*. (2025) 25:406–21. doi: 10.4314/ahs.v25i1.31

309. Lang J, Cao H, Wei A. Comparative study on effect of *Panax notoginseng* and ticlid in treating early diabetic nephropathy. *Chin J Integr Tradit West Med*. (1998) 18:727–9.

310. Guo D-Z, Wang Y-H, Chen Z-Q. Effect of treatment in 39 patients with diabetic nephropathy by safflor yellow and benazepril in combination. *Chin J Integr Tradit West Med*. (2008) 28:360–3.

311. Zhao Y, Zhang X. Effect of Tongxinluo capsule on plasma endothelin in patients with diabetic nephropathy. *Chin J Integr Tradit West Med*. (2005) 25:131–3.

312. Ge Y, Xie H, Li S, Jin B, Hou J, Zhang H, et al. Treatment of diabetic nephropathy with *Tripterygium wilfordii* Hook F extract: a prospective, randomized, controlled clinical trial. *J Transl Med*. (2013) 11:134. doi: 10.1186/1479-5876-11-134

313. Yonghui M. WCN26-2234 a randomised controlled trial of *Tripterygium wilfordii* polyglycoside in the treatment of stage IV diabetic nephropathy. *Kidney Int Rep*. (2026) 11:105993. doi: 10.1016/j.ekir.2026.105993

314. Aghadavod E, Khodadadi S, Baradaran A, Nasri P, Bahmani M, Rafieian-Kopaei M. Role of oxidative stress and inflammatory factors in diabetic kidney disease. *Iran J Kidney Dis*. (2016) 10:337–43.

315. Sharma M, Gupta S, Singh K, Mehndiratta M, Gautam A, Kalra OP, et al. Association of glutathione-S-transferase with patients of type 2 diabetes mellitus with

and without nephropathy. *Diabetes Metab Syndr*. (2016) 10:194–7. doi: 10.1016/j.dsx.2016.06.006

316. McClelland AD, Herman-Edelstein M, Komers R, Jha JC, Winbanks CE, Hagiwara S, et al. miR-21 promotes renal fibrosis in diabetic nephropathy by targeting PTEN and SMAD7. *Clin Sci*. (2015) 129:1237–49. doi: 10.1042/CS20150427

317. Mahmoodnia L, Aghadavod E, Beigrezaei S, Rafieian-Kopaei M. An update on diabetic kidney disease, oxidative stress and antioxidant agents. *J Renal Inj Prev*. (2017) 6:153–7. doi: 10.15171/jrip.2017.30

318. Chen H-W, Yang M-Y, Hung T-W, Chang Y-C, Wang C-J. *Nelumbo nucifera* leaves extract attenuate the pathological progression of diabetic nephropathy in high-fat diet-fed and streptozotocin-induced diabetic rats. *J Food Drug Anal*. (2019) 27:736–48. doi: 10.1016/j.jfda.2018.12.009

319. Filla LA, Edwards JL. Metabolomics in diabetic complications. *Mol Biosyst*. (2016) 12:1090–105. doi: 10.1039/C6MB00014B

320. Jin H, Wang Y, Wang D, Zhang L. Effects of Qingshen granules on the oxidative stress-NF/kB signal pathway in unilateral ureteral obstruction rats. *Evid Based Complement Alternat Med*. (2018) 2018:4761925. doi: 10.1155/2018/4761925

321. Singh A, Kukreti R, Saso L, Kukreti S. Mechanistic insight into oxidative stress-triggered signaling pathways and type 2 diabetes. *Molecules*. (2022) 27:950. doi: 10.3390/molecules27030950

322. Volpe CMO, Villar-Delfino PH, Dos Anjos PMF, Nogueira-Machado JA. Cellular death, reactive oxygen species (ROS) and diabetic complications. *Cell Death Dis*. (2018) 9:119. doi: 10.1038/s41419-017-0135-z

323. Satirapoj B, Adler SG. Comprehensive approach to diabetic nephropathy. *Kidney Res Clin Pract*. (2014) 33:121–31. doi: 10.1016/j.krcp.2014.08.001

324. Mallipattu SK, Uribarri J. Advanced glycation end product accumulation: a new enemy to target in chronic kidney disease? *Curr Opin Nephrol Hypertens*. (2014) 23:547–54. doi: 10.1097/MNH.0000000000000062

325. Steenbeke M, Speeckaert R, Desmedt S, Glorieux G, Delanghe JR, Speeckaert MM. The role of advanced glycation end products and its soluble receptor in kidney diseases. *Int J Mol Sci*. (2022) 23:3439. doi: 10.3390/ijms23073439

326. Jha JC, Banal C, Chow BSM, Cooper ME, Jandeleit-Dahm K. Diabetes and kidney disease: role of oxidative stress. *Antioxid Redox Signal*. (2016) 25:657–84. doi: 10.1089/ars.2016.6664

327. Mason RM, Wahab NA. Extracellular matrix metabolism in diabetic nephropathy. *J Am Soc Nephrol*. (2003) 14:1358–73. doi: 10.1097/01.ASN.0000065640.77499.D7

328. Xu Z, Cai K, Su S-L, Zhu Y, Liu F, Duan J-A. Salvianolic acid B and tanshinone IIA synergistically improve early diabetic nephropathy through regulating PI3K/akt/NF-κB signaling pathway. *J Ethnopharmacol*. (2024) 319:117356. doi: 10.1016/j.jep.2023.117356

329. Ji T, Wang J, Xu Z, Cai H-D, Su S, Peng X, et al. Combination of mulberry leaf active components possessed synergetic effect on SD rats with diabetic nephropathy by mediating metabolism, Wnt/β-catenin and TGF-β/Smads signaling pathway. *J Ethnopharmacol*. (2022) 292:115026. doi: 10.1016/j.jep.2022.115026

330. Alebiosu CO, Ayodele OE. The global burden of chronic kidney disease and the way forward. *Ethn Dis*. (2005) 15:418–23.

331. Alebiosu CO. Detrimental effects of late referral for dialysis. *Afr J Health Sci*. (2001) 8:89–92.

332. Salako B, Ayodele O, Kadiri S, Arije A. Assessment of blood pressure control in a black African population. *Cardiol Trop*. (2002) 28:3–6.

333. Ubhenin AE, Ikebuiro JO, Anura F, Idris RI, Erharuyi O. Novel approaches and therapeutic targets for diabetic nephropathy: advances and promising strategies. *Trop J Pharm Res*. (2024) 22:2567–74. doi: 10.4314/tjpr.v22i12.21

334. Rafieian-Kopaei M. Medicinal plants for renal injury prevention. *J Renal Inj Prev*. (2013) 2:63–5. doi: 10.12861/jrip.2013.21

335. Gupta RC, Chang D, Nammi S, Bensoussan A, Bilinski K, Roufogalis BD. Interactions between antidiabetic drugs and herbs: an overview of mechanisms of action and clinical implications. *Diabetol Metab Syndr*. (2017) 9:59. doi: 10.1186/s13098-017-0254-9

336. Sugahara M, Pak WLW, Tanaka T, Tang SCW, Nangaku M. Update on diagnosis, pathophysiology, and management of diabetic kidney disease. *Nephrology*. (2021) 26:491–500. doi: 10.1111/nep.13860

337. Koo N, Lee EJ, Chang J-E, Lee K-R, Chae Y-J. Pharmacokinetic landscape and interaction potential of SGLT2 inhibitors: bridging *in vitro* findings and clinical implications. *Pharmaceutics*. (2025) 17:1604. doi: 10.3390/pharmaceutics17121604

338. Kamiyama M, Iijima K, Okuzawa R, Kawata R, Kimura A, Shinohara Y, et al. Mechanisms of the effects of polyphenols on diabetic nephropathy. *Curr Issues Mol Biol*. (2025) 47:735. doi: 10.3390/cimb47090735

339. Liu S, Gui Y, Wang MS, Zhang L, Xu T, Pan Y, et al. Serum integrative omics reveals the landscape of human diabetic kidney disease. *Mol Metab*. (2021) 54:101367. doi: 10.1016/j.molmet.2021.101367

340. Vanaie A, Shahidi S, Iraj B, Siadat ZD, Kabirzade M, Shakiba F, et al. Curcumin as a major active component of turmeric attenuates proteinuria in patients with overt diabetic nephropathy. *J Res Med Sci Off J Isfahan Univ Med Sci*. (2019) 24:77. doi: 10.4103/jrms.JRMS_1055_18

341. Zhao C, Pu Z, Gao J, Liu C, Xing J, Lang W, et al. Multiomics" analyses combined with systems pharmacology reveal the renoprotection of mangiferin monosodium salt in rats with diabetic nephropathy: focus on improvements in renal ferroptosis, renal inflammation, and podocyte insulin resistance. *J Agric Food Chem*. (2023) 71:358–81. doi: 10.1021/acs.jafc.2c05595

Glossary

DM	diabetes mellitus
DN	diabetic nephropathy
ESRD	end-stage renal disease
GFR	glomerular filtration rate
T2DM	type 2 diabetes mellitus
T1DM	type 1 diabetes mellitus
DKD	diabetic kidney disease
CKD	chronic kidney disease
HbA1c	glycated hemoglobin
LDL	low-density lipoprotein
VLDL	very low-density lipoprotein
TG	triglyceride
HDL	high-density lipoprotein
GSP	glycated serum protein
MMPs	matrix metalloproteinases
ECM	extracellular matrix
MCP-1	monocyte chemoattractant protein-1
NGAL	neutrophil gelatinase-associated lipocalin
NAG	N-acetyl-beta-glucosaminidase
NO	nitric oxide
IGF-1	insulin-like growth factor-1
VEGF	vascular endothelial growth factor
SGLT2	sodium-glucose cotransporter-2
GLP-1	glucagon-like peptide-1
NLRP3	NOD-like receptor family pyrin domain containing 3
PRRs	pattern recognition receptors
TLRs	Toll-like receptors
NLRs	NOD-like receptors
DAMPs	damage-associated molecular patterns
RAAS	renin–angiotensin–aldosterone system
ACEIs	angiotensin-converting enzyme inhibitors
ARBs	angiotensin II receptor blockers
NSAIDs	nonsteroidal anti-inflammatory drugs
AGEs	advanced glycation end products
RAGE	receptor for advanced glycation end products
sRAGE	soluble receptor for advanced glycation end products
ROS	reactive oxygen species
LPO	lipid peroxidation
NF-κB	nuclear factor kappa B
PKC	protein kinase C
TGF-β1	transforming growth factor beta-1
MAPK	mitogen-activated protein kinase
IL-1	interleukin-1
IL-1β	interleukin-1 beta
IL-6	interleukin-6
IL-8	interleukin-8
IL-18	interleukin-18
IL-33	interleukin-33
TNF-α	tumor necrosis factor-alpha
SOD	superoxide dismutase
CAT	catalase
MDA	malondialdehyde
BUN	blood urea nitrogen
DPPH	2,2-diphenyl-1-picrylhydrazyl
DPP-IV	dipeptidyl peptidase-4
RBC	red blood cell
TBARS	thiobarbituric acid reactive substrates
GSH	glutathione
GST	glutathione S-transferase
GPx	glutathione peroxidase
GR	glutathione reductase
MPO	myeloperoxidase
OGTT	oral glucose tolerance test